

Seeds of Market-Based Capitalism: Christian Missionaries and China's Modernization *

First version: November 2024

This version: May 2026

RICCARDO DI CATO[†]

JIACHEN LI[‡]

Abstract

This paper examines the historical and institutional roots of China's market-based capitalism through the lens of financial development. We study how Christian missionaries facilitated the adoption of modern banks and the establishment of enterprises between 1857 and 1936, and how this historical exposure interacted with the market-oriented reforms of the 1980s to shape growth through 2022. Using a staggered difference-in-differences design for the pre-1936 period, we find that missionary presence increased modern banks by 31% and enterprises by 27%. Our evidence points to two mechanisms: practical education and the diffusion of useful knowledge through publications and translations. Missionaries, especially Protestants, introduced training in subjects such as mathematics, bookkeeping, economics, and English, helping local actors operate modern financial institutions. They also translated Western texts into vernacular Chinese, helping diffuse useful knowledge relevant to modern economic and financial practices. Using an instrumental variable strategy for long-run exposure, we document that these effects weakened between 1949 and 1978 but re-emerged after the market reforms. Overall, the results suggest that missionaries transmitted not only human capital, but also institution-specific knowledge that helped plant the seeds of market-based capitalism in China.

*We thank Ruixue Jia, Ruiming Liu, Samuel Bazzi, Sara Lowes, Paul Niehaus, Gaurav Khanna, Craig McIntosh, Daryl Ireland, Laura Muñoz Blanco, Martin Fernandez Sanchez, as well as seminar and conference participants at UC San Diego, Boston University, University of Bologna, Xiamen University, and ASREC for helpful feedback.

[†]University of California San Diego. PhD Candidate of Economics. Email: rdicato@ucsd.edu

[‡]Hong Kong Baptist University. Email: jiachenli@hkbu.edu.hk

1 Introduction

What explains the origins and takeoff of market-based capitalism? This question has long been central to economics, but it remains especially important for understanding China. Since the market-oriented reforms of 1978, China has become increasingly integrated into markets, private enterprise, and modern finance, with GDP growth averaging over 9 percent annually and nearly 800 million people lifted out of poverty ([World Bank, 2024](#)). Yet many of the institutional and human capital foundations that made this transition possible were formed much earlier. This paper studies one historical source of these foundations: Christian missionaries in China.

The role of religious institutions in capitalist development is debated. Some work views religion as central to Europe’s economic rise ([Weber, 2001](#); [Becker and Woessmann, 2009](#); [Becker et al., 2016](#)), while other work shows how specific religious laws can constrain financial and institutional change ([Kuran, 2012, 2018](#); [Rubin, 2011](#)). Although China has been somewhat peripheral to this debate, it experienced notable exposure to Western religious institutions and capitalist ideology before 1949, and later went through a sharp period of institutional reversal under Communist rule before reopening to markets in the 1980s.

This paper offers a new perspective on China’s modernization by examining how Christian missionaries influenced the adoption of modern banking and enterprises between 1857 and 1936, and how this historical exposure interacted with the 1980s reforms to shape recent growth. We focus on one channel: financial innovation. Missionaries did not simply expose China to Western religion. They also brought schools, books, translations, and practical knowledge that could help local actors understand and operate new economic institutions. We argue that this mattered for the spread of modern banks, which represented a new way of organizing the Chinese economy around corporations, deposits, formal governance, and capital markets.

Following the Opium Wars, China was forced to open to the West. Trade expanded, foreign banks entered treaty ports, and Chinese reformers began to build modern financial institutions. This was a major break from the traditional financial system. Traditional banks, including *Piaohao* and *Qianzhuang*, were largely based on kinship, personal networks, and unlimited liability. These institutions were trusted locally and worked well for remittances and local lending, but their scale was limited. Modern banks, by contrast, were organized as joint-stock corporations with limited liability, larger deposit mobilization, formal governance, and standardized accounting. Aggregate historical data show this transition clearly: traditional banks initially dominated but eventually declined, while modern banks

grew rapidly over time.¹

To study these dynamics, we combine newly available data from the China Historical Christian Database (CHCD) with historical data on modern banks, enterprises, railways, treaty ports, government investments, and bankers’ biographies. The CHCD allows us to construct detailed measures of exposure to recorded Christian churches, schools, hospitals, orphanages, publishing houses, and other institutions from 1550 to 1950. To our knowledge, this is the first economics paper to use this source to construct a long panel of Christian institutions in China. The panel structure is important because common missionary atlases tend to oversample prominent missions established early (Jedwab et al., 2022). In our data, early missionary entry is positively selected into more central and developed places, but this selection becomes much weaker after the initial treaty-port period and does not appear in the variation used in our panel design.

Our first empirical strategy uses a staggered difference-in-differences design that exploits the yearly expansion of missionaries across prefectures from 1857 to 1936. A prefecture becomes treated in the year missionaries first arrive and remains treated afterward. The design compares each treated prefecture to places that have not yet received missionaries and to places never treated, while controlling for prefecture and year fixed effects. We find no evidence of pre-trends in the main outcomes or in economic and geographic predictors of modernization, including enterprises, modern banks, traditional banks, railways, treaty ports, and government investments.

Missionary presence led to a large and persistent increase in modern banking. After missionary arrival, treated prefectures experienced a 31% increase in modern banks relative to comparison areas. The effect is visible in event studies, where treated and control locations follow similar trends before missionary arrival and then diverge afterward. The results are robust to alternative measures of banking activity, including deposits and loans, to different functional forms, and to excluding early periods and treaty ports. Missionary presence also increased private enterprises by around 27%, consistent with modern banks supporting broader market activity.

These results raise the question of why missionary presence affected the adoption of modern banks. We provide evidence for two main mechanisms: the formation of practical human capital and the diffusion of useful knowledge.

First, Christian education helped create the human capital needed for modern finance. China already had an advanced education system, but it was largely oriented toward the civil service examination and classical learning (Bai, 2019). Missionary schools, especially

¹Appendix Figure A7 plots the aggregate modern-bank series.

Protestant ones, introduced a different form of human capital, teaching subjects such as mathematics, science, bookkeeping, economics, and English. These were the skills needed to work in, manage, and expand modern financial institutions. We digitize bankers’ curricula for the historical period of our difference-in-differences design and find that around 30% of bankers were affiliated with Christianity through schooling or religious ties. In places with missionary presence, these “Christian bankers” established more banks. The effect of missionary presence is also stronger where missionaries founded at least one school and is driven by Protestant missions, with no comparable effect for Catholic missions. This is consistent with Protestant missionaries’ stronger emphasis on mass education and practical training (Bai, 2019; Bai and Kung, 2015).

Second, missionaries diffused useful knowledge through publications and translations. Missionaries produced and translated Western texts into Chinese, including educational, scientific, and economic materials, often in vernacular Chinese or bilingual formats. These texts made modern economic and institutional practices more accessible outside the classroom. We find that the effect of missionary presence on modern banks is stronger in locations where missions engaged in publication activity. This supports the idea that missionaries mattered not only by locating in a prefecture, but also by transmitting practical knowledge that helped local actors adopt new financial technologies.

We also show that missionaries contributed to China’s transition from traditional to modern finance. We digitize data on two major financial institutions predating modern banks and find that missionaries first boosted these older institutions, yet modern banks soon rose in the same locations and eventually replaced them. Only those traditional banks focused on local lending and enjoying established networks and trust survived longer, suggesting possible complementarities with modern banks.² The effect of missionaries on modern banks is stronger where traditional institutions were already present, indicating that modern banks emerged where financial demand was higher.

We then examine whether historical missionary exposure shaped outcomes after 1949. This part of the paper moves from a panel design to a cross-sectional framework, using the total number of missions in each prefecture as the measure of historical exposure. Because cumulative missionary presence is endogenous, we follow the logic of Sequeira et al. (2020) and instrument missionary exposure with the interaction of railway expansion timing and variations in missionary migration flows.

We compare prefectures with similar lengths of railway connection and number of railway

²In fact, this type of traditional bank, called *QianZhuang*, often collaborated with modern banks, which initially lacked local connections and trust.

lines but different connection years, which either coincided with a surge or a lull in missionary migration. Prefectures connected to the railway just before a national missionary surge became more exposed to missionaries than otherwise similar prefectures connected just after the surge. This approach assumes that the external “push” factors driving missionary booms or lulls are uncorrelated with the exact timing of local railway construction. By matching prefectures in this way, we account for railway endogeneity. While railways can influence modern banks and enterprises through migration and productivity, our identification hinges on the assumption that, conditional on railway connection and railway lines, the missionary migration cycles themselves are not systematically aligned with local rail development at the yearly level.

We first validate the IV design by applying it to the final six years of the historical panel, from 1930 to 1936. The IV estimates reproduce the historical positive effect on modern banks and are similar in magnitude to the difference-in-differences estimates. We then extend the analysis to 1949–2022. The effect of missionary exposure weakens during the Communist period, when market institutions were suppressed. It then re-emerges after the market reforms of the 1980s. Event-study estimates show that the effects become especially pronounced after major land and banking reforms, and we find a sustained positive impact on per capita GDP from 1992 onward, with a larger increase after China joined the WTO in 2001. Together, these results suggest that places with historical missionary exposure had a comparative advantage when China reopened to markets.

Our first contribution is to the literature on the historical roots of development and the debate over institutions and human capital (North, 1990; Acemoglu et al., 2001, 2002; Glaeser et al., 2004; Nunn, 2008; Dell and Olken, 2020; Lowes and Montero, 2021). We show that missionary exposure was not only a source of schooling. It also helped transmit institution-specific knowledge: the practices, networks, and organizational forms that allowed modern banks to take root. Modern banks were an institutional technology, changing how capital was mobilized, allocated, and governed; missionary education and publications supplied some of the skills needed to use it. By tracing these effects from the late Qing and Republican periods through the Communist period and the reform era, we show how early institutional exposure can persist even after decades of suppression. Our reform-era results also connect to work on China’s post-1978 reforms (Lin, 1992; Song et al., 2011; Xu, 2021; Erten and Leight, 2021), showing that these policies were especially effective in places with earlier missionary exposure.

Second, the paper contributes to work on missionaries and development. Prior studies show that missions shaped education (Nunn et al., 2014; Wantchekon et al., 2015; Va-

lencia Caicedo, 2019), democracy (Woodberry, 2012), and cultural change (Henrich, 2020; Bergeron, 2019). Yet less is known about how missionary activity affected finance. The related literature on religion and financial development has mostly studied how religious law, political authority, and community organization shaped credit and financial institutions in Islamic, Christian, and Jewish contexts (Kuran, 2012, 2018; Rubin, 2011; Pascali, 2016). We add evidence that missionaries could also facilitate financial innovation by diffusing practical skills and institutional knowledge. This provides a new pathway through which religious institutions can affect development: not only by changing beliefs or schooling, but also by helping local actors adopt new organizational forms.

Third, we add to the literature on missionaries in China. Existing work shows that missions contributed to enterprise growth and urbanization (Bai and Kung, 2015), women’s education (Ma et al., 2022b), and higher GDP today (Chen et al., 2014). Related work also shows that anti-Christian conflict inhibited modernization and foreign investment (Huang et al., 2024), and that early Jesuit missions affected international trade by transmitting information about China to Europe (Chen et al., 2025b). We provide a finance-based channel that helps connect these findings. By linking pre-1949 missionary exposure to the reform-era response of banks and GDP, we show how these historical seeds interacted with the policies of the 1980s to shape China’s later market orientation and growth. We also document the transition from traditional to modern finance, showing that modern banks eventually substituted for *Piaohao*, while *Qianzhuang* remained more complementary because of their local networks. China is an especially demanding setting because missionaries operated without formal colonial rule, faced conflict and repression, and remained small players in a large education market. We argue that they could still have large effects because modern banking had high economic leverage: a small amount of institution-specific knowledge could affect capital allocation, enterprise formation, and later responses to market reforms. This also relates to work on the historical rise of modern banks in China, which has emphasized legal institutions, culture, new technologies, railways, and upper-tail human capital.³ Leveraging newly compiled panel data from the China Historical Christian Database, we identify the timing and intensity of missionary exposure and trace its effects before 1949, during the period of suppression, and after markets reemerged.

³See, among others, Levine et al. (2023) on legal institutions, Sun et al. (2024) on patriotism, Chen et al. (2022b) on Confucianism, Lin et al. (2021) on telegraphs, Zhao et al. (2023) on railways, and Lin et al. (2024) and Chen et al. (2025a) on upper-tail human capital.

2 Conceptual and Historical Background

2.1 Conceptual Framework

The paper studies how historical foreign exposure shaped the adoption of market-based institutions, and why its effects could disappear and later reemerge. A simple formulation is

$$Y_{it} = O_t \times S_i, \quad S_i = F_i \times T_i, \quad (1)$$

where Y_{it} denotes market-based economic activity in location i and period t , O_t denotes the openness of the economic and institutional environment, and S_i is the stock of market-relevant skills, practical knowledge, and transnational ties. This stock depends on access to foreign knowledge, organizations, and networks, F_i , and the ability to translate and adapt them to local conditions, T_i . The multiplicative notation captures complementarity rather than a specific functional form. The first relation makes the return to historically accumulated skills and connections regime-dependent. The second states that foreign access alone is insufficient if knowledge cannot be translated into forms that local actors can use.

The first act is formation. China's forced opening after the Opium Wars increased the value of institutions able to operate across distance, mobilize capital at scale, and connect Chinese economic activity to international markets. This interpretation echoes the broader framework in [Greif et al. \(2025\)](#). They contrast the lineage-based cooperation that developed in China with the European development of corporations among unrelated individuals. The point is not that traditional Chinese institutions were inherently inferior. Institutions such as *Piaohao* and *Qianzhuang* were well suited to a relatively closed economy in which finance relied heavily on personal relationships, reputation, and local information. Opening changed the economic environment. As international trade, industrial technologies, and large-scale enterprise expanded, so did the demand for standardized and scalable organizational forms ([Spence, 1990](#)). Modern banks were one important response to this new demand.

Missionary exposure could help Chinese actors adopt these new forms. Mission organizations gave local communities access to foreign knowledge and institutions through translations, publications, schools, and international connections. These channels formed a sequence. Translations and publications made foreign practical and economic knowledge accessible. Education, English-language instruction, and training in subjects such as mathematics and bookkeeping gave students the skills needed to use it. These skills, in turn, helped Chinese students and elites study abroad, work with foreign organizations, and enter transnational networks. Mission organizations were particularly well placed to perform

this bridging role because they combined foreign connections with Chinese-language skills, long local residence, and extensive Chinese participation.⁴ Chinese students, translators, entrepreneurs, bankers, and officials then selected, adapted, and expanded the knowledge and organizational practices they encountered.

The output of this process was therefore more specific than education in general. Missionary exposure helped produce high-skilled people with foreign-language and commercially useful knowledge, familiarity with foreign organizations, and connections to universities, firms, and individuals abroad. We focus on modern banks because they provide a specific and measurable form of market-based organization: they mobilize capital and allocate credit. We also examine modern enterprises, trade-related activity, and long-run economic outcomes to capture the broader development of market institutions. Modern banks are thus one observable component of market-based capitalism, not a complete definition of it.

The second act is suppression. After 1949, Maoist planning sharply reduced the market return to these skills, knowledge, and international ties. Foreign connections were severed, markets were suppressed, and finance was reorganized around state allocation. The state could expand formal banking presence without reproducing the internationally oriented skills and relationships relevant to market finance. The disappearance of the historical banking differential during this period therefore need not imply that every earlier effect was erased. The original mission schools, publications, and organizations could disappear while some of their outputs remained embodied in people or preserved through relationships within China and abroad. Historical advantages may therefore remain latent when the institutional environment no longer rewards them.

The third act is reactivation. Reform restored access to markets and international connections, raising the return to the skills, knowledge, and relationships created before 1949. The original schools and publications no longer needed to operate. Their role was in the formation of people who could move between Chinese and international systems and of relationships that could later reconnect particular locations to the outside world. The long-run analysis therefore focuses on two channels consistent with this process: the return of internationally exposed people and the reemergence of trade and foreign-investment links. These outcomes capture the possibility that an earlier geography of international connection survived political closure and became economically relevant again when China reopened.

⁴[Appendix E](#) provides historical evidence on this point, including systematic language training, long inland residence, and the extensive use of Chinese and female personnel. It also discusses the relationship between mission education and comprador brokerage. The argument is not that missions monopolized local knowledge, but that they could combine foreign access with the local knowledge needed to translate and adapt it.

The framework therefore generates the paper’s up–down–up pattern. Missionary exposure first helped create market-relevant skills, practical knowledge, and transnational ties. Maoist institutions reduced their observable economic return. Reform then allowed historically exposed places to draw again on internationally connected people and relationships. Figure 1 summarizes this sequence.

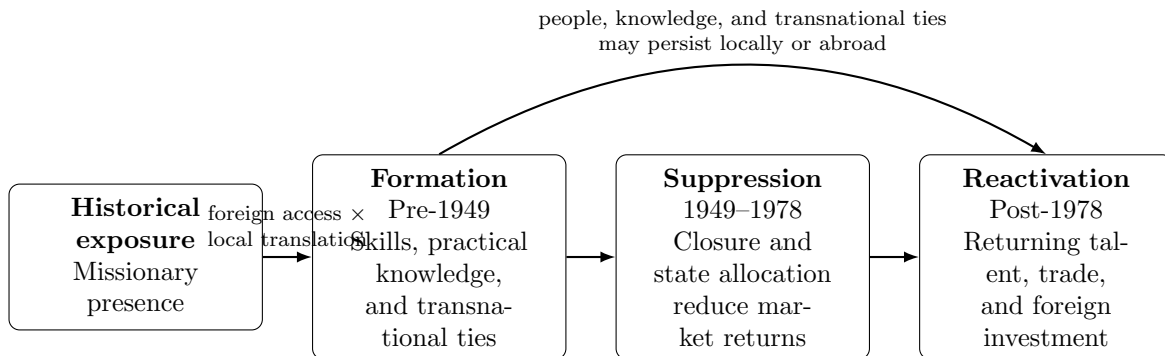


Figure 1: Conceptual Framework: Formation, Suppression, and Reactivation

Notes: The figure summarizes the paper’s organizing framework. The arrows represent the proposed sequence of formation, suppression, and reactivation; they are not separately identified causal “effects.”

The following subsections provide the historical background for each element of this framework.

2.2 China’s Financial Institutions

The institutional transition at the center of this paper is the movement from traditional Chinese finance to modern banking. Before the First Opium War, China’s financial market was dominated by two traditional institutions: *Piaohao* (ticket houses) and *Qianzhuang* (money shops). These institutions provided remittances, loans, and currency exchange services. *Piaohao* managed long-distance remittances for the Qing government and large merchants through extensive branch networks (Cheng, 2003). *Qianzhuang* were smaller, locally operated money shops that accepted deposits, exchanged currencies, extended short-term merchant credit, and connected local merchants to the broader financial system.

Modern banks entered China after the First Opium War. The first were privately owned foreign banks established in treaty ports, where they were active in import–export finance, political loans, and other commercial ventures (Moazzin, 2022).⁵ Chinese entrepreneurs

⁵Several elements of the European banking model later introduced into China developed in late medieval and Renaissance Europe, including deposit banking, book-entry transfers, bills of exchange, and merchant credit (de Roover, 1948; Spufford, 1988). Joint-stock ownership and limited-liability corporate forms devel-

and government officials soon recognized the value of these institutions. As foreign trade expanded and modern enterprises became more common, Chinese-run modern banks began to appear. The first Chinese-owned modern bank, the Imperial Bank of China, was founded in 1897 under official sponsorship. It was organized as a joint-stock corporation with limited liability and formal rules governing deposits, loans, and banknote issuance (Cheng, 2003).⁶

The key difference between traditional and modern banks was not that traditional finance lacked value. Traditional banks relied more heavily on kinship and native-place networks, local information, and relational enforcement. These features supported trust, including across the long-distance branch networks of *Piaohao*. But dependence on owners' capital and inside networks constrained their scalability. Modern banks included joint-stock corporations and state banks. They mobilized deposits and capital at larger scale, used more formal governance, and adopted standardized reporting, contracts, collateral practices, and telegraphic transfers (Cheng, 2003; Lin et al., 2021). Telegraphic transfers accelerated settlement, while rapid reporting improved the monitoring of distant branches. In joint-stock banks, limited shareholder liability facilitated risk sharing and larger institutions. At the same time, weaker personal exposure to losses and delegated management introduced risk-taking and agency trade-offs. These arrangements expanded banks' capacity not only to mobilize funding, but also to transform it into trade and industrial credit. Together, these features made modern banks better suited for an economy increasingly connected to trade, industry, and larger capital markets. Table 1 summarizes these differences.

Between 1911 and 1936, modern banks expanded rapidly in response to rising financial demand. The collapse of the Qing government weakened *Piaohao*, whose business depended heavily on official remittances and ties to the state. Modern banks replaced some functions of *Piaohao* by offering lower-cost remittance services. At the same time, modern banks often lent to and through *Qianzhuang*, which survived longer because their local information and networks of trust complemented the more standardized operations of modern banks. Figure 2 shows the aggregate transition: modern banks rose quickly, while *Piaohao* declined. Beginning in 1937, many banks relocated to the southwestern interior to escape the Japanese invasion, which is why our main historical panel stops in 1936.

oped later.

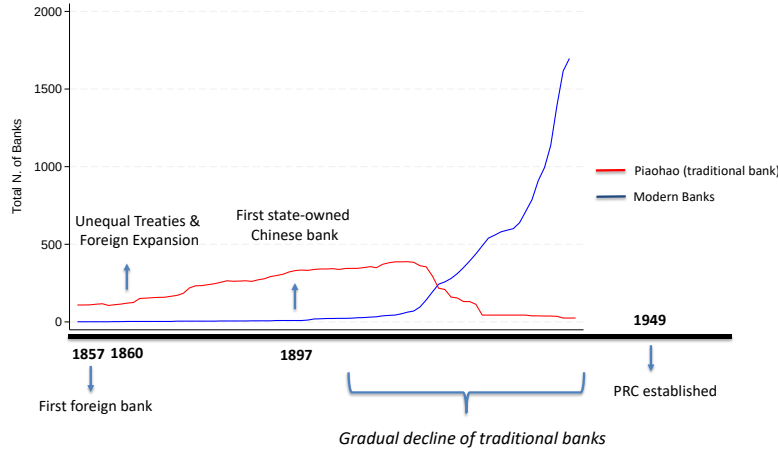
⁶The Imperial Bank was officially sponsored but not fully state-owned. The first state bank, the Bank of the Board of Revenue, followed in 1904. Modern Chinese banks subsequently included private joint-stock, state, and mixed public-private institutions.

Table 1: Traditional and Modern Banks in China

Feature	Traditional banks	Modern banks
Organization and governance		
Institutional form	Proprietorships; kin-based partnerships	Joint-stock corporations; state banks
Owners' liability	Unlimited	Generally limited
Agency and control	Owner- and network-based monitoring	Boards, managers, and formal reporting
Financial intermediation		
Funding base	Owners' capital; short-term funds	Share capital; larger deposit base
Core intermediation	Remittances, exchange, merchant credit	Payments, trade finance, industrial credit
Screening and security	Reputation, guarantors, local information	Accounts, contracts, and collateral
Transfers and reporting	Mail transfers; branch correspondence	Telegraphic transfers; rapid reporting
Economic trade-offs		
Main advantage	Relational enforcement; local knowledge	Scale, diversification, standardization
Main constraint	Capital and network dependence	Agency costs; risk-taking incentives

Notes: The traditional-bank category pools *Piaohao* and *Qianzhuang*; “kin-based” refers broadly to family, lineage, and native-place ties rather than exclusively to family ownership. Entries describe broad tendencies rather than mutually exclusive practices. *Piaohao* operated long-distance remittance networks, whereas *Qianzhuang* were more locally embedded and sometimes intermediated between modern banks and local merchants. Traditional institutions also used sophisticated accounting, guarantees, and, in some cases, collateral; modern banks also continued to rely on relationships. The central distinction is the greater standardization and scalability of modern ownership, reporting, and lending practices. Appendix Table A6 separates the two traditional institutions.

Figure 2: Traditional and Modern Banks During China’s Opening



Notes: The figure plots the aggregate number of modern banks and *Piaohao* over time. The x-axis is calendar year and the y-axis is the total number of banks. Modern bank data come from the Bank of China Economic Research Office and the *National Banking Yearbook*. *Piaohao* data come from Huang (2002) and Liu (2018).

2.3 Missionary Expansion in China

Christian missions expanded in China during the same period as this transition to modern institutions.⁷ Before the treaty-port era, missionary activity had little formal protection and depended on Qing and local approval (Liu et al., 1980). The treaty-port system created after the First Opium War gave missionaries greater access to coastal cities. The treaties concluded around the Second Opium War marked a further turning point: after 1860, missionaries obtained broader rights to enter inland regions and to build churches, residences, hospitals, and schools (Fairbank and Liu, 1980; Chen et al., 2014; Ma et al., 2022a).⁸

Missionary expansion was not smooth. Christianity remained culturally and politically contentious, and anti-Christian conflict culminated in the Boxer Rebellion of 1900, in which over 240 foreign missionaries and 20,000 Chinese Christians were killed (Ren, 1903). Part

⁷Christianity had a much earlier Jesuit presence in China. Most prominently, Matteo Ricci entered China in the late sixteenth century and established himself in Beijing in 1601. This presence later receded: the Yongzheng emperor proscribed Christianity in 1724, missionaries outside Beijing were expelled or forced to operate clandestinely, and the papacy suppressed the Society of Jesus in 1773. Chinese Catholic communities nevertheless survived, although foreign missionary infrastructure remained sparse until the renewed missionary expansion of the 1840s (Bays, 2011; Mungello, 2012).

⁸The *Convention of Peking* of 1860 confirmed and expanded missionary rights. By the late nineteenth century, around 1,500 missionaries and more than 80,000 Protestant converts were recorded in China. Conversion remained the principal missionary objective. Schools, hospitals, and material support also served educational, medical, and evangelical purposes and could attract students and potential converts. Some mission schools offered food, housing, medical care, clothing, or stipends alongside religious and academic instruction (Spence, 1990; Nunn, 2010).

of this hostility reflected the political implications of missionary work: missionaries introduced unfamiliar ideas and acted both as proselytizers and as intermediaries with the West, heightening official concerns about foreign influence and domestic political change.⁹ After the uprising was suppressed, new treaty protections and reparations strengthened the security of missionary activity. Between 1900 and 1920, the number of Christian believers quadrupled from 95,943 to 366,527, and missionary presence reached 94% of China’s counties (Liang, 1988; Stauffer, 1922).

Missionary traditions also differed in ways that matter for our interpretation. Catholic missions often required converts to renounce traditional practices such as ancestor worship, while Protestant missions placed more emphasis on schools, vernacular publications, and practical training, and were more closely associated with ideas of hard work and material success (Fairbank and Liu, 1980; Bays, 2011; Liang, 1988). These differences motivate the Protestant–Catholic splits in the mechanism analysis.

This expansion generated the spatial and temporal variation used in the empirical analysis. Missionaries were initially concentrated in treaty ports and a small number of coastal locations, and gradually moved inland.¹⁰ Figure 3 illustrates this staggered expansion in 1857, 1900, 1905, and 1936. After the founding of the People’s Republic of China in 1949, most foreign missionaries left, and Christian schools, hospitals, and other institutions were brought under state control.

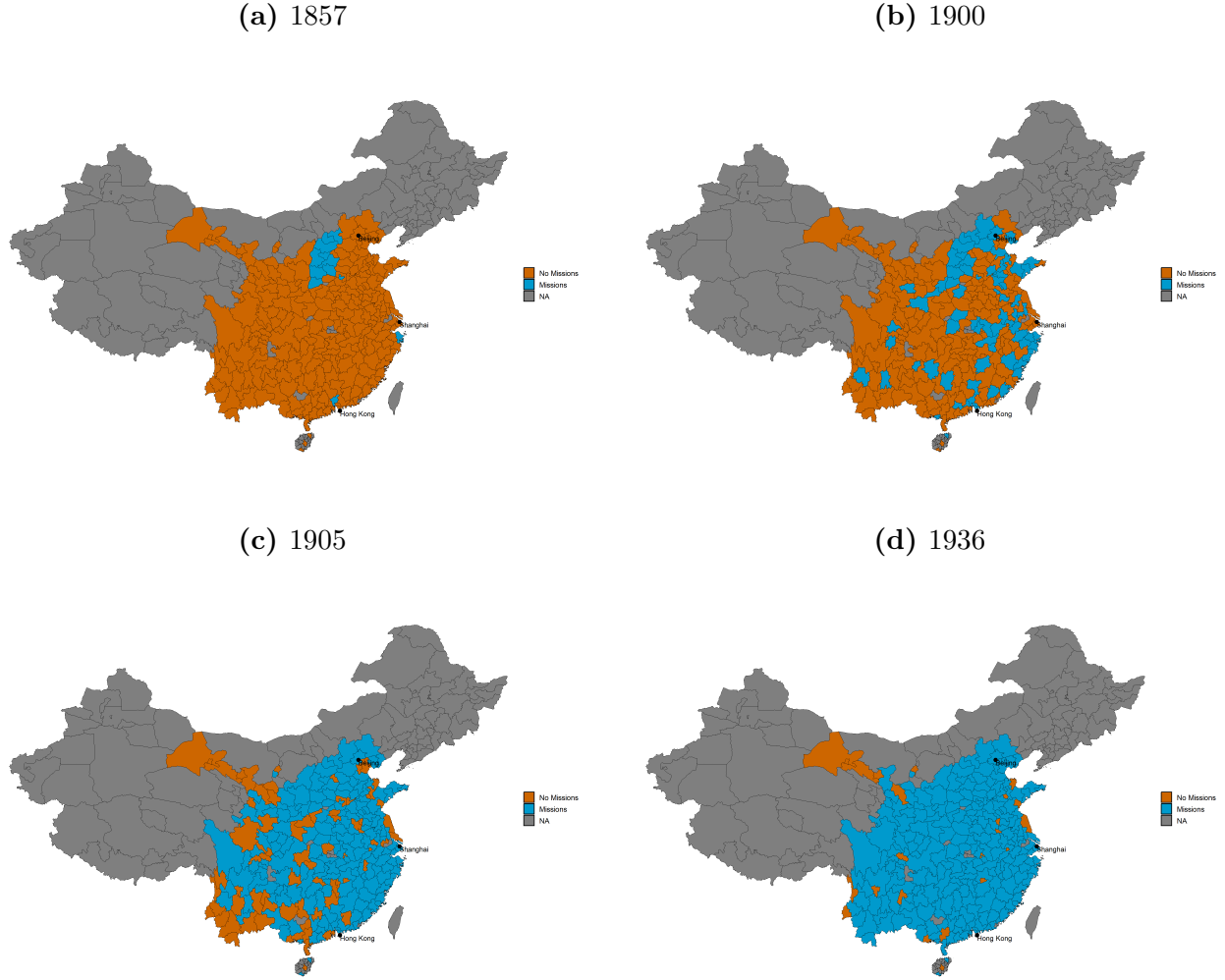
2.4 Missionary Education and the Diffusion of Useful Knowledge

Missionaries could affect modern banking and other new organizations because they supplied skills and knowledge useful for operating them. China already had an advanced education system, but much formal schooling was oriented toward the civil service examination, a system of bureaucratic recruitment that had operated for more than 1,300 years (Bai and Jia, 2016; Bai, 2019). This system emphasized classical texts, literary training, and preparation for official careers. It produced a highly educated elite, but it was less directly connected

⁹China’s defeats in the Opium Wars exposed Western advantages in military technology, while the Qing’s later reliance on Western-officered forces such as the Ever-Victorious Army made the value of foreign military expertise especially visible. Western science and industry attracted similar interest. The political implications were also apparent in specific individuals and institutions: Protestant tracts shaped Hong Xiuquan’s religious vision and the Taiping movement; Sun Yat-sen, the founder of the Republic of China (ROC) and the Kuomintang (KMT), encountered Christianity, democracy, and republican government in mission schools; and mission-trained figures and missionary translators contributed to Self-Strengthening projects through machinery procurement, foreign-language education, and technical translation (Spence, 1990).

¹⁰According to Gu (2010), missionaries arrived at major coastal ports by ship and then ventured inland using railroads and other local transport. We discuss these transport data in Section 3.

Figure 3: Expansion of Missionaries



Notes: The maps show missionary expansion in selected years. Prefectures with missionary presence are shown in blue, those without missionaries in orange, and prefectures excluded from the main analysis in gray. Panel A shows missionary presence by 1857, Panel B by 1900, Panel C by 1905, and Panel D by 1936. Figure A3 presents more detailed maps of missionary diffusion. Mission data come from the CHCD, and prefecture boundaries are based on CHGIS 1911.

to the practical skills used in modern banks and enterprises, such as accounting, foreign languages, commercial law, and applied mathematics.

Missionary education introduced a different type of human capital. Protestant missionaries in particular established institutions ranging from village primary schools to high schools, professional schools, practical-training programs, colleges, and universities. For our mechanism, the most relevant margin was not basic literacy alone, but the upper-secondary, higher-education, and applied-training institutions that prepared students for work in new organizations. By 1935, Protestant missionaries had founded 16 of the 80 colleges operating in China—20% of the total—and Protestant colleges offered especially broad training in the natural sciences and applied fields (Ma et al., 2022a). Mission high schools, colleges, and training programs taught subjects such as mathematics, science, bookkeeping, economics, and English (Bai, 2019; Bays, 2011; Bai and Kung, 2015). These skills were useful for working in, managing, and expanding modern banks and enterprises. English training was particularly valuable because it provided access to foreign texts and prepared students for employment in foreign firms and banks, including as intermediaries known as compradors.¹¹ Missionary schools were small relative to China’s entire education market, but they were disproportionately important in modern higher education and specialized practical training. They could therefore have high economic value at a time when new financial and commercial institutions were being adopted.¹²

Missionaries also diffused useful knowledge through publications and translations. Missionary presses produced religious texts, school materials, newspapers, scientific works, and bilingual or vernacular publications. A concrete example is the American Presbyterian Mission Press, also known as the Huamei Press. Founded in 1844 and moved to Shanghai in 1860, it trained Chinese workers in modern printing and integrated typefounding, typesetting, printing, binding, sales, and distribution. Skills and personnel from this ecosystem later contributed to Chinese-owned commercial publishing. The founding group of the Commercial Press, established in Shanghai in 1897, came from the missionary-school and mission-press network and drew on the Huamei Press model.¹³ These materials made foreign knowledge

¹¹Compradors were Chinese agents who connected foreign firms and banks to Chinese clients and markets through local knowledge and networks. Some later used this experience in Chinese enterprises and financial institutions. See Spence (1990) and Appendix Table E4.

¹²As qualitative evidence, we reviewed 44 interviews with Christian missionary workers held by the Claremont Colleges Special Collections. The interviews describe mission schools and curricula, including English instruction, girls’ education, adult and practical training, and graduates entering Chinese banks. The interviews are used as qualitative illustrations rather than as a representative sample. Appendix E presents this evidence alongside historical photographs of classroom instruction, a college library, adult worker training, and practical field instruction; see Figures E1 and E2.

¹³Appendix Figure E3 reproduces an 1895 photograph of the American Presbyterian Mission Press. The Commercial Press’s founding group included Xia Ruifang, Bao Xian’en, Bao Xianchang, and Gao Fengchi.

more accessible to Chinese readers and helped transmit ideas about science, education, commerce, and modern institutions (Liang, 1988; Stauffer, 1922; Dennis et al., eds, 1911).¹⁴

This publication channel complements the education channel. Schools trained students directly, while translations and printed materials allowed knowledge to circulate beyond the classroom. Together, these channels explain why missionary exposure could affect the local adoption of modern banks. Missionary organizations did not simply transplant foreign institutions. They helped Chinese students, workers, and entrepreneurs access and adapt external knowledge, creating skills and networks that could support banks and other modern organizations.

3 Data and Descriptive Statistics

This section describes the datasets used in our analysis, focusing on missionary diffusion and modern banks, and presents additional data sources. All data are aligned with 1911 prefectural boundaries from CHGIS (2016), a standard reference in this field (Bai and Jia, 2016; Lin et al., 2021; Dong and Zhang, 2024). Table A3 reports summary statistics for outcomes from the historical difference-in-differences samples, Table A4 reports missionary treatment, mechanism measures, and the principal predetermined controls used in the balance tests, and Table A5 reports the variables in the main long-run instrumental-variables samples.

3.1 Missionary Institutions

We use the *China Historical Christian Database* (CHCD) for detailed geographic data on Christian institutions in China from 1550 to 1950. This dataset, a collaborative effort by theology researchers at the Center for Global Christianity and Mission at Boston University, merges data from various sources for a comprehensive study of Christianity in China. For more details, see Mayfield et al. (2024).¹⁵

The Bao brothers studied at the Presbyterian Qingxin school and worked at the Huamei Press. Gao came from the same school and printing network and remained at the mission press during the Commercial Press's early years. Xia learned English and printing at Qingxin before working as a typesetter in foreign-run presses. The Commercial Press's official history describes the firm as deliberately drawing on the Huamei Press model. We interpret this link as the diffusion of personnel, skills, and organizational practices, not as missionary ownership of the Commercial Press (McIntosh, 1895; Lu, 2017).

¹⁴This information channel is related to recent work on Jesuit missions and international trade, which emphasizes missionary letters and publications as information flows from China to Europe. Our mechanism instead concerns the internal diffusion of skills and knowledge useful for financial modernization (Chen et al., 2025b).

¹⁵Here is the complete list of CHCD sources. Appendix Figure A1 describes the CHCD data structure. Appendix Table A2 reports a short external check against CMS/Stauffer Protestant missionary counts in

The current baseline CHCD v3.0.0 extract contains 20,213 institution-location records. After matching to 1911 prefectures and counting each institution once, it covers 5,885 unique institutions. For supplementary descriptive and school-heterogeneity measures, we also add 971 new prefecture-matched school records digitized with CHCD collaborators from historical school and mission directories.¹⁶

The CHCD records the locations of missionary institutions and, when available, the start and end years of each institution-location relationship. We use those locations and dates to build a prefecture-year panel. Our main treatment is the first year in which any missionary institution is recorded in a prefecture.¹⁷ Appendix Table A1 details the distribution of institution types, and Appendix Figure A2 shows the cumulative expansion of missionary institutions over time. In the main analysis, we treat all missionary institutions equally, while using school data to explore heterogeneity.

Figure 3 shows the expansion of missionary presence at the prefecture level. When a missionary institution is established, the prefecture is marked as “treated.” Prefectures shown in gray are excluded due to limited data for remote areas.¹⁸ Before 1900, missions were concentrated along the coast and the Yangtze River. After 1900, they spread rapidly inland, and by 1936 only a few prefectures lacked missionary presence. We call these “never treated” prefectures. Our results remain robust even when these are excluded.

The CHCD also includes person, corporate-entity, and event nodes that can be linked to institutions and prefectures. We use these data mainly to describe missionary flows and to recover subdenomination detail. For the broad Protestant–Catholic split, we instead use the institution tradition field, because the CHCD team treats institution affiliation as more stable than individual affiliation.¹⁹

1920.

¹⁶The directory sources documented in the CHCD handoff material include Educational Directory of China snapshots for 1914, 1916, 1917, 1918, and 1920, as well as 1936 handbooks. The added records mainly affect education descriptives and school heterogeneity; the any-mission arrival treatment is essentially unchanged because most affected prefectures already had earlier CHCD missionary presence.

¹⁷We use institutions because they capture a stable local organizational presence. In the language used by CHCD collaborators, a mission station is a place where a mission organization sent missionaries for evangelization. It is less formal than a hospital or university, but still represents a local base where missionaries lived or worked, not only individual movement. In the 5,885 prefecture-matched institution records, 1,112 lack a raw relationship-level start year. After supplementing the raw field with institution-node dates and the linked-relation date recovery used in the cleaned data, 966 records (16.4%) still lack a usable start-year

¹⁸See Figure A3 for more detailed maps of missionary diffusion.

¹⁹The CHCD v3.0.0 node file contains 37,394 person nodes, 1,213 corporate-entity nodes, and 163 event nodes. Among the 5,560 prefecture-matched institutions classified as either Protestant or Catholic, 5,001 are Protestant (89.9%) and 559 are Catholic (10.1%). Recent work also emphasizes that measuring Catholics in China is difficult and may understate Catholic presence (Zhao and Ma, 2025). See Appendix Table B1 for the institution-based broad split, Appendix Table B2 for people-based subdenominations, Figures B4 and B5, and Appendix Figure B3.

In the linked person data, 54% of named missionary personnel are women. [Appendix E](#) places this composition alongside historical evidence on the broader Chinese mission workforce.

3.2 Modern Banks in China

For our historical analysis (1857–1936), we obtain bank data primarily from the *National Banking Yearbook* by [Bank of China Economic Research Office \(1937\)](#). The Republic of China government conducted a nationwide bank survey from 1934 to 1937, recording banks and bank offices listed by the survey, along with some older entries that had already ceased operations. The establishment-year field reaches back to 1857.²⁰ The dataset includes bank names, establishment dates, prefectures, addresses, equity ownership (Chinese government-owned, private, government-private joint, and foreign banks), and details on head offices and branches.

We supplement our analysis with foreign bank data from [Huang \(1995\)](#). Since the census mainly covers Chinese-owned banks, Huang’s collection of enterprise directories (1861–1948) fills in gaps for foreign banks by providing details on bank names, operational dates, countries of origin, main branch locations, assets, and founders.²¹ We used the Google API to obtain geographic coordinates for each modern bank’s location, manually verified the data, and matched these coordinates to 1911 prefectures defined in [CHGIS \(2016\)](#).

The data show a rapid but uneven spatial expansion of modern banking. As shown in [Figure 2](#), modern banks grew quickly as *Piaohao* declined. In the full historical panel, 9 of 247 prefectures had a modern bank by 1897. This increased to 57 by 1915, 82 by 1926, and 168 by 1936. [Appendix Figure A5](#) maps this expansion using 1911 prefecture boundaries. Banks were initially concentrated in treaty-port and lower-Yangtze prefectures, then spread more widely inland during the Republican period.

For the long-term analysis, we collected data from the China Financial Supervision and Administration Bureau website, which provides detailed information on financial institutions

²⁰Direct inspection of the cleaned bank files shows that the data are not a pure survivor sample. The survey includes some older establishments that had already closed, as well as all older establishments that were still operating when the survey was conducted. The notes field flags about 5% of bank rows as ceased operations, in liquidation, or otherwise closed, and a smaller number include explicit closure or reorganization years. At the same time, the files do not contain a systematic closure-date column, so banks and branches that failed before the survey are still likely underrecorded. We therefore interpret the historical bank outcomes as total recorded modern-bank establishments, not as a complete opening–closure panel.

²¹[Huang \(1995\)](#) collected dozens of enterprise lists published in major Chinese cities between 1861 and 1948, such as the North China Desk Hong List, The Shanghai Directory, and others.

that have received government-issued licenses from 1949 to 2022.²² For each institution, we record the establishment date, prefecture of location, and closure date (if applicable).²³

3.3 Other Data

Additional details on these data sources are provided in the [Data Appendix](#).

Historical Biographies (1857–1949). We hand-collected and coded biographies of directors, founders, and managers of Chinese-funded banks from *Jindai Zhongguo Yinhangye Jigou Renming Dacidian* (Jiang, 2014). The resulting person-level data link banking positions to education, Christian connections, comprador experience, and careers across Chinese and foreign financial institutions.²⁴ We also use the Republican China Elite Database assembled by Wang et al. (2026) from Xu’s *Minguo Renwu Dacidian* (Xu, 2007). The source volume was converted by optical character recognition, manually cleaned, parsed into person-level fields, and linked by native place to 1911 prefectures. Its 16,445 biographies record education, overseas study, and careers, but represent prominent individuals rather than the population as a whole.

Historical Financial Records (1823–1940). We digitized two complementary sources of modern-bank balance sheets. Republican Ministry of Agriculture and Commerce tables for the 1910s list head offices and branches separately; we geocode each location to 1911 prefectures and aggregate deposits, capital, banknote issuance, and reserves (Ministry of Agriculture and Commerce, 1912–1924). The *National Banking Yearbook* provides richer bank-level accounts for 1932–1936, including loans, assets, income, cash, securities, and paid-in capital (Bank of China Economic Research Office, 1937). We also reconstruct branch histories of *Piaohao* from Huang (2002) and Liu (2018), and geocode *Qianzhuang* records from the *National Banking Yearbook*. The dated opening and closing records for *Piaohao* allow us to track entry and exit rather than only surviving institutions.

Other Historical Outcomes (1840–1936). We geocode domestic-enterprise head offices from Du (1991, 2019) and complement them with foreign-enterprise directories from Huang (1995). The domestic source records registered, relatively capitalized firms and therefore

²²See <https://www.cbirc.gov.cn>.

²³According to the Ministry of Justice of the People’s Republic of China, financial institutions include policy banks, commercial banks, financial asset management companies, credit unions, postal savings institutions, trust and investment companies, financial leasing companies, and foreign financial institutions; see https://www.moj.gov.cn/pub/sfbgw/flfggz/flfggzbmzg/200308/t20030813_143502.html. The recent registry’s institutional structure and definition may therefore differ from those of the historical modern-bank records. Using both sources nevertheless provides a broad sense of financial activity across the historical and reform periods, without implying that their institution counts are perfectly comparable.

²⁴[Appendix Figure B2](#) illustrates how narrative biographies were converted into coded fields.

underrepresents smaller businesses.²⁵ Prefecture-level domestic and international trade outcomes come from an unusually detailed 1934–1936 census of product types, production, prices, and sales areas. The Republican government commissioned local postal offices to collect the underlying information at the county and township levels ([Ministry of Communications, 1937](#)).

Historical Controls. We geocode railway stations and opening years from [Ma et al. \(1983\)](#) and reconstruct prefecture-level railway access over time. Geographic and economic controls—including historical population, coastal and river access, rice suitability, provincial-capital status, and treaty-port status—come from [Bai and Jia \(2016\)](#); selected specifications add genealogy-book counts from [Dong and Zhang \(2024\)](#) as a measure of clan intensity.

Long-Run Economic Outcomes. Beyond the post-1949 financial-license records described above, we use 1-km gridded GDP for 1992–2019 from [Chen et al. \(2022a\)](#), summing grid cells directly within 1911 prefecture polygons rather than passing through modern administrative units. Industrial-enterprise outcomes for 2007–2013 come from the National Bureau of Statistics’ Annual Survey of Industrial Production. We assign geocoded firms directly to 1911 prefectures, using county-centroid and area-weighted city matches only when a direct spatial match is unavailable. Foreign direct investment in 2017 comes from the *China City Statistical Yearbook*; population adjustments use historical population and the National Bureau of Statistics’ 2000 Population Census and 2015 one-percent population sample survey.

Shaanxi Provincial Records (1949–1998). We transcribed the Shaanxi Provincial Bureau of Statistics’ county compilation for 1949–1990, which provides education, fiscal, investment, infrastructure, enterprise, and market outcomes unavailable in a comparable national panel ([Shaanxi Provincial Bureau of Statistics, 1991](#)). We harmonize changing county, district, and county-level-city names to 1911 counties using reviewed Chinese-name aliases. We separately digitize dated references to Christianity, closures, and repression from Shaanxi county and city gazetteers through 1998. These data permit a finer county-level replication within one province, although they do not substitute for the national analysis.

Other Long-Run Measures. The Chinese Engineers Relational Database provides structured biographies, education histories, jobs, and locations for engineers in Republican China and postwar Taiwan ([Pelzer et al., 2025](#)); the Republican-era elite biographies described above measure foreign exposure and later institutional roles. Contemporary education, religious affiliation, values, and financial behavior come from the 2000 Population Census, the

²⁵The source includes shipping, industrial, and mining enterprises capitalized above 10,000 silver dollars and financial enterprises capitalized above 50,000 silver dollars.

China Family Panel Studies (Xie and Hu, 2014), and the 2007 [Spiritual Life Study of Chinese Residents](#). We link their reported locations to the corresponding historical geography; the [long-run mechanism appendix](#) describes the samples and specifications.

4 Empirical Strategy

This section outlines the empirical exercise, which is divided into two parts. First, we apply a historical staggered difference-in-differences (DiD) analysis, examining the impact of missionary exposure on the development of modern banks and enterprises from 1857 to 1936. Next, we explore the long-run effects by studying the impact of historical missionary exposure on contemporary bank establishments, using an instrumental variable (IV) strategy.

4.1 Historical Analysis: Staggered DiD

4.1.1 Estimating Equations

Our baseline historical panel includes 247 prefectures between 1857 and 1936, of which 232 receive missionaries during the sample period.²⁶ We exploit the gradual expansion of missionaries described in Section III and define treatment based on the timing of missionary arrival. We follow Sant’Anna and Zhao (2020) to construct heterogeneous-robust estimates in an event-study regression. For simplicity, consider first the following static standard TWFE model, valid under heterogeneous treatment effects:

$$Y_{it} = \alpha_i + \theta_t + D_{it}\beta_{post} + \varepsilon_{it} \quad (2)$$

Where i and t stand for prefecture and year, respectively. D_{it} is a dummy variable that takes the value 1 for all years after the first missionary institution entered that prefecture, and 0 otherwise. D_{it} is always 0 for prefectures where missionary institutions never entered (i.e., “never treated”). Y_{it} represents the total number of modern banks or enterprises. The parameter of interest, β_{post} , measures the change in the outcome variables of the treated prefectures compared to the yet-to-be-treated and never-treated prefectures, conditional on the set of prefecture and year fixed effects. By design, we exclude prefectures that were already treated before 1857 (i.e., “always treated”), and we cluster standard errors at the prefecture level.

²⁶This is not the total number of Chinese prefectures, which ranged between 250 and 300 at the time. We included prefectures in the sample for which we have data on modern banks, enterprises, missionaries, and covariates. This yields coverage similar to that of other historical studies of China, due to data availability.

In practice, our estimation is more closely represented by a TWFE event-study model like the following:

$$Y_{it} = \alpha_i + \sum_{r \neq 0} 1[R_{i,t} = r] \beta_r + \varepsilon_{it} \quad (3)$$

Where we define $G_i = \min\{t : D_{i,t} = 1\}$ to be the earliest period at which unit i receives treatment, $R_{i,t} = t - G_i + 1$ is the time relative to treatment (e.g., $R_{i,t} = 1$ in the first treated period for unit i), and the summation runs over all possible values of $R_{i,t}$ except for 0. However, even this specification can encounter “forbidden comparison” issues under heterogeneity across cohorts (Sun and Abraham, 2021). Given that in our setting we are estimating ATTs over an 80-year period and for 42 cohorts, we believe that the doubly-robust Sant’Anna and Zhao (2020) estimator is suitable for our baseline model, as it will allow us to correctly estimate β_r even if the treatment effect for a cohort treated in 1861 is very different from that of a cohort treated in 1934.²⁷

In short, our estimator is calculating many group-time average treatment effects on the treated, $ATT(g, t) = \mathbb{E}[Y_{i,t} - Y_{i,g-1} \mid G_i = g] - \mathbb{E}[Y_{i,t} - Y_{i,g-1} \mid G_i = g']$, for any $g' > t$, which gives the average treatment effect at time t for the cohort first treated in time g . These ATT’s are then aggregated to obtain event-study estimates as an average of the treatment effect l periods after adoption across different adoption cohorts (Roth et al., 2023):

$$ATT_l^\omega = \sum_g \omega_g ATT(g, g + l) \quad (4)$$

Intuitively, our estimator can be viewed as an aggregation of the multi-period analog of equation 3, with comparison groups being the not-yet treated in subsequent periods and the never-treated. The distribution G_i is shown in Appendix Figure C1.

4.1.2 Balance Test

A key assumption in our model is parallel trends between control and treated units in the staggered setting. More precisely:

$$\mathbb{E}[Y_{i,t}(0) - Y_{i,t'}(0) \mid G_i = g] - \mathbb{E}[Y_{i,t}(0) - Y_{i,t'}(0) \mid G_i = g'] \quad (5)$$

²⁷Note that since our estimation does not include covariates, we are assuming unconditional parallel trends and not taking advantage of the outcome model and propensity score matching. This makes this estimator very general and similar to other estimators proposed in the literature.

For all $t, t' \geq g_{min}$, where g_{min} is the first period in which a unit is treated, this imposes that, in the counterfactual scenario where treatment had not occurred, the average outcomes for all adoption groups would have evolved in parallel in the post-period. In our case, this means that for each cohort of treated prefectures, we would expect the total number of modern banks (or enterprises) to have evolved similarly to not-yet-treated cohorts if they had not received treatment. While we do not observe the counterfactual outcome, we can examine how prefectures that receive missionaries in a given time frame compare to the next-treated cohort of prefectures. If we observe differences in economic and social indicators across these prefectures, it could suggest that they were already following distinct trends.

A key concern from the literature on missionaries is that missions might have been established in places that were positively selected (Jedwab et al., 2022). However, it is important to note that most studies on missionaries lack access to panel data, so we know relatively little about the path of expansion and the relative selection year-to-year, especially when considering economic dynamics rather than static levels. Given the tumultuous history of missionaries in China—ranging from being confined to treaty ports, to concessions allowing expansion into the interior, and events like the Boxer Rebellion—we need to ensure we can address some of these concerns and rule out potential biases.

A first simple exercise is to examine what predicts the establishment year of missions in the cross-section (the variable that defines treatment cohorts G_i). We regress this variable on a series of predetermined economic and social indicators at the prefecture level, including province fixed effects. The time-varying variables in this specification are measured before the relevant missionary entry window. They include modern banks, enterprises, government investment during the Self-Strengthening Movement, railways, and treaty ports, together with geographic and economic controls from Bai and Jia (2016).²⁸

The results from these regressions are shown in Table 2. Appendix Figures C2 and C3 report the corresponding one-by-one coefficient plots. Columns 1 and 2 reveal significant selection when considering the entire time period and all prefectures, except for those always treated. Prefectures with more railways, treaty ports, population, or provincial capital status are more likely to have received missionaries earlier. This finding aligns with historical accounts and the missionary literature, which indicate that missionaries were initially confined to a few major cities and treaty ports, with their freedom of movement only being established after the Second Opium War in 1860. Following this, missionaries aimed to reach as many prefectures as possible to convert all ethnic groups, eventually extending their reach to more

²⁸Modern banks, enterprises, and government investments are observed in 1880, treaty ports in 1860, and railways in 1890. For the post-1900 specification, time-varying controls are measured in 1900. Population and provincial-capital status are measured using the earliest available pre-period values.

Table 2: Predictors of Mission Establishment Timing and Status: Prefecture-Level Estimates

Variables	(1)	(2)	(3)	(4)	(5)
	Establish. Year All	Post 1900	Cohort Comparisons		
			1860-1900	1900-1904	1904-1920
Total Modern Banks	9.156 (8.755)	-1.899 (5.715)	-0.196 (0.244)	-0.069 (0.364)	0.106 (0.251)
Total Enterprises	0.851 (2.399)	0.096 (0.522)	-0.051 (0.038)	0.024 (0.034)	-0.020 (0.023)
Ln (investment gov. factory)	0.016 (0.413)	0.332 (0.696)	0.026 (0.018)	-0.086* (0.045)	0.063 (0.056)
Total railways	-5.237* (3.004)	-2.163 (3.654)	0.837*** (0.230)	-0.056 (0.174)	0.025 (0.099)
Total Treaty Ports	-25.857*** (9.315)	-0.629 (2.666)	0.695*** (0.158)	0.152 (0.171)	0.319 (0.275)
Whether there's main river	-1.008 (1.814)	0.195 (1.371)	0.058 (0.059)	0.073 (0.088)	-0.072 (0.107)
Provincial capital	-9.202*** (2.391)	-4.765** (2.241)	0.190** (0.078)	0.195 (0.146)	0.257 (0.217)
Crop suitability: Rice	-0.660 (1.225)	0.861 (1.002)	-0.014 (0.040)	-0.051 (0.064)	0.068 (0.081)
Ln population	-2.464** (1.128)	-1.428 (0.883)	0.039 (0.037)	0.116** (0.057)	0.034 (0.068)
Coastal	-0.240 (3.100)	0.260 (3.421)	-0.054 (0.101)	0.072 (0.213)	0.062 (0.475)
Province FE	Yes	Yes	Yes	Yes	Yes
Obs.	214	156	215	157	77
Adj. R sq.	0.330	0.105	0.342	0.0752	0.208
Mean Dep. Var.	1901	1907	0.250	0.448	0.635
S.D. Dep. Var.	13.75	8.173	0.433	0.498	0.482
P-value F-stat.	1.56e-05	0.452	3.59e-08	0.180	0.494

Notes: The historical panel includes 247 prefectures, of which 232 receive missionaries during the sample period. Regressions are at the prefecture level. Column 1 regresses G_i , the variable defining treatment (i.e., the establishment year of missions), on a series of covariates. The dependent variable for column 2 is the same but only for prefectures where missions were established post-1900. For columns 3-5, the dependent variable is whether a mission was established in the given range of years, and the sample includes only prefectures receiving missionaries in that time frame and in the subsequent period. Controls are from [Bai and Jia \(2016\)](#) and are measured in the pre-period. Banks, enterprises, and government investments are observed in 1880, treaty ports in 1860, and railways in 1890 (all measured in 1900 for column 2). Obs. refers to the number of observations; Adj. R^2 is the adjusted R^2 . Fixed effects may vary; refer to the table. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Appendix Figures [C2](#) and [C3](#) report the corresponding one-by-one coefficient plots.

remote and less populated areas ([Bays, 2011](#)). Column 2 focuses on prefectures treated after 1900, when selection is much weaker. Due to this observation, for robustness, we will later restrict the sample to post-1900 periods only.

Next, following [Colonnelli and Prem \(2022\)](#), we compare cohorts of prefectures that

receive treatment within a given range of years to those that will receive it in the future and those never treated. We regress the indicator of whether a missionary institution entered a prefecture for different treatment cohorts on the same X 's, now measured at the beginning of the time period of the treated cohorts. Columns 3–5 of Table 2 split the sample into four time periods and confirm the patterns observed in columns 1 and 2. While the estimation provides a weighted average of group treatment effects across different time periods, which is difficult to exactly replicate in a balance table, we can also generate narrower cohorts to compare a given treated cohort with the next cohort. Table 3 uses a finer partition and splits the sample into eight narrower cohort comparisons: 1860–1880, 1880–1890, 1890–1900, 1900–1903, 1903–1904, 1904–1905, 1905–1910, and 1910–1920. Appendix Figure C4 reports the corresponding one-by-one coefficient plot.

Table 3 shows that treatment and control groups are generally balanced, with only a few variables statistically different from zero. We could be facing power issues due to the limited sample size, but the results suggest that the selection we previously observed was largely driven by comparisons over long periods, when prefectures treated early were very different from those treated decades later. When comparing cohorts closer in time, treated and not-yet-treated prefectures appear more similar. One interpretation is that mission expansion depended on the interaction between local suitability and external missionary supply. Local factors, such as a new railway connection or demand for schools, could make several prefectures plausible mission targets around the same time. Which of these places received missionaries first could then depend on external factors, such as missionary inflows, denominational budgets, conferences, revivals, wars, or funding cycles, that were not driven by the local economic trends of a specific prefecture. This interpretation is related to the source of variation used in the long-run IV analysis below, although the DiD estimates themselves do not rely on that instrument.

4.1.3 Results

We begin with modern banks, our main historical outcome. Panel A of Figure 4 displays the raw data over a window of $[-15, 15]$ years surrounding the arrival of missionaries, using the 247-prefecture historical panel between 1857 and 1936. The control group consists of the next cohort of treated prefectures and the never-treated group for each year. While not perfect, this comparison closely mimics the variation underlying our estimator. Treated and control prefectures follow similar trends before missionary arrival and then begin to diverge after treatment. Panel B plots this difference directly.

To identify these dynamics more precisely, Figure 5 plots the event-study estimates from

Table 3: Predictors of Mission Establishment by Cohort: Prefecture-Level Estimates

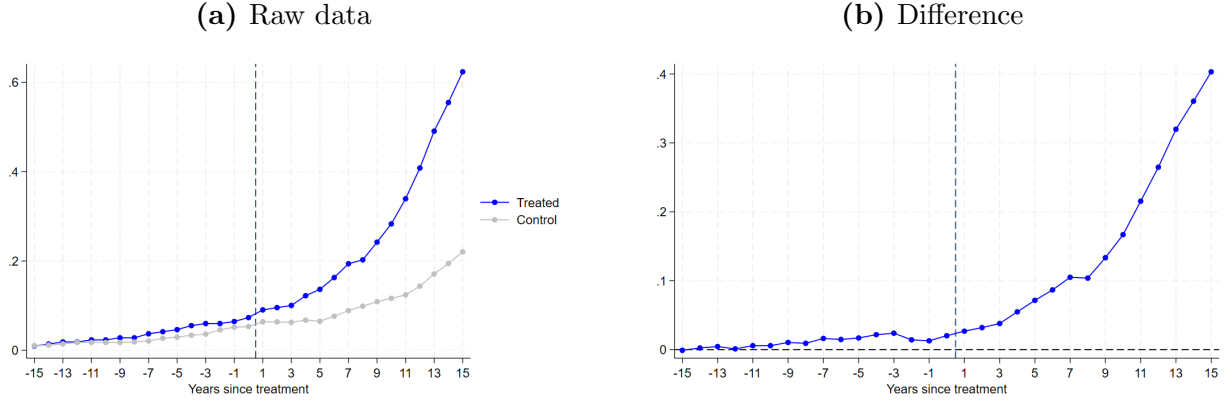
VARIABLES	(1) 1860-1880	(2) 1880-1890	(3) 1890-1900	(4) 1900-1903	(5) 1903-1904	(6) 1904-1905	(7) 1905-1910	(8) 1910-1920
Total Modern Banks	-0.830 (0.664)	-0.873 (1.874)	0.732 (1.313)	-0.086 (0.407)	0.107 (0.667)			1.359 (0.824)
Total Enterprises	0.840 (0.540)	-1.115 (0.780)	-0.085 (0.116)	-0.006 (0.047)	0.218 (0.265)	-0.057 (0.102)	-0.018 (0.042)	0.053 (0.285)
Ln (investment gov. factory)	0.005 (0.040)	-0.000 (0.045)	0.046 (0.030)	0.106 (0.105)	-0.096 (0.086)	0.076 (0.087)		
Total railways		0.414 (0.396)	-0.016 (0.353)	0.120 (0.204)	-0.588 (0.628)	-0.117 (0.158)	0.096 (0.201)	-0.034 (0.288)
Total Treaty Ports	0.491 (0.297)	0.252 (0.381)	0.164 (0.374)	0.492 (0.354)		0.093 (0.560)	0.665 (0.488)	-0.330 (1.117)
Whether there's main river	0.035 (0.221)	0.254 (0.329)	0.106 (0.110)	-0.034 (0.118)	-0.117 (0.246)	0.069 (0.183)	0.203 (0.189)	-0.026 (0.205)
Provincial capital	0.257 (0.215)	0.094 (0.304)	-0.018 (0.171)	-0.185 (0.167)	0.156 (0.306)	-0.118 (0.433)	0.609 (0.376)	-0.989* (0.507)
Crop suitability: Rice	0.201 (0.124)	-0.178 (0.239)	-0.009 (0.094)	-0.018 (0.091)	0.182 (0.209)	-0.196 (0.182)	0.140 (0.161)	-0.026 (0.117)
Ln population	0.106 (0.142)	-0.372** (0.151)	0.135* (0.069)	0.031 (0.082)	0.026 (0.177)	-0.031 (0.131)	-0.004 (0.113)	0.028 (0.141)
Coastal	-0.282 (0.200)	0.574 (0.346)	-0.217 (0.221)	-0.002 (0.250)	0.898 (1.518)	0.532 (0.677)	0.122 (0.606)	
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	42	36	68	79	38	44	36	30
Adj. R sq.	0.272	0.133	0.214	0.185	-0.279	-0.0376	0.218	0.510
Mean Dep. Var.	0.317	0.481	0.157	0.591	0.390	0.339	0.473	0.286
S.D. Dep. Var.	0.466	0.500	0.364	0.492	0.488	0.474	0.500	0.452
P-value F-stat.	0.0543	0.263	0.0414	0.896	0.926	0.913	0.670	0.350

Notes: The historical panel includes 247 prefectures, of which 232 receive missionaries during the sample period. Regressions are conducted at the prefecture level. In each column, the dependent variable is whether a mission was established in the given range of years, with the sample including only prefectures that received missionaries during that time frame and the subsequent period. Controls are from [Bai and Jia \(2016\)](#) and are measured in the pre-period for each column. For example, banks are measured in 1860 in column 1, meaning we compare how the number of banks in 1860 differs between prefectures with missionary presence in 1860-1880 and those with presence in 1880-1890. Obs. refers to the number of observations; Adj. R² is the adjusted R². Fixed effects may vary; refer to the table. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Appendix Figure [C4](#) reports the corresponding one-by-one coefficient plot.

equation [3](#). The pre-treatment coefficients are close to zero, while the effect becomes positive and grows gradually after missionary arrival. The post-period average ATT is 0.22, which corresponds to a 31% increase over the mean number of modern banks in the sample.

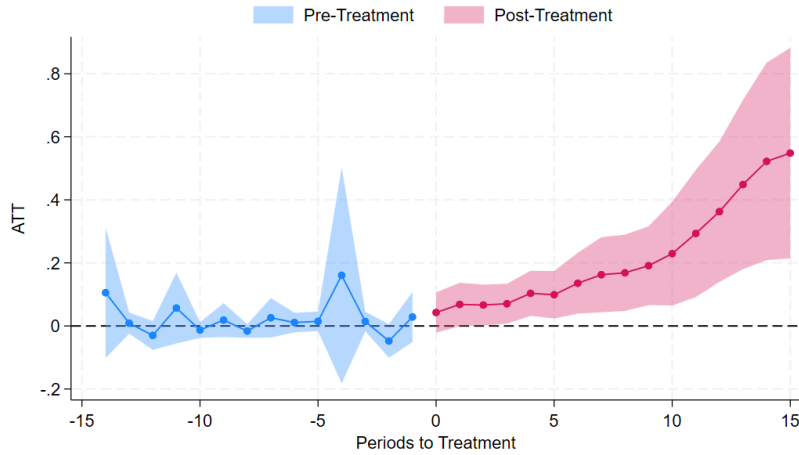
One concern is that modern banks are rare in the early part of the sample, so flat pre-trends may partly reflect limited pre-period variation. Appendix [D.1](#) addresses this concern in two ways. First, a design-based minimum detectable effect exercise uses the estimated pre-period coefficients and covariance matrix to ask how large hidden pre-trends would need to be detected. The exercise shows that the design is informative against several smooth pre-trend shapes, although very late pre-trend activation is harder to detect. Second, the result is robust to alternative outcome transformations, including $\log(y + 1)$, growth rates,

Figure 4: Raw Data: Total Modern Bank Dynamics



Notes: Panel A displays the raw data for total modern banks over a window $[-15, 15]$ years surrounding missionary arrival, using the 247-prefecture historical panel between 1857 and 1936. The control group consists of the next cohort of treated prefectures and the never-treated group for each year. Panel B plots the treated-control difference in the raw data.

Figure 5: Missionary Presence and Total Modern Banks: 15 Periods



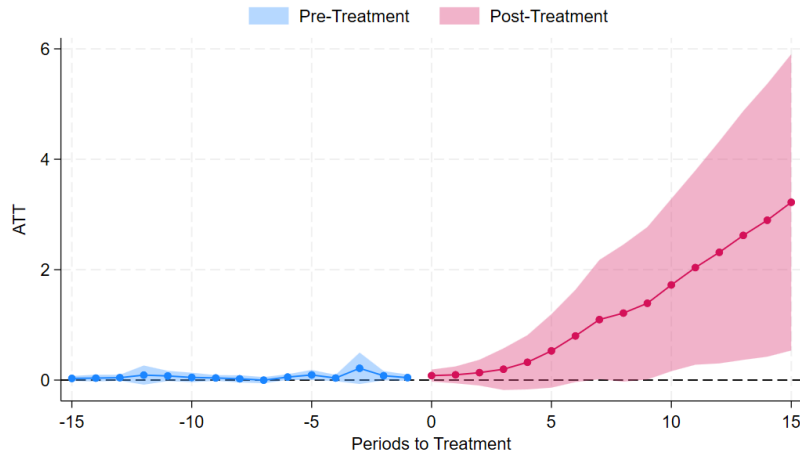
Notes: The figure reports event-study estimates from Equation 3, using 15 periods before and after missionary arrival. The dependent variable is the total number of modern banks. The average post-treatment ATT is 0.22. The dependent-variable mean is 0.7 overall, 1.7 in 1920, and 5.8 in 1936 (Figure A7 reports the mean by year). Appendix Table D1 reports the corresponding pre- and post-treatment averages and alternative bank outcomes.

an extensive-margin indicator, new-bank flows, and province-conditioned outcomes.

The effect is also not limited to the aggregate number of banks. Appendix D.2 shows similar patterns for private, government, and foreign banks. Supporting balance-sheet evidence also points in the same direction: missionary exposure is associated with larger deposits, loans, assets, and related measures of bank size. These results suggest that missionary presence affected the scale of modern banking activity, not only the recording of bank establishments.

We next ask whether missionary exposure also affected downstream market activity. The number of enterprises, particularly private ones, is a natural outcome because modern banks could relax financial constraints and support enterprise formation. Figure 6 presents the results for total enterprises over the same 15-period window. As with modern banks, there are no visible pre-trends before missionary arrival. After treatment, missionary presence has a delayed positive effect on enterprises. This delayed timing is one reason we interpret enterprises as a downstream outcome rather than the first mechanism through which missionaries mattered. The event-study aggregation reported in Appendix Table D1 gives a post-treatment ATT of 1.29, about 27% of the mean.

Figure 6: Missionary Presence and Total Enterprises: 15 Periods



Notes: The figure reports event-study estimates from Equation 3, using 15 periods before and after missionary arrival. The dependent variable is the total number of enterprises. The average post-treatment ATT is 1.29. The dependent-variable mean is 4.8 overall, 7 in 1920, and 17 in 1936 (Figure A8 reports the mean by year). Appendix Table D1 reports the corresponding pre- and post-treatment averages.

Appendix D.3 reports additional downstream outcomes. Missionary exposure is associated with stronger growth in trade-oriented activity and exports in 1936, consistent with modern banks supporting broader market exchange. The trade effects also appear with a

lag, which reinforces the interpretation of trade as a downstream market response. Appendix D.4 then studies the transition from traditional to modern finance. The clearest substitution pattern is for *Piaohao*: missionary arrival is followed by the rise of modern banks and the eventual decline of these older remittance institutions. Results for *Qianzhuang* are reported separately in the appendix, since their local networks appear to have been more complementary with modern banks.

Finally, Appendix D.5 reports the remaining identification checks. The results are robust to restricting the sample to the post-1900 and post-1905 periods, excluding treaty ports, using longer event windows, balancing the number of pre- and post-treatment periods, adding railway controls, and examining group-specific ATTs. We also find no pre-trends in other predetermined outcomes, including enterprises, government investment, railways, treaty ports, traditional banks, and trade.

4.1.4 Mechanisms

Why would exposure to missionary institutions contribute to the development of modern banks? We focus on two related mechanisms, supported by available data: practical education and the diffusion of useful knowledge through publications and translations. Both channels speak to the same idea. Modern banks were a new institutional technology in China, and missionary institutions helped form the skills and knowledge that made this technology usable locally. The exercises below should be interpreted as suggestive mechanism evidence, since missionary schools, publications, and other mission activities often arrived together.

Education One of the primary channels through which Christianity can foster market-oriented development is through the dissemination of ideas and education (Becker and Woessmann, 2009; Bryan et al., 2020). Jedwab et al. (2022) survey fifty studies on missionaries published over the past decade and find that nearly all of them associate colonial Christian missions with improved human capital outcomes. Education has also been identified as a key mechanism in other studies that find positive effects of missions on Chinese development (Bai and Kung, 2015; Ma et al., 2022b). In our setting, the relevant channel is a specific form of practical education. Missionary schools, especially Protestant ones, taught subjects such as mathematics, science, bookkeeping, economics, and English. These were skills needed to work in, manage, and expand modern financial institutions. We explore the importance of education in four ways.

First, we focus on skilled education by using our banker dataset to assess how missionary institutions influenced the rise of Christian-affiliated bankers. If missionary education had a

significant effect on the financial sector, we would expect a higher share of bankers who were either affiliated with Christianity or educated in Christian schools in areas with missionary presence.²⁹

To test this, we digitized a book that compiled the resumes of notable bankers and created a variable to represent their relationship with Christianity (Jiang, 2014). We coded bankers as “Christian-affiliated” if they were born into a Christian family, converted to Christianity, attended a church school, or worked in a Christian institution. We then matched the names of the banks where these bankers were employed with our bank data for the period 1857–1936.³⁰ The presence of Christian bankers is significant, even though the data include only Chinese individuals and exclude foreigners: across the years, there are 693 Christian bankers out of 2,576 total bankers, yielding a ratio of Christian bankers to total bankers of 26.90%. For intuition on the magnitude of this ratio, we compare it with the proportion of Christian converts. According to data provided by Mattingly and Chen (2022), the average number of Christians in China’s prefectures in 1920 was 2,730, with an average population of 1,371,070, implying that Christians made up 0.20% of the population. Appendix Figure E4 shows the ratio of Christian bankers to the total number of bankers over time.

Figure 7 then estimates the effect of missionary presence on the share of Christianity-related bankers. The share increases after missionary arrival, consistent with the idea that missions helped train or connect individuals who later worked in modern banking.

Second, we turn to Republican-era elites to ask whether missionary presence contributed to high-skill Chinese human capital and connections to foreign education and networks. Among the 14,763 biographies matched to our historical prefectures, 1,282 (8.7%) have a coded Christian-school background, our proxy for Christian educational affiliation. Appendix Figure B1 reports its level and share across school-age cohorts. Because the biographies do not consistently report when and where schooling occurred, we construct two prefecture-year proxies.³¹

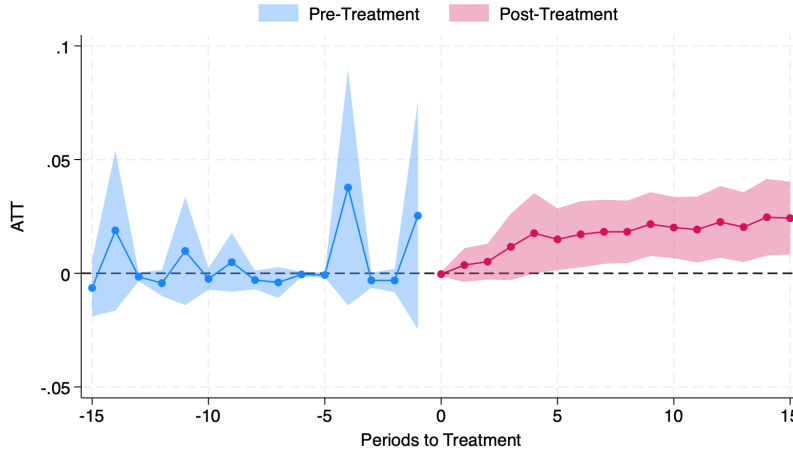
Panel A of Figure 8 assigns elites with a Christian-school background to their native prefecture at age 15, proxying for school-age mission exposure. Panel B dates foreign study using the reported or parsed departure year. The separately estimated event studies show

²⁹For example, Lin Zifeng (1892–1971) was a prominent banker and Baptist church leader in the Republic of China. His father was a Baptist pastor, and Lin became a Christian at the age of 14, attending the Baptist Jieyang Truth High School. In 1922, Lin founded the K. W. Bank in Guangzhou, which was reorganized as the K. W. Savings Bank in June of the same year. Between 1924 and 1931, he opened a branch in Shanghai with a share capital of one million Hong Kong dollars.

³⁰Currently, the year is therefore imputed from the bank data.

³¹The Data Appendix describes the construction of the Christian-school indicator and foreign-study departure-year proxy in detail.

Figure 7: Missionary Presence and the Share of Christianity-Related Bankers



Notes: The figure reports event-study estimates from Equation 2, using the share of Christianity-related bankers as the outcome. The average post-treatment ATT is 0.016. The dependent-variable mean is 0.025 overall, 0.052 in 1920, and 0.113 in 1936.

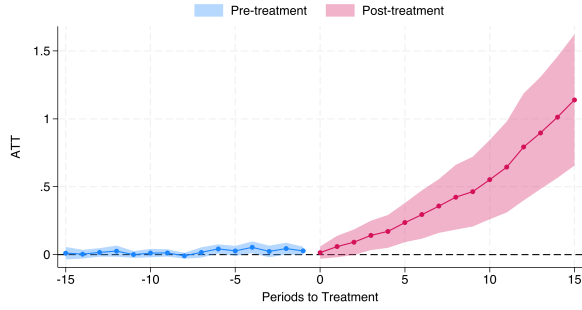
limited differential accumulation before missionary arrival and sustained increases afterward in both proxies. Subject to the age and location assumptions in Panel A, the results are consistent with missions helping form highly skilled Chinese intermediaries and strengthening access to foreign education and networks.

Third, we consider education more broadly by distinguishing missionary exposure that included schools from exposure that did not. With the expanded school data, the effect on modern banks is positive both in prefectures with missionary schools and in prefectures without them. This suggests that education is important, but it is not the only channel. The timing is also informative: the school difference becomes larger later in the event window, which is consistent with the time needed for education to affect the local supply of bankers and managers. For downstream enterprises, the school channel carries most of the bite, suggesting that practical education mattered especially for the broader diffusion of market activity. Appendix Table E3 summarizes these magnitudes, and Appendix Figure E8 reports coefficient-by-coefficient difference tests for the main school and school-publication splits.

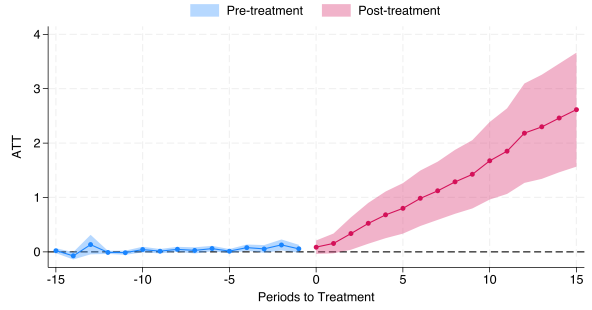
Fourth, using the CHCD institution tradition field, we categorize mission presence as Protestant or Catholic. If the effect operates through practical education, we expect it to be driven mainly by Protestants, who placed stronger emphasis on schooling and conversion through education (Bai, 2019; Bai and Kung, 2015). Figure 10 supports this interpretation. Protestant missionary arrival is associated with a positive effect on modern banks, while

Figure 8: Missionary Presence and Chinese Elite Formation

(a) Christian-school elites



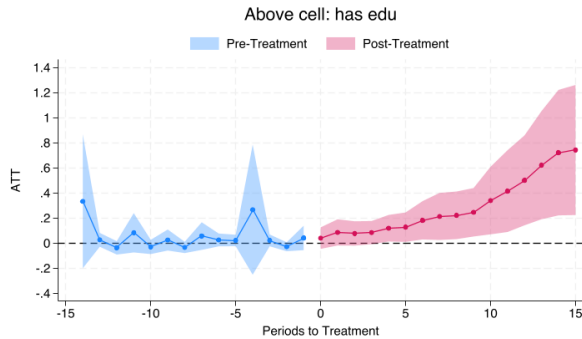
(b) Study abroad



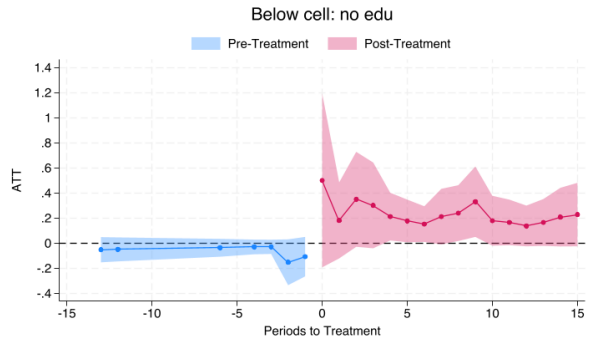
Notes: The treatment is the first year of missionary presence in the elite's native prefecture, harmonized to 1911 boundaries. Both panels report cumulative prefecture-year stocks and are estimated separately using Equation 2. Panel A cumulates biographies with a coded Christian-school background, assigning each to the native prefecture at birth year plus 15. This is a proxy for school-age mission exposure, not a direct observation of local school attendance. Panel B cumulates biographies coded as having studied abroad, assigning each to the first four-digit departure year parsed from the study-abroad year field. The post-window increases are 158% and 134% in Panels A and B, respectively.

Figure 9: Missionary Presence and Modern Banks: Places With and Without Missionary Schools

(a) Modern banks on missions with schools



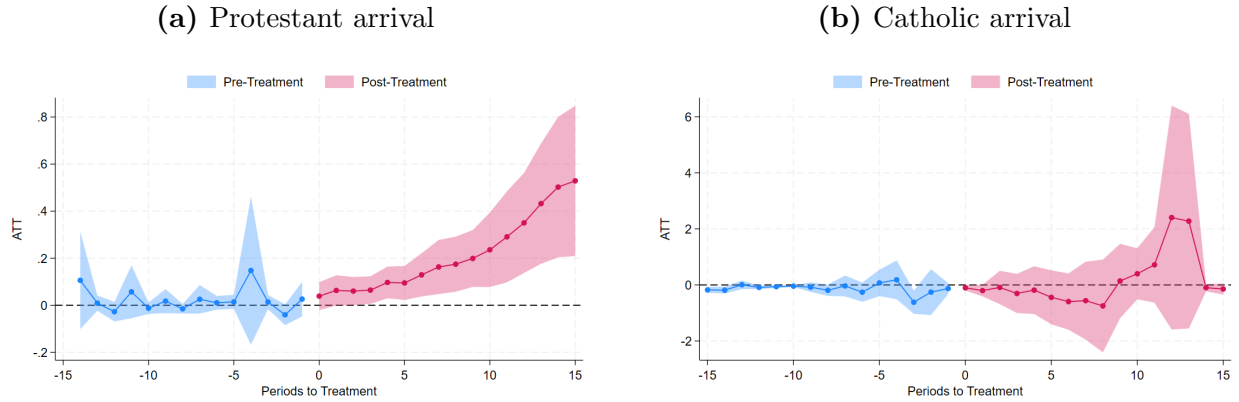
(b) Modern banks on missions without schools



Notes: The figure uses the expanded school definition that includes the additional digitized missionary schools. Panel A includes treated prefectures with at least one missionary school; Panel B includes treated prefectures without a missionary school. Never-treated prefectures are used as controls in both panels.

Catholic missionary arrival is not.

Figure 10: Missionary Presence and Modern Banks: Protestant and Catholic Missions



Notes: The treatment in Panel A is defined as the year of arrival of a Protestant missionary institution. The treatment in Panel B is defined as the year of arrival of a Catholic missionary institution. The outcome is the total number of modern banks.

Publications and translations The second mechanism is the diffusion of useful knowledge through publications and translations. Missionaries produced and translated Western texts into Chinese, including religious texts, school materials, newspapers, scientific works, and economic or educational materials. Many of these publications were written in vernacular or romanized Chinese, making them accessible beyond the classical elite.³² This channel complements the education channel: schools trained students directly, while translations and printed materials helped knowledge circulate outside the classroom.

We exploit publication data from 677 missionary publications. Table 4 summarizes the main features of these publications. About 38% were in vernacular or romanized Chinese, 32% included a Western language, 17% included English, and 16% were educational or student-oriented. Figure 11 plots the growth of missionary publications over time.

We then test whether locations with missionary publication activity experienced stronger effects on modern banks. Figure 12 splits treated prefectures by whether they ever had a missionary publication in circulation. The effect is much stronger in places with publication activity and flatter in places without publications. This is consistent with publications and translations helping diffuse the knowledge needed to adopt and operate modern financial institutions.

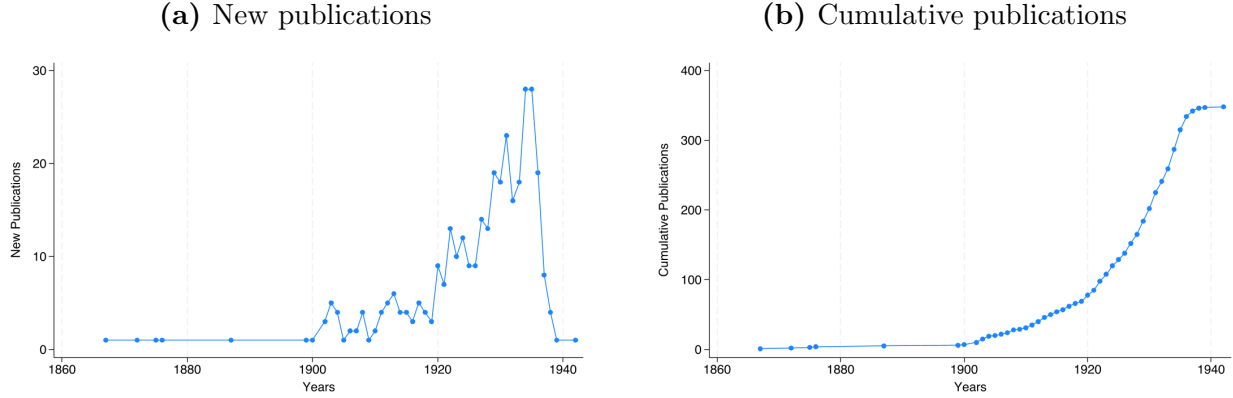
³²This emphasis was especially pronounced among Protestants. Their doctrine placed particular weight on lay believers reading Scripture directly, rather than receiving it primarily through clergy as in Catholicism, which encouraged Bible translation, vernacular printing, and literacy. (Gallego and Woodberry, 2010; Woodberry, 2012).

Table 4: Features of Missionary Publications

Publication feature	Freq.	Percent
Religious / missionary focus	461	70.2%
Vernacular / romanized Chinese	260	38.4%
High circulation (>900)	260	38.4%
Western language present	214	31.6%
English present	112	16.5%
Educational / student-oriented	108	16.4%
High frequency (24+ issues per year)	101	14.9%
Bilingual (Chinese + Western)	55	8.1%
Scientific / technical	29	4.4%

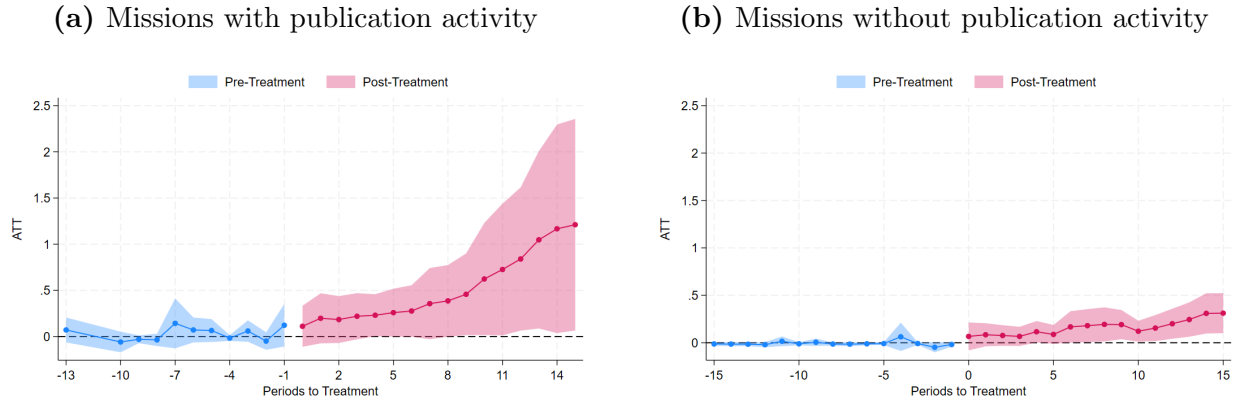
Notes: The table reports descriptive features of the missionary publication data used in the mechanism exercises. The categories are not mutually exclusive.

Figure 11: Missionary Publications Over Time



Notes: Panel A plots the number of new missionary publications by start year. Panel B plots the cumulative number of missionary publications over time.

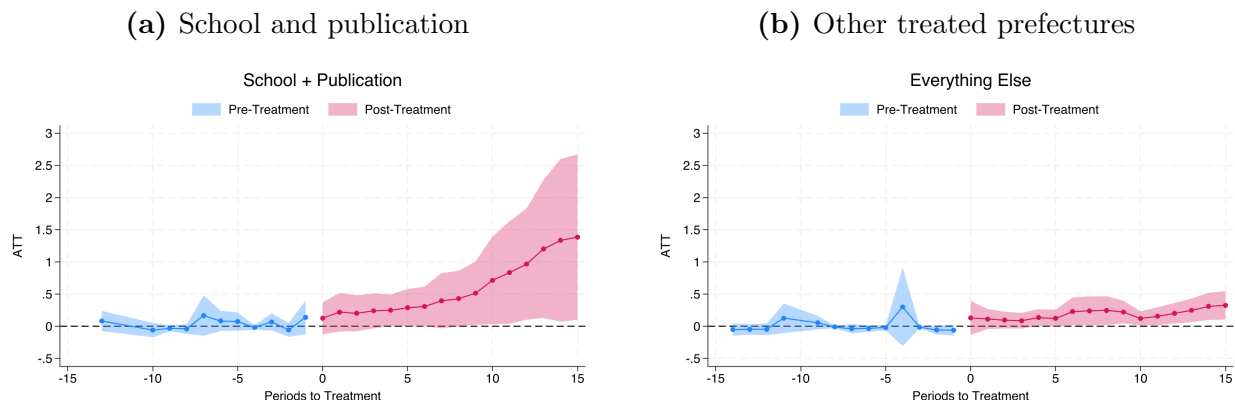
Figure 12: Missionary Publication Activity and Modern Banks



Notes: The outcome is the total number of modern banks. Panel A includes prefectures with at least one missionary publication in circulation. Panel B includes prefectures without missionary publication activity. Never-treated prefectures are included in both panels.

The education and publication channels also appear complementary. In the split by schools and publications, the largest effect is in prefectures that had both a missionary school and missionary publication activity. Prefectures with neither still show a positive effect, but the effect is smaller. This reinforces the interpretation that education and publications amplified missionary exposure, while also indicating that they do not fully exhaust the channels through which missionaries affected modern banks. Appendix Figure E7 reports the four separate cells, and Appendix Table E3 gives the corresponding average pre- and post-treatment differences.

Figure 13: School and Publication Complementarity



Notes: The figure compares prefectures with both a missionary school and missionary publication activity to the remaining treated prefectures. Never-treated prefectures are used as controls in both panels. Appendix Figure E7 reports the four separate school-by-publication cells.

Additional channels and alternative explanations The mechanism evidence should not be read as only a human-capital story. The broader institutions literature distinguishes explanations based on skills from explanations based on rules, organizations, and institutional practices (Acemoglu et al., 2001, 2002; Glaeser et al., 2004). In this setting, missionary schools and publications are one way institution-specific knowledge moved, but other channels could also matter.

We examine two additional channels. The first is a competing explanation based on clans. Recent work argues that Confucian clan institutions and financial markets can act as substitutes (Chen et al., 2022b), and missionary activity may weaken kin-based institutions (Henrich, 2020; Schulz et al., 2019). This raises the possibility that our results reflect missionaries weakening clans, rather than directly facilitating modern banks. Appendix Figure E9 shows that the effect on modern banks is similar across high- and low-clan prefectures. Appendix Tables E5 and E6 then report a horse-race exercise between missionary exposure and clan intensity, using the IV strategy introduced in Section 4.2. The missionary effect

remains positive, while the clan effect becomes small and statistically insignificant.

The second is a possible network channel through compradors. Compradors were Chinese intermediaries who worked with foreign firms and banks and could transmit foreign financial practices into Chinese institutions. Our banker biography data provide descriptive evidence consistent with this channel. Appendix Table E4 shows that comprador-background bankers are a small group, but disproportionately connected to Christianity and to Chinese finance: 28% are Christianity-related, compared with 15% among other bankers, 63% later founded at least one Chinese bank, *Qianzhuang*, *Yinhao*, or *Piaohao*, and 72% worked at both foreign and Chinese banks.³³ We interpret this as suggestive evidence that missionary exposure may also have operated through networks of intermediaries connecting foreign and Chinese finance, although we do not treat this as a separate causal mechanism.

Overall, the mechanism evidence suggests that missionaries mattered through more than one channel. Education created practical human capital, especially for banking and downstream enterprise formation. Publications and translations diffused useful knowledge about modern economic and financial practices. The clan exercises suggest that the results are not simply driven by missionaries weakening kin-based institutions, while the comprador evidence points to a possible intermediary network connecting foreign and Chinese finance. Taken together, the evidence is consistent with missionaries transmitting institution-specific knowledge: not just schooling, but also practices, networks, and organizational models that helped modern banks take root.

4.2 Long-Term Analysis: IV Estimates

Can the early influence of missionaries, who introduced capitalist practices between 1857 and 1936, create lasting advantages that contribute to China’s modernization from 1949 to 2022? To test this hypothesis, we examine the relationship between historical missionary exposure and bank development over this later period. Specifically, we estimate the following regression model:

$$Y_{i,p} = \xi_p + \psi \text{TotMiss Inst}_{i,p} + \nu_{i,p}$$

Where i denotes a prefecture inside a province p . Our outcome of interest is the total number of modern banks in a given recent year, and our treatment of interest is a measure of exposure to missionaries over the whole time period, $\text{TotMiss Inst}_{i,p}$, which is the total

³³In levels, Sheet 1 of *Bankers CV.xlsx* contains 1,041 bankers, of whom 43 are flagged as comprador-related.

number of missions ever built or established between 1850 and 1936.³⁴ This cross-sectional regression presents obvious endogeneity issues that we cannot overcome directly by exploiting panel techniques, as we did using the difference-in-difference analysis. Specifically, over the whole time period, the total number of missions established in a given prefecture is endogenous and likely to be correlated with both the error term and the outcome. Our solution is to construct an instrumental variable aimed at predicting the total number of missionary institutions. Following [Sequeira et al. \(2020\)](#), we exploit the interaction between national missionary migration flows and railway connections to instrument for missionary exposure. In this section, we show the estimating equations, explain the logic behind the instrument both graphically and descriptively, provide some examples, and then show the results.

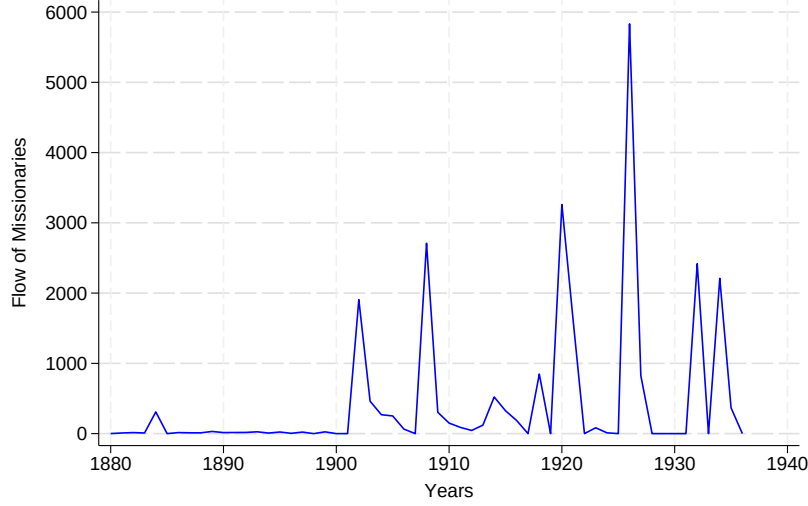
4.2.1 Estimating Equations

This IV strategy is based on two key observations regarding missionary immigration patterns during this period. Upon arriving in China, missionaries’ preferred modes of transportation, whenever available, were rivers and the railway system to travel inland to their intended destinations. Consequently, a prefecture’s connection to the railway network affected the number of missionaries settling in that prefecture.³⁵ The second observation relates to the significant year-to-year fluctuations in the total number of missionary immigrants to China within our study period, as shown in [Figure 14](#). As a result, prefectures connected to the railway network during periods of high missionary immigration typically experienced an increase in missionary arrivals and, consequently, in the total number of missions established over the entire time period.

³⁴This approach ensures consistency with the period used in the DiD analysis and allows us to examine outcomes in the 1930s. The results remain unchanged when using the broader timeframe of 1850-1950 instead.

³⁵The impact of railroads on the number of missionaries is unlikely to confound the DiD estimates. In the DiD analysis, we assume that the number of railroads does not significantly affect the timing of missionary entry in the cohorts of prefectures that receive treatment within a given range of years, relative to those that will receive treatment in the future and those that are never treated. The balance tests in [Tables 2](#) and [Table 3](#) suggest that this assumption holds.

Figure 14: Annual Flow of Missionaries in China, 1880–1936



Notes: Flow of Missionaries at time t is defined as the total number of missionaries present in China in $t + 1$ minus the total present in t . Negative changes are set to zero, and the series is winsorized at the 1st and 99th percentiles to match the missionary-flow shifter used in the zero-stage regression. Source: China Historical Christian Database (CHCD).

To better understand the source of variation in the instrument and to construct it, our analysis begins by estimating the following zero-stage equation:

$$\begin{aligned} \text{MissInst Stock}_{i,t} = & \alpha_t + \alpha_i + \delta \text{RRstock}_{i,t-1} \\ & + \beta \text{Missionary Flow}_{t-1} \times \text{RRstock}_{i,t-1} + \varepsilon_{i,t} \end{aligned} \quad (6)$$

Where i indexes prefectures and t years (1850-1936). $\text{MissInst Stock}_{i,t}$ is the total missionary institutions in prefecture i and year t . $\text{Missionary Flow}_{t-1}$ is the flow of missionaries arriving in China between $t - 1$ and t and $\text{RRstock}_{i,t-1}$ is total number of railways in year $t - 1$. α_i is prefecture fixed effects and α_t is year fixed effects.

Central to our identification strategy is the interaction between the aggregate flow of missionaries into China and whether a prefecture was connected to the railway network:

$$\beta \text{Missionary Flow}_{t-1} \times \text{RRstock}_{i,t-1}.$$

This interaction captures the differential effect that railway connection had on immigrant settlement during periods of high aggregate missionary immigration compared to periods of low aggregate immigration. Consequently, we expect the estimate of β in the equation to be positive.

The two variables that comprise the interaction term are also included in equation (6), capturing the effect of being connected to a positive number of railway lines when no missionaries are entering China, and the general effect of missionary flow (absorbed by year fixed effects).

Then, to aggregate the zero stage at the prefecture level, we conduct the following exercise. After estimating equation (6), we calculate the missionary stock in each prefecture and year as predicted by the interaction between the aggregate inflow of missions and connection to the railway network:

$$\widehat{\text{MissInst Stock}}_{i,t} = \widehat{\beta} \text{Missionary Flow}_{t-1} \times \text{RRstock}_{i,t-1}$$

where $\widehat{\beta}$ is the estimate of β from equation (6). We then sum this prefecture year-specific predicted missionary stock over the whole time period (1850-1936):

$$\widehat{\text{TotMiss Inst}}_i = \sum_{t=1}^T \widehat{\text{MissInst Stock}}_{i,t}$$

Where T is the total number of periods. Finally, our 2SLS equations are equation (7), the first stage, and equation (8), the second stage:

$$\begin{aligned} \text{TotMiss Inst}_{i,p} = & \zeta_p + \mu \widehat{\text{TotMiss Inst}}_{i,p} \\ & + \omega \text{RR Duration}_{i,p} + \text{RR Density}_{i,p} + \varepsilon_{i,p} \end{aligned} \quad (7)$$

$$\begin{aligned} Y_{i,p} = & \xi_p + \psi \text{TotMiss Inst}_{i,p} \\ & + \pi \text{RR Duration}_{i,p} + \text{RR Density}_{i,p} + \nu_{i,p} \end{aligned} \quad (8)$$

Where i indexes prefectures and p indexes provinces, ζ_p and ξ_p are province fixed effects, and $Y_{i,p}$ is our outcome of interest, measured in a given recent year. We use the total number of banks for each year in separate regressions from 1949 to 2022, and GDP estimated through night lights from 1992 to 2019. $\text{RR Duration}_{i,p}$ denotes the number of years a prefecture has been connected to the railway network as of 2000, while $\text{RR Density}_{i,p}$ is defined as: $\frac{\sum_{t=1}^T \text{RRstock}_{i,t}}{\text{Area in km}^2}$, representing the total number of railway lines per km^2 . This metric serves as the prefecture-level equivalent of the same term included in the zero-stage equation.

We include these controls, already introduced in the zero-stage, because the railway network influenced missionary numbers beyond its interaction with missionary flows. For instance, prefectures connected to the railway early could have a higher number of missions established by the end of the study period. Similarly, a greater number of railway lines could

be associated with an increased presence of missions over time.

For this reason, our IV strategy cannot rely on railway stock and length of connection as sources of variation, but instead needs to directly control for these factors. In fact, conditional on the length of connection to the railways and the total number of railway lines per km^2 , we are identified by the fact that prefectures connected prior to high migration periods received more missions than those connected prior to a lull. What matters is therefore the coincidental timing of railway construction in interaction with missionary flows. To be more precise, our source of exogenous variation comes from the zero-stage interaction: number of railways $t-1 \times$ national missionary flows *from $t-1$ to t* .

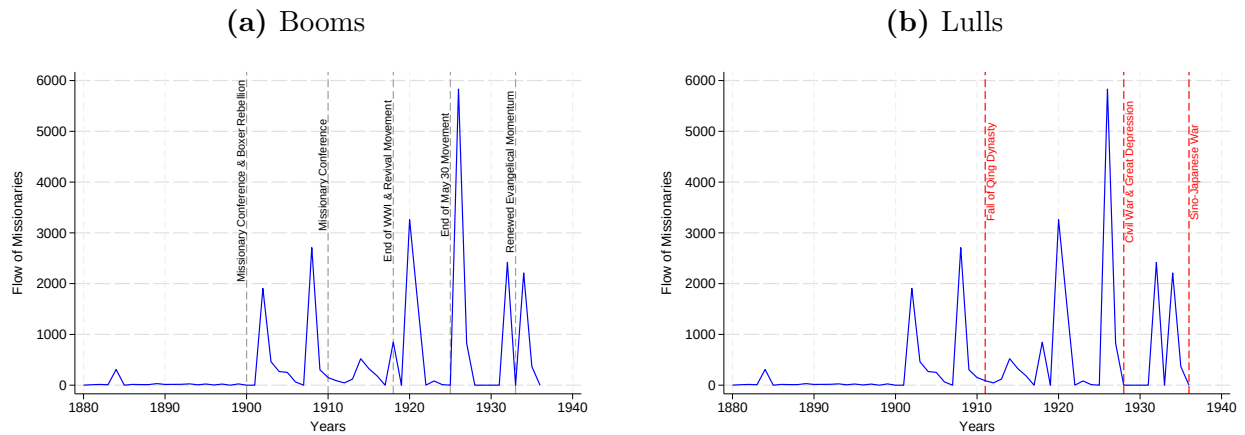
The exclusion restriction is satisfied if this interaction affects our outcome of interest, such as the total number of banks in 2000, solely through its impact on the total number of missions. An example of a violation of this assumption would occur if a prefecture decides to build a railway line in period $t-1$ specifically to attract more missionaries in period t . Note that internal movements of people and resources due to the railway do not pose an issue because we are comparing places with the same exposure to the railway over the entire time period, with the only variable being the exact timing of railway connections. For this reason, this IV can only be used when it is possible to aggregate a long historical time period to study more recent outcomes. Potential issues could arise from economic or social conditions within China that affect both the exact timing of railway construction in a given prefecture and the flow of missionaries in a given year. For example, a war might delay railway expansion and missionary flows. However, what matters is that the war or event specifically shifts railway construction from $t-1$ to t , or from t to $t-1$, rather than causing general delays. In the next sections, using a combination of balance and descriptive statistics, we will argue that this is unlikely to occur.

4.2.2 Sources of Variation for Identification

The first source of variation used for identification is the yearly missionary migration flows. While multiple factors influence these flows, we can identify some key push and pull factors that may contribute to the spikes and lulls observed in Figure 14. Figure 15 illustrates events that may correlate with missionary migration. We note that some of these spikes coincide with factors related to the Christian church's priorities, such as the conclusion of missionary conferences aimed at boosting missionary presence overseas or major revival movements.

In fact, during this period, missionary migration was primarily driven by religious revivalist movements and economic conditions in the missionaries' home countries, rather than by the economic situation in China. In the nineteenth century, Western countries, particularly

Figure 15: Missionary Booms and Lulls: Annual Missionary Flow, 1880–1936



Notes: The figure marks potential push and pull factors associated with missionary migration flows between 1880 and 1936. The displayed series is the regression-consistent missionary-flow shifter used in the zero stage: negative annual changes are set to zero and the series is winsorized at the 1st and 99th percentiles. Source: China Historical Christian Database (CHCD).

in Europe and the United States, experienced several religious revivalist movements, such as the Second Great Awakening, which fueled a strong desire for overseas missionary work (Shelley, 2004). Around 1830, there was a prevailing belief in British society that, as a Christian nation, it had a responsibility to spread Christianity to overseas territories (Shelley, 2004). Similarly, between 1880 and 1920, American society viewed overseas missions as a means to demonstrate American power, leading to an unprecedented wave of student volunteers for missionary work abroad (Wang, 1997). The end of World War I saw a significant missionary revival movement, especially in countries with sufficient resources to send missionaries abroad. A similar pattern followed the end of the Great Depression.

Additionally, missionary migration was constrained by the economic power of their home countries. A stable and continuous source of funding was essential for missionaries to sustain their work abroad. For instance, prior to World War I, British missions were the predominant force in China. However, during the war and the subsequent decline of British influence, the strength of British missions diminished. By 1930, the American Missionary Society's assets in China had grown to \$41.9 million, four times those of the British, reflecting the shift in missionary power due to war efforts (Dan, 1996).

Other foreign immigrants in China followed a different trajectory than the missionaries. During this period, foreigners in China also included merchants, foreign government officials, and explorers. Due to security concerns, treaty restrictions, and economic incentives, merchants and government officials primarily operated within treaty port areas, while the only foreigners who ventured into the inland were a small number of explorers.

Nevertheless, it is important to note that other factors may have influenced these flows. For example, another pattern that emerges is the presence of spikes following the end of conflicts against Christians, such as the Boxer Rebellion or the May 30th Movement. Additionally, panel b shows that some of the lowest missionary migration periods coincide with the onset of political and social instability in China. These factors may pose a problem for identification insofar as they are correlated with the exact timing (year by year) of railway line completion.

The second source of variation is the number of new railway lines built annually in China. Figure 16 overlays this data with missionary flows to graphically illustrate the interaction used in the IV. Panel A illustrates the type of comparisons our IV makes, where an X represents a prefecture that gets connected to a new railway line the year before a missionary migration boom, while a circle represents a prefecture connected the year after. Implicitly, our IV relies on these comparisons, within provinces. As an example, consider the 1934 boom, which involved 2,210 new missionaries. In our data, Jiangning Prefecture in Jiangsu Province, connected just before the boom, received 57 new missionaries (2.58% of the total) and established 5 new missions (a 12% increase from a base of 42 missions). In contrast, Haizhou Prefecture in the same province, connected just after the boom, received 1 new missionary (0.05% of the total) and established 0 new missions (a 0% increase from a base of 2 missions).

Since only 30% of Chinese prefectures were connected to the railway by 1950, many comparisons also resemble those depicted in Panel B. In Panel B, the same prefecture, represented by the color black, is first connected prior to a boom, then receives a new railway line prior to a lull, and another one before a subsequent boom. The same logic applies to the “orange” prefecture. Essentially, these comparisons are made for each year and each prefecture that gets a new railway line, and then they are aggregated at the prefecture level.

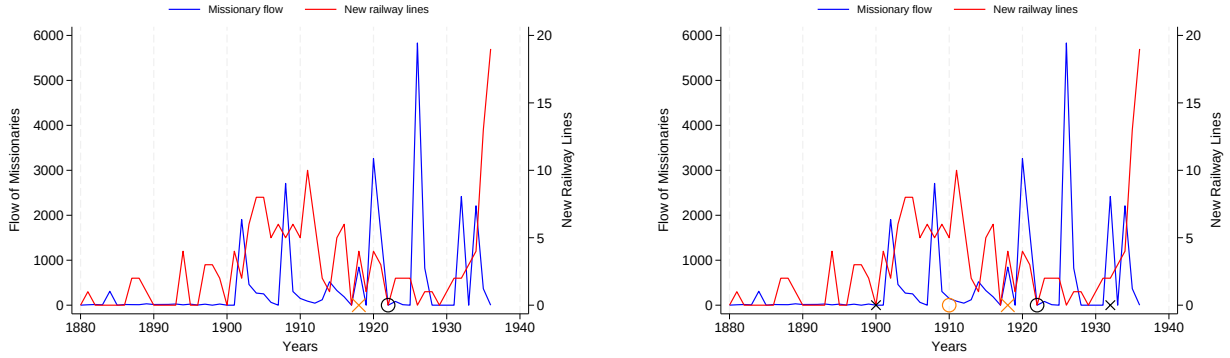
4.2.3 Threats to Identification

A first threat to our identification strategy is the correlation between missionary flows and both early railway connections and railway density. To account for this, in the construction of the instrument, we control for years of connection (as of 2000) and the number of railway lines per square kilometer. For example, Jiangning Prefecture has been connected to the railway network for 92 years with 7 railway lines, whereas Haizhou Prefecture has been connected for 95 years with 6 railway lines. The difference between these two prefectures, 3 years of connection and 1 railway line, will be accounted for in the first stage regression allowing the comparison to rely solely on the exact timing of railway expansion and missionary flows.

Figure 16: New Railway Lines and Missionary Flow, 1880–1936

(a) Across-prefecture timing

(b) Within-prefecture timing



Notes: The legend identifies the blue national missionary-flow series and the red number of new railway lines. A new railway line is either an added stop on an existing line or a newly constructed line. Black markers represent one prefecture and orange markers another prefecture in the same province; X's denote boom-year connections and circles denote lull-year connections. The missionary-flow series is the regression-consistent shifter used in the zero stage. Sources: China Historical Christian Database (CHCD) and Ma et al. (1983).

Secondly, the two interaction terms (connection date and railway density) could be correlated with each other. For example, in Jiangning Prefecture, the coincidental timing of railway expansion was driven by a deliberate desire to attract more missionaries. Moreover, political unrest in China might affect both the timing of railway construction and missionary flows. To address this issue, we can implement several strategies. First, we examine the balance between “boom Prefectures” and “lull Prefectures”. We define boom Prefectures as those connected to a railway line one or two years prior to when missionary flows are above the median, and lull Prefectures as those connected one or two years after. Table 5 presents the results of a pairwise t-test difference test using OLS, with clustering at the prefecture level (since the same prefecture may appear multiple times). All variables are measured at the beginning of the period, in 1880. We note that there are no significant differences between boom and lull places.

Another important consideration is our reliance on the exact year of completion for each railway line. Even if there are factors that could bias our estimates, these factors would need to simultaneously affect both railway construction and missionary flows with precise timing. Given the likely unforeseen delays in railway construction - our data indicate an average completion time of three years with a standard deviation of six years - such precise alignment is improbable. For example, we would have to assume that the Chinese Civil War of 1927, which likely correlates with a lull in migration between 1928 and 1931, specifically delayed the completion of a railway in a prefecture from 1926 to 1927 (or vice versa).

Table 5: Balance Between Boom and Lull Prefectures

Variable	(1) Lull Prefecture		(2) Boom Prefecture		(1)-(2) Pairwise t-test	
	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	N/Clusters	Mean difference
Total Modern Banks	45	0.133	105	0.114	150	0.019
	25	(0.098)	38	(0.083)	48	
Total Enterprises	45	0.578	105	0.381	150	0.197
	25	(0.316)	38	(0.267)	48	
Ln (investment gov. factory)	45	1.815	105	2.726	150	-0.911
	25	(0.925)	38	(1.043)	48	
Total railways	47	8.149	120	17.100	167	-8.951
	27	(5.990)	43	(10.063)	53	
Total Treaty Ports	45	0.133	105	0.133	150	0.000
	25	(0.069)	38	(0.061)	48	
Whether there's main river	47	0.617	120	0.658	167	-0.041
	27	(0.110)	43	(0.080)	53	
Provincial capital	46	0.370	120	0.450	166	-0.080*
	26	(0.111)	43	(0.089)	52	
Crop suitability: Rice	47	2.082	120	2.075	167	0.007
	27	(0.251)	43	(0.185)	53	
Ln population	44	5.307	105	5.412	149	-0.104
	24	(0.144)	38	(0.148)	47	
Coastal	47	0.447	120	0.283	167	0.163
	27	(0.115)	43	(0.084)	53	

Notes: Boom prefectures are prefectures connected to a railway line one or two years before national missionary flows are above the median. Lull prefectures are prefectures connected one or two years after such high-flow years. Variables are measured at the beginning of the period, in 1880, except time-invariant geographic variables. The table reports means for each group and the pairwise difference from an OLS regression with standard errors clustered at the prefecture level, since a prefecture can appear more than once. N/Clusters reports the number of prefecture-year observations and distinct prefecture clusters. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Finally, the timing of railway connections may be correlated with a prefecture's geographic location. To address this, we include prefecture fixed effects in our zero-stage regression and subsequently control for latitude and longitude as a robustness check in the first stage.

4.2.4 Results

We begin by presenting our results from the zero-stage estimating equation (6) and the first stage equation (7) in table 6.³⁶

³⁶The baseline winsorizes the missionary stock, missionary-flow shifter, railway variables, and outcomes at the 1st and 99th percentiles. The findings remain similar under the older 0.5th–99.5th percentile convention and when excluding extreme railway outliers. With no winsorization, the coefficient path remains positive but the first stage is weak. For detailed tables and figures, refer to the robustness section.

Table 6: Zero-Stage and First-Stage Estimates: Constructing the Instrument

Variables	Zero-stage (1) MissInst Stock	First stage (2) Tot Miss Inst
RRstock (t-1)	1.5672*** (0.3465)	
Miss Flow (t-1) x RRstock (t-1)	0.00003738** (0.00001456)	
Predicted Tot Miss Inst		3.3997*** (0.8816)
RR Duration		0.0690** (0.0279)
RR Density		-685.2861*** (128.9182)
Prefecture FE	Yes	No
Time FE	Yes	No
Province FE	No	Yes
Obs.	21,242	247
Adj. R sq.	0.722	0.240
Mean Dep. Var.	3.10	10.28
S.D. Dep. Var.	6.06	9.06
F-stat	6.59	14.87

Notes: MissInst Stock $_{i,t}$ is the total number of missionary institutions in prefecture i and year t . Missionary Flow $_{t-1}$ is the flow of missionaries arriving in China between $t-1$ and t , and RRstock $_{i,t-1}$ is the total railway stock in year $t-1$. Predicted Tot Miss Inst is the prefecture-level sum over time of predicted missionary institutions from the zero-stage regression in Equation 6. RR Duration $_{i,p}$ is the number of years connected to the railway network as of 2000, and Density $_{i,p}$ is $\left(\sum_{t=1}^T \text{RRstock}_{i,t}\right) / (\text{area in km}^2)$. The missionary stock, flow shifter, and railway variables use the 1st–99th percentile convention. Fixed effects are reported in the table. Robust standard errors are in parentheses. The Kleibergen–Paap first-stage F-statistic is 14.87. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

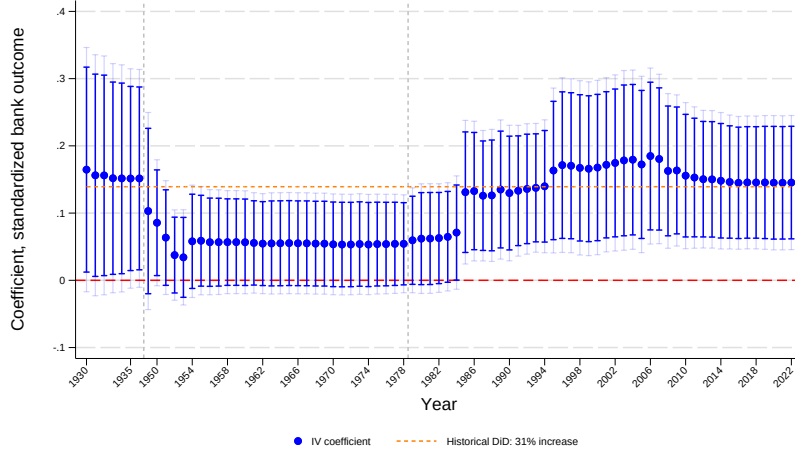
We see that our interaction of interest, Missionary Flow $_{t-1} \times \text{RRstock}_{i,t-1}$, is positive and significant in our zero-stage estimating equation. This indicates that increased missionary flow in China the year before a railway line is constructed predicts the establishment of more missionary institutions. Column (2) shows the result of the first stage, which is also positive and significant. Our predicted measure of missionary institutions at the prefecture level, derived solely from the effect of railway and missionary flows, remains positively associated with the actual number of missions in a given prefecture over the entire time period. The

Kleibergen–Paap F-statistic of the instrument is 14.87.

Next, we proceed to our second-stage regressions using the last six years of our historical modern-bank dataset, previously utilized in the DiD estimation, along with data on total banks from 1949 to 2022. Given the substantial variation in the total number of banks over this period (see Appendix, Figure F1), we standardize the outcome within each year to facilitate comparisons of magnitudes.³⁷ We use a portion of the historical data to assess whether our IV approach yields results similar to those in the previous section. Results from individual cross-sectional regressions for each year are presented in Figure 17, and Appendix Table F1 reports the headline years in table form. The historical estimates from 1930 to 1936 are positive and significant at the 10% level. In raw-bank units, they imply increases of approximately 51% in 1930 and 34% in 1936. The 1936 magnitude is close to the 31% historical DiD estimate, while the less precise 1930 estimate is larger.

³⁷Results using the total number of banks, not standardized, are in Figures F2 and F3.

Figure 17: 2SLS Estimates: Standardized Modern and Total Banks, 1930–1936 and 1949–2022



Notes: The figure reports coefficients from Equation 8 estimated separately by year, using the historical modern-bank data used in the DiD design and recent county- and district-level bank counts assigned to 1911 prefectures by each modern unit’s centroid. The annual outcome and IV-side variables use the 1st–99th percentile convention. Blue points are IV estimates; dark and light intervals show 90% and 95% confidence intervals. The red dashed horizontal line marks zero, the orange dashed horizontal line maps the historical DiD estimate of a 31% proportional increase into 1936 standardized-outcome units (0.139 standard deviations), and the vertical dashed lines mark the establishment of the People’s Republic in 1949 and the beginning of market reforms in 1978. Historical modern-bank means are 2.4 in 1930 and 5.8 in 1936, so the raw-unit IV estimates imply increases of approximately 51% and 34%, respectively. Recent total-bank means are 145 in 1990 and 707 in 2022, so the corresponding increases are approximately 14.2% and 14.7%. Appendix Table F1 reports the 1936 and 2020 estimates in regression-table form. Source: China Financial Supervision and Administration Bureau.

We observe a positive and significant effect of missionary exposure on total banks starting in the 1980s, coinciding with the country’s economic reforms.³⁸ During the period between the 1949 and 1978 vertical markers, private and foreign banks were closed, merged, or brought under state control as the financial system was nationalized. The muted coefficients during this period are therefore consistent with the institutional environment suppressing private market activity. This evidence suggests that regions with a historical presence of missionaries, and consequently an increase in modern banks, possessed a form of “latent capitalism” that was ready to take off once modern banks were permitted and profitable again. The early development of a modern financial system likely set the stage for the adoption of the same technology and capitalist practices once the Chinese government allowed it.

³⁸The 1990 estimate is significant at the 5% level and the 2022 estimate at the 1% level. In raw units, they imply increases in the total number of banks of approximately 14.2% and 14.7%, respectively.

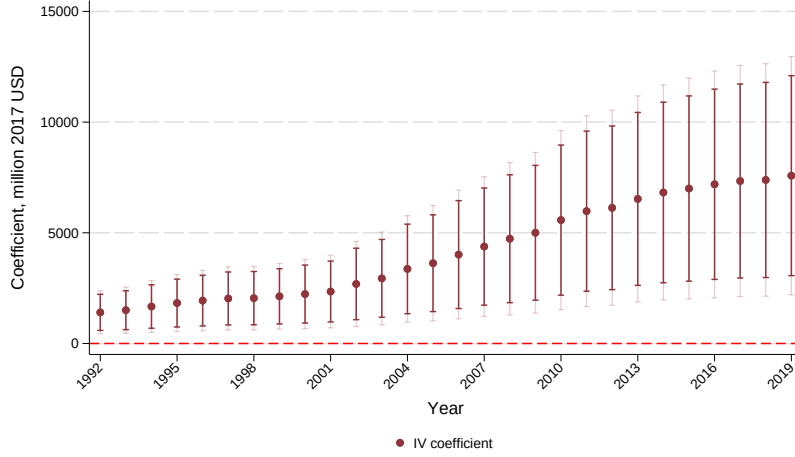
The bank result is robust to adding alternative population measures as controls and to expressing bank outcomes per capita (Appendix Table F2 and Figure F4).³⁹

These results are also reflected in GDP, as shown in Figure 18, which uses gridded revised real GDP and electricity consumption data from 1992 to 2019 based on night-light observations. The annual coefficients are positive and statistically significant throughout the sample period. Although they rise in levels as average output expands, their magnitude relative to mean prefecture GDP is stable: the average estimate is approximately 16.1% of the annual mean, and the 2019 estimate is 16.5%.⁴⁰ The enterprise evidence points in the same direction. Historical missionary exposure predicts more industrial enterprises after reform, with similar 2007–2013 standardized estimates for private and all enterprises. The increase in private enterprises is also positive in industries with high and low external-finance dependence; the directly estimated high-minus-low outcome remains positive. Appendix Figures F6 and F7 report these results, and Appendix F explains the enterprise mapping and external-finance-dependence classification.

³⁹We do not use these population-adjusted specifications as our baseline because missionary exposure may itself affect population growth, migration, and urban concentration, making later population a potential outcome of treatment. Moreover, no single population measure is an obviously preferable denominator for the per-capita outcomes, and reliable prefecture-level population estimates are particularly scarce for the Mao period.

⁴⁰Appendix Table F3 reports corresponding 2019 IV estimates with two alternative population controls and with two standardized per-capita GDP outcomes. Figure F5 reports the annual per-capita path using the fixed Census 2015 denominator.

Figure 18: 2SLS Estimates: Real GDP, 1992–2019 (Millions of 2017 USD)



Notes: The figure reports coefficients from Equation 8 estimated separately by year. GDP is constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons. The mission stock, instrument, railway controls, and annual GDP outcomes use the 1st–99th percentile convention. Maroon points and dark/light intervals show the IV estimates and 90%/95% confidence intervals. All 28 annual estimates are positive and statistically significant at the 1% level. The average estimate is approximately 16.1% of annual mean prefecture GDP, and the 2019 estimate is 16.5% of that year’s mean. Appendix Table F1 reports the 2019 estimate in regression-table form. Source: Chen et al. gridded revised real GDP and electricity consumption, 1992–2019, based on night lights.

As a within-province validation, we reproduce the long-run bank exercise at the county level in Shaanxi, using historical-prefecture fixed effects and the same railway-based source of variation. The county data are substantially thinner than the national prefecture panel, especially for financial institutions in the early Mao years. The Mao-period IV is therefore not estimable, but the post-1978 association between historical missionary exposure and standardized financial institutions is positive in both OLS and IV. We view this exercise as supporting the national result rather than as an independent research design. Appendix Figure F8 reports the coefficient path and Appendix Table F4 summarizes the pooled estimates.

4.2.5 Robustness Checks

We perform several robustness checks on our main result. First, since constructing the instrument requires two stages and the exact functional form of the railway numbers in the first stage - which is cross-sectional rather than panel - is unknown, we modify how we control for railways in the first stage. We believe that our main specification, which includes railway

duration and railway density, already precisely captures the railway duration introduced in the zero-stage. However, the results remain robust when adding non-linear railway controls (such as railway bins - Figures F18 and F19), mileage of railway lines per km^2 (Figure F20), cubic polynomials of latitude and longitude (Figure F21), the number of railways in 1936 (instead of the sum over time - Figure F22), and replacing density with bins (Figure F23).

Second, we alter the variation introduced in the zero-stage or exclude certain observations from the sample. For example, we exclude all treaty ports (Figure F24), drop the top three railway outliers (instead of winsorizing - Figure F25), use different or no winsorization (Figure F27 and F28), and use all the years in the zero-stage instead of the ones used in the DiD (1850-1950 instead of 1850-1936 - Figure F26).

Third, we use a recentering exercise following [Borusyak and Hull \(2023\)](#) to check that the IV is not simply capturing average exposure to missionary inflows through the railway network. The IV gives more predicted missions to prefectures whose railways were present in years with larger national missionary inflows. A related concern is that prefectures with longer or denser railway histories would have received more exposure to any average flow process, independent of the realized timing of missionary inflows. We therefore simulate counterfactual histories of national missionary flows, holding each prefecture’s railway path fixed, and subtract the exposure each prefecture would receive under these simulated flow histories. This removes the average-flow-times-railway component and keeps the part coming from the realized alignment between missionary-flow spikes and railway connections. [Appendix F](#), in the recentered-IV subsection, shows that the long-run estimates are very similar: for standardized banks in 2020, the coefficient falls from 0.145 to 0.127 under the full-flow permutation and remains significant at the 1% level. For GDP in 2019, recentering reduces the coefficient from 7,581 to 6,548 under the full-year permutation and 6,616 under the 10-year block permutation; both estimates remain significant at the 1% level. The first stage is also not weakened: the K-P F-statistic rises from 14.87 to about 16.7 for banks and from 15.01 to about 16.8–16.9 for GDP.

5 Interaction with Openness Policies of the 1980-2000s

Our objective in this section is to examine whether missionary exposure interacts with the significant policy changes that occurred in China during the 1980s. To achieve this, we use our instrumental variable estimation within an event study framework. This approach allows us to compare the effects of missionary exposure on our outcome of interest both before and after the implementation of nationwide policy changes. Essentially, this is a more direct test

of what we observe graphically in Figure 17, where we argue that the effect begins to take off around the 1980s. We are now going to look directly at how the missionary treatment effect appears a few years before and after a given policy.

The key policies we consider in this analysis include:

Land Reform (1978-1984): A series of measures aimed at redistributing land and restructuring agricultural practices, moving away from collective agriculture and towards privatization. This policy was staggered over time and space. However, for the moment, we would treat it as a nationwide shock and test it both at the onset (1978) and at the end (1984).

Provisional Regulations on Banking Administration (1986): Reform intended to modernize and stabilize the banking sector by limiting government involvement and guaranteeing minimal government control.

World Trade Organization (WTO) Accession (2001): China’s entry into the WTO, marking a significant step towards integrating its economy with global markets.

WTO Banking Sector Commitments (2006): Specific commitments related to the banking sector as part of China’s WTO accession. This policy signifies the moment when banks finally began taking full advantage of the WTO accession, as well as allowing foreign banks to operate in the country.

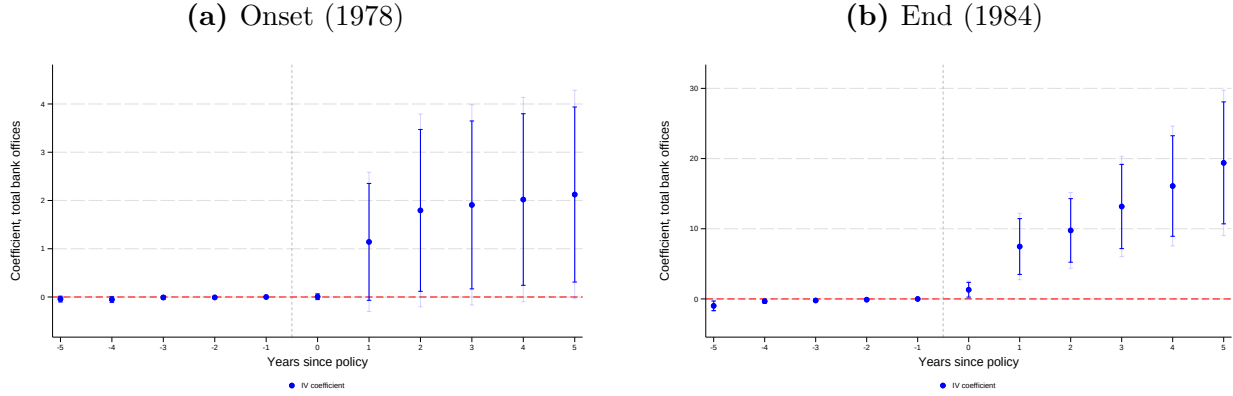
Our empirical model is specified as follows:

$$Y_{it} = \alpha + \sum_{k=-5}^5 \beta_k \widehat{\text{TotMiss Inst}}_i \times D_{t=k} + \delta_t + \mu_i + \varepsilon_{it} \quad (9)$$

In this equation, Y_{it} represents the total number of banks for prefecture i at time t . The term $\widehat{\text{TotMiss Inst}}_i \times D_{t=k}$ captures the interaction between our instrumented total missionary institutions and the time prior to or post a policy change, using a five-year window. The coefficients β_k measure the differential impact of missionary exposure around the time of policy implementation. Additionally, δ_t accounts for time fixed effects, μ_i controls for prefecture fixed effects, and ε_{it} denotes the error term. Standard errors are clustered at the prefecture level, given that the panel structure of this estimation involves stacking cross-sectional regressors on a panel outcome, resulting in repeated observations for the same prefecture multiple times.

The event study using the onset and end of the land reform as time zero is shown in Figure 19. Around the 1978 onset, the missionary-exposure gradient becomes positive and increasingly distinguishable from zero over the following years. The 1984 endpoint shows a sharper post-reform increase, consistent with the long-run bank gradient strengthening as

Figure 19: Missionary Exposure and Land Reform: Total Banks



Notes: The figure reports event-study estimates from Equation 9 for the five years before and after the policy change, using total banks as the outcome. Dark and light intervals show 90% and 95% confidence intervals. Standard errors are clustered at the prefecture level.

implementation spread.

When examining banking regulations specifically, as shown in Appendix Figure F16, we observe a discrete positive shift after the 1986 regulations and continued strengthening around the 2006 WTO banking commitments. The negative coefficients before these later policy dates reflect that each event study is normalized to the year immediately before the policy even though the bank gradient was already evolving after earlier reforms. Finally, Appendix Figure F17 analyzes WTO accession and the differential effect of places with more missionary exposure on GDP. The rebuilt direct-grid estimates show a clear positive jump after accession, while the negative pre-period coefficients reflect the same normalization to an already rising outcome path.

6 Long-Run Mechanisms: What Reactivated After Reform?

The disappearance and subsequent re-emergence of the bank gradient associated with historical missionary exposure is not a simple story of physical persistence. Banking offices did not vanish after 1949. Existing institutions were nationalized, merged, or reorganized, and the state expanded banking offices across China. These offices, however, operated within a planned financial system rather than as the market-oriented institutions studied before 1949. In the bank data, historically mission-exposed prefectures lost their earlier relative advantage during the Mao period and regained it only after reform. The continuous presence of banking infrastructure therefore cannot by itself account for the pattern in Figure 17.

We interpret this pattern as a complementarity between national openness and local capabilities formed before 1949. The reforms raised the return to internationally useful skills, organizational experience, and connections to outside markets. Missionaries were one set of intermediaries through which these inputs became accessible: they combined foreign networks with long residence, Chinese-language ability, and local knowledge. Chinese students, bankers, entrepreneurs, officials, and publishers then selected, adapted, and expanded the practices they encountered. The relevant legacy is therefore not a foreign institution preserved intact, but knowledge and relationships embedded in Chinese people and places. We examine the two channels for which the evidence is strongest—internationally oriented people and networks, and foreign-facing commercial connections—before considering alternative explanations. These exercises provide mechanism evidence rather than a formal causal decomposition. Additional results and data details appear in the [long-run mechanism section of Appendix F](#).

Internationally oriented people and networks The first channel is the survival of human capital and relationships across political and geographic boundaries. Mission schools and related organizations expanded access to practical subjects, foreign languages, and institutions abroad. After 1949, the people who acquired these skills did not remain in a single location. Some continued in mainland institutions, while others moved to Taiwan, Hong Kong, or elsewhere. Reactivation therefore need not take the form of mass return migration to the same prefecture. It can operate through physical return, professional collaboration, information flows, or relationships connecting mainland organizations to people outside China.

We use two biographical sources to examine this channel. The first links records of Chinese engineers and scientists in mainland and Taiwan sources. It identifies 4,640 mainland-origin individuals who are observed in Taiwan in or after 1974. Table F5 asks whether historical missionary exposure in the place of origin predicts this near-reform Taiwan presence. Origin in a prefecture with a mission by 1880 is associated with a 2.45 percentage-point increase, relative to a dependent-variable mean of 3.0 percent. Origin in a prefecture with a mission school by 1900 is associated with a 5.36 percentage-point increase. These estimates show that mission-exposed origins are overrepresented in the technical diaspora. They do not identify return to mainland China.

The second source follows Republican-era elites from their places of origin and foreign exposure to later institutional roles in mainland China. In separate period regressions, the interaction between mission-exposed origin and foreign exposure is 14.77 percentage points during the Mao period and 54.64 percentage points during the reform period (Table F6).

The implied association between mission-exposed origin and a mainland institutional role among foreign-exposed elites is 7.72 percentage points during Mao and 30.53 percentage points after reform. The reform-period estimate is therefore substantially larger, consistent with internationally experienced elites from mission-exposed places becoming more visible in mainland institutions after reopening. Because the biographies form a selected sample and the separate columns do not by themselves establish a causal difference across periods, we interpret this evidence cautiously.

Together, the two sources identify complementary margins of the people channel. Technical human capital from mission-exposed origins remained active outside mainland China, while foreign-exposed elites from those origins were more strongly represented in mainland institutions after reform. The evidence supports continuity and reactivation through people and their networks, without requiring uninterrupted residence or large-scale return to the prefecture of origin.

Foreign-facing commercial connections The second channel is the ability of local economies to reconnect to foreign markets and capital. Before 1949, the historical analysis shows that mission-exposed prefectures developed more modern banks, enterprises, and trade-oriented activity. Missionary organizations, Chinese converts, translators, and commercial intermediaries also linked local communities to foreign organizations, practices, and information. These links had little market return when private activity and international exchange were restricted. They could again reduce information and matching costs once China reopened to trade and foreign investment.

Foreign direct investment provides the sharpest test of this implication. Historical missionary exposure predicts higher foreign direct investment in 2017 in both OLS and IV (Table F9). The IV coefficient is 0.129 for log foreign direct investment and 0.060 for log foreign direct investment per capita; the corresponding first-stage F-statistic is 14.22. The per-capita result makes it less likely that the association simply reflects larger contemporary prefectures.

A complementary exercise adds historical channel variables to the 2022 bank regression one block at a time. Historical banks and enterprises absorb 61.1 percent of the baseline OLS coefficient, more than the education, publication, institutional, or trade-and-opening blocks (Table F8). This attenuation is descriptive: historical market development may itself be an outcome of missionary exposure and cannot be treated as a predetermined control. Taken together, the FDI and attenuation results are consistent with historically exposed places retaining a greater capacity to reconnect to foreign capital. They do not establish

that the same firms, investors, or bilateral relationships persisted continuously through the Mao period.

Alternative explanations We find less support for three alternative explanations. The first is direct persistence of mission schools and publications. The historical analysis shows that these organizations initially transmitted practical skills and useful knowledge, but this does not imply that the original facilities remained the long-run carriers. The school and publication interaction estimates for 2022 banks are positive but imprecise, and the corresponding interaction IVs are weak (Tables F14 and F15). IV estimates from the 2000 Census also show small and imprecise effects on educational attainment, while the Shaanxi education results are unstable across OLS and IV (Figures F12 and F13). We therefore do not find affirmative evidence for direct institutional persistence, although the available tests are not precise enough to rule it out. This does not contradict the historical mechanism: schools and publications may have formed people and networks that later persisted after the original organizations disappeared.

The second alternative is broad religious or cultural persistence. Historical missionary exposure predicts slightly higher Christian affiliation in some survey specifications and is associated with greater ownership of savings and financial assets. Measures of religious participation, the importance of religion, prayer, donations, trust, and broader values, however, do not move consistently (Tables F10–F13). We therefore find little evidence that the long-run bank result reflects a general Protestant ethic or a broad change in preferences.

The third alternative is differential state capacity or public infrastructure after 1949. We study this possibility using county-level data from Shaanxi provincial yearbooks, which report fiscal capacity, public investment, infrastructure, enterprises, and market activity during the Mao and early reform periods. The IV estimates for fiscal expenditure, fiscal revenue, fixed-asset investment, electricity, machinery, and health infrastructure are generally null or negative. Positive estimates are concentrated in industrial enterprises, retail sales, and township enterprise activity (Table F16). Because these results come from one province and cover many outcomes, they should not be read as a precise ranking of public and private capacity. They do, however, provide little support for a simple explanation based on greater state investment or physical infrastructure in historically mission-exposed places.

Taken together, the evidence points most strongly to two related carriers of the historical legacy: internationally oriented people and networks, and local connections to foreign-facing commercial activity. Schools and publications appear to have been important formation technologies before 1949, but the evidence does not suggest that the original facilities them-

selves drove the post-reform reactivation. Mao-era closure reduced the return to the skills and relationships formed earlier; reform made them useful again. The long-run pattern is therefore better understood as translation and local adaptation than as the transplantation and uninterrupted survival of foreign institutions.

7 Conclusion

This paper studies the historical and institutional roots of China’s market-based capitalism through the lens of financial development. Using newly compiled data on Christian missionary institutions, modern banks, enterprises, traditional financial institutions, and long-run banking outcomes, we show that missionary exposure helped China adopt modern banks before 1936 and shaped the response of local economies to the market-oriented reforms of the 1980s.

Our historical analysis exploits the staggered expansion of missionaries across Chinese prefectures between 1857 and 1936. We find that missionary presence increased modern banks by 31% and enterprises by 27%. These effects are not driven by pre-existing trends in banks, enterprises, railways, treaty ports, government investment, or traditional banks. The results are also robust to alternative transformations of the bank outcome, different bank types, balance-sheet measures, and a range of sample restrictions.

The evidence points to two main mechanisms. First, missionaries, especially Protestants, supplied practical education in subjects such as mathematics, bookkeeping, economics, science, and English. These skills were useful for operating modern financial institutions. Consistent with this channel, we find a high share of Christian-affiliated bankers, stronger effects in places with missionary schools, and larger effects for Protestant missions. Second, missionaries helped diffuse useful knowledge through publications and translations. The effects on modern banks are stronger in locations where missions engaged in publication activity, suggesting that printed materials and translations helped make modern economic and institutional practices more accessible.

We also document how missionary exposure interacted with China’s transition from traditional to modern finance. Modern banks expanded especially in places with pre-existing traditional financial activity, suggesting that they emerged where financial demand was already present. Over time, they substituted for *Piaohao*, whose remittance functions became less central after the fall of the Qing state. By contrast, *Qianzhuang* survived longer and appear more complementary to modern banks, likely because they retained local information and networks of trust that modern banks initially lacked.

The long-run analysis shows that these historical effects did not disappear permanently. Using an instrumental variable based on the interaction between railway expansion and national missionary inflows, we find that missionary exposure predicts modern banking development before 1949, weakens during the Maoist period, and re-emerges after the market reforms of the 1980s. We also find effects on per capita GDP after 1992, with a stronger increase after China joined the WTO. Suggestive mechanism evidence points most strongly to internationally exposed people and elite networks and to the persistence of commercial links with foreign capital; we find less support for direct persistence through mission schools, publications, or state infrastructure. These patterns are consistent with the idea that missionary exposure created locally embedded market capabilities that became valuable again once the policy environment allowed markets and private financial activity to expand.

Overall, the results suggest that missionaries transmitted more than human capital. They also helped diffuse institution-specific knowledge: the skills, practices, networks, and organizational forms that made modern banks usable in the Chinese context. In this sense, missionaries helped plant the seeds of market-based capitalism, not by replacing Chinese institutions, but by interacting with local conditions and helping new financial institutions take root. This helps explain why some places were better positioned to respond to China's reform-era opening and why the roots of China's modern market economy partly predate the reforms themselves.

References

- Acemoglu, Daron, Simon Johnson, and James A Robinson**, “The colonial origins of comparative development: An empirical investigation,” *American economic review*, 2001, *91* (5), 1369–1401.
- , – , and **James A. Robinson**, “Reversal of Fortune: Geography and Institutions in the Making of the Modern World Income Distribution,” *The Quarterly Journal of Economics*, 2002, *117* (4), 1231–1294.
- Bai, Limin**, *Fusion of East and West: Children, Education and a New China, 1902-1915*, Vol. 5, Brill, 2019.
- Bai, Ying and James Kai sing Kung**, “Diffusing knowledge while spreading God’s message: Protestantism and economic prosperity in China, 1840–1920,” *Journal of the European Economic Association*, 2015, *13* (4), 669–698.
- and **Ruixue Jia**, “Elite Recruitment and Political Stability: The Impact of the Abolition of China’s Civil Service Exam,” *Econometrica*, 2016, *84* (2), 677–733.
- Bank of China Economic Research Office**, *National Banking Yearbook*, Hanwen Regular Script Printing House, 1937.
- Bays, Daniel H**, *A new history of Christianity in China*, John Wiley & Sons, 2011.
- Becker, Sascha O. and Ludger Woessmann**, “Was Weber Wrong? A Human Capital Theory of Protestant Economic History,” *The Quarterly Journal of Economics*, 05 2009, *124* (2), 531–596.
- Becker, Sascha O, Steven Pfaff, and Jared Rubin**, “Causes and consequences of the Protestant Reformation,” *Explorations in Economic History*, 2016, *62*, 1–25.
- Bergeron, Augustin**, “Religion and the Scope of Morality: Evidence from Exposure to Missions in the DRC,” Technical Report, Technical report. <https://www.augustinbergeron.com/research> 2019.
- Borusyak, Kirill and Peter Hull**, “Nonrandom Exposure to Exogenous Shocks,” *Econometrica*, 2023, *91* (6), 2155–2185.
- Bryan, Gharad, James J Choi, and Dean Karlan**, “Randomizing Religion: the Impact of Protestant Evangelism on Economic Outcomes,” *The Quarterly Journal of Economics*, 06 2020, *136* (1), 293–380.

- Caicedo, Felipe Valencia**, “The Mission: Human Capital Transmission, Economic Persistence, and Culture in South America,” *The Quarterly Journal of Economics*, 2019, 134 (1), 507–556.
- Chen, Jiandong, Ming Gao, Shulei Cheng, Wenxuan Hou, Malin Song, Xin Liu, and Yu Liu**, “Global 1 km $\times$ 1 km gridded revised real gross domestic product and electricity consumption during 1992–2019 based on calibrated nighttime light data,” *Scientific Data*, 2022, 9 (1), 202.
- Chen, Ting, Chong Liu, Shuang Shi, and Yuling Zhu**, “The Intellectual Origin of Legal Institution and Modern Banking,” *Available at SSRN 5083231*, 2025.
- Chen, Yuyu, Hui Wang, and Se Yan**, “The long-term effects of Protestant activities in China,” *Available at SSRN 2186818*, 2014.
- Chen, Zhiwu, Chicheng Ma, and Andrew J Sinclair**, “Banking on the Confucian clan: why China developed financial markets so late,” *The Economic Journal*, 2022, 132 (644), 1378–1413.
- , **Xinhao Li, and Chicheng Ma**, “Christian Missionaries and International Trade, 1580–1936,” 2025. *Available at SSRN 5195597*.
- Cheng, Linsun**, *Banking in modern China: entrepreneurs, professional managers, and the development of Chinese banks, 1897-1937*, Cambridge University Press, 2003.
- CHGIS**, “CHGIS,” 2016.
- ching Liu, Kwang, John King Fairbank, and Denis Crispin Twitchett**, *The Cambridge History of China: Late Ch’ing, 1800-1911*, Cambridge University Press, 1980.
- Colonnelli, Emanuele and Mounu Prem**, “Corruption and firms,” *The Review of Economic Studies*, 2022, 89 (2), 695–732.
- Dan, Cui**, “The cultural contribution of British Protestant missionaries to China’s national development during the 1920s.” PhD dissertation, London School of Economics and Political Science 1996.
- de Roover, Raymond**, *Money, Banking and Credit in Mediaeval Bruges: Italian Merchant-Bankers, Lombards and Money-Changers: A Study in the Origins of Banking*, Cambridge, MA: Mediaeval Academy of America, 1948.

- Dell, Melissa and Benjamin A Olken**, “The development effects of the extractive colonial economy: The dutch cultivation system in java,” *The Review of Economic Studies*, 2020, 87 (1), 164–203.
- Dennis, James Shepard, Harlan Page Beach, Charles Harvey Fahs, and John George Bartholomew**, eds, *World Atlas of Christian Missions: Containing a Directory of Missionary Societies, a Classified Summary of Statistics, an Index of Mission Stations, and Maps Showing the Location of Mission Stations Throughout the World*, New York: Student Volunteer Movement for Foreign Missions, 1911.
- Dong, Baomin and Yu Zhang**, “Confucianism and science,” *Explorations in Economic History*, 2024, 91, 101517.
- Du, Xuncheng**, *Mingzu Ziben Zhuyi Yu Jiuzhongguo Zhengfu (1840-1937) (Domestic Capitalism and Old Chinese Government, 1840-1937)*, Shanghai Shehui Kexueyuan Chubanshe, 1991.
- , *Zhongguo de Minzu Ziben Zhuyi (1927-1937) (Domestic Capitalism, 1927-1937)*, Shanghai Caijing Daxue Chubanshe, 2019.
- Erten, Bilge and Jessica Leight**, “Exporting out of agriculture: The impact of WTO accession on structural transformation in China,” *Review of Economics and Statistics*, 2021, 103 (2), 364–380.
- Fairbank, John K and Kwang-Ching Liu**, *The Cambridge History of China: Late Ch'ing, 1800-1911, Part 2*, Vol. 2, Cambridge University Press, 1980.
- Gallego, Francisco A. and Robert D. Woodberry**, “Christian Missionaries and Education in Former African Colonies: How Competition Mattered,” *Journal of African Economies*, 2010, 19 (3), 294–329.
- Glaeser, Edward L., Rafael La Porta, Florencio Lopez de Silanes, and Andrei Shleifer**, “Do Institutions Cause Growth?,” *Journal of Economic Growth*, 2004, 9 (3), 271–303.
- Greif, Avner, Joel Mokyr, and Guido Tabellini**, *Two Paths to Prosperity: Culture and Institutions in Europe and China, 1000–2000*, Princeton, NJ: Princeton University Press, 2025.
- Gu, Weimin**, *Christianity and Modern Chinese Society*, Shanghai People’s Publishing House, 2010.

- Henrich, Joseph**, *The WEIRD people in the world: How the West became psychologically peculiar and particularly prosperous*, Penguin UK, 2020.
- Huang, Guangyu**, *The Universal Dictionary of Foreign Business in Modern China*, Sichuan: Sichuan People's Publishing House, 1995.
- Huang, Jianhui**, *Historical Materials of Shanxi Ticket Office (Updated Edition)*, Shanxi Economic Publishing House, 2002.
- Huang, Yujing, Chicheng Ma, and Felipe Valencia Caicedo**, "Religion and Conflict: Evidence from China, 1860–1911," 2024. Available at SSRN 4948527.
- Jedwab, Remi, Felix Meier zu Selhausen, and Alexander Moradi**, "The economics of missionary expansion: evidence from Africa and implications for development," *Journal of Economic Growth*, 2022, 27 (2), 149–192.
- Jiang, Jianqing**, *Jindai Zhongguo Yinhangye Jigou Renming Dacidian (Dictionary of Institutions and Peoples of Banking in Modern China)*, Shanghai Gugi Chubanshe, 2014.
- Kuran, Timur**, "The long divergence: How Islamic law held back the Middle East," 2012.
- , "Islam and economic performance: Historical and contemporary links," *Journal of Economic Literature*, 2018, 56 (4), 1292–1359.
- Levine, Ross, Chen Lin, Chicheng Ma, and Yuchen Xu**, "The legal origins of financial development: Evidence from the shanghai concessions," *The Journal of Finance*, 2023, 78 (6), 3423–3464.
- Liang, Jialin**, *Fortune in China: Ten Lectures on the History of the Church in Modern China (in Chinses)*, Tien Dao Bookstore, 1988.
- Lin, Chen, Chicheng Ma, Yuchen Sun, and Yuchen Xu**, "The telegraph and modern banking development, 1881–1936," *Journal of Financial Economics*, 2021, 141 (2), 730–749.
- , —, —, and —, "The allocation of talent and financial development, 1897 to 1936," *Management Science*, 2024.
- Lin, Justin Yifu**, "Rural reforms and agricultural growth in China," *The American economic review*, 1992, pp. 34–51.
- Liu, Jianmin**, *Collection of Historical Materials on Shanxi Businessman*, Shanghai: The Commercial Press, 2018.

- Lowes, Sara and Eduardo Montero**, “The legacy of colonial medicine in Central Africa,” *American Economic Review*, 2021, 111 (4), 1284–1314.
- Lu, Renlong**, “Searching for a Missing Figure in Modern Publishing History: Xia Ruifang, Founder of the Commercial Press,” Commercial Press official history; originally published in *Dushu* 2017. In Chinese.
- Ma, L, Y Lu, K Wang, and X Wang**, *Zhongguo Tielu Biannian Jianshi (A Briefly and Chronical History of Railroads in China)*, Zhongguo Tiedao Chubanshe, 1983.
- Ma, Ningning, Se Yan, and Yiling Zhao**, “Becoming ‘Co-Ed’: A Protestant Gift to China,” *Available at SSRN 4309533*, 2022.
- , —, and —, “The Mission to Educate Women: a Protestant legacy in China,” 2022.
- Mattingly, Daniel C. and Ting Chen**, “The Missionary Roots of Nationalism: Evidence from China,” *The Journal of Politics*, 2022, 84 (3), 1638–1651.
- Mayfield, Alex, Margaret Frei, Daryl Ireland, and Eugenio Menegon**, “The China Historical Christian Database: A Dataset Quantifying Christianity in China from 1550 to 1950,” *Data*, 2024, 9 (6), 76.
- McIntosh, Gilbert**, *The Mission Press in China: Being a Jubilee Retrospect of the American Presbyterian Mission Press*, Shanghai: American Presbyterian Mission Press, 1895.
- Ministry of Agriculture and Commerce**, *Agricultural and Commercial Statistical Tables of the Republic of China*, Ministry of Agriculture and Commerce, Republic of China, 1912–1924.
- Ministry of Communications**, *China Postal and Local Products Gazetteer*, Shanghai: The Commercial Press, 1937.
- Moazzin, Ghassan**, *Foreign banks and global finance in modern China: Banking on the Chinese frontier, 1870-1919*, Cambridge University Press, 2022.
- Mungello, David Emil**, “Reinterpreting the history of Christianity in China,” *The Historical Journal*, 2012, 55 (2), 533–552.
- North, Douglass C.**, *Institutions, Institutional Change and Economic Performance*, Cambridge University Press, 1990.
- Nunn, Nathan**, “The long-term effects of Africa’s slave trades,” *The Quarterly journal of economics*, 2008, 123 (1), 139–176.

- , “Religious conversion in colonial Africa,” *American Economic Review*, 2010, *100* (2), 147–152.
- , **Emmanuel Akyeampong, Robert Bates, and James A Robinson**, “Gender and missionary influence in colonial Africa,” *African development in historical perspective*, 2014.
- Pascali, Luigi**, “Banks and development: Jewish communities in the Italian Renaissance and current economic performance,” *Review of Economics and Statistics*, 2016, *98* (1), 140–158.
- Pelzer, Thorben et al.**, “Chinese Engineers Relational Database,” Zenodo dataset 2025.
- Rajan, Raghuram G. and Luigi Zingales**, “Financial Dependence and Growth,” *American Economic Review*, 1998, *88* (3), 559–586.
- Ren, Tingxu**, *Tribulations of the Church in China, A.D. 1900 Natives and Foreigners*, Guang xue hui [kan], Mei hua shu guan bai yin, 1903.
- Roth, Jonathan, Pedro HC Sant’Anna, Alyssa Bilinski, and John Poe**, “What’s trending in difference-in-differences? A synthesis of the recent econometrics literature,” *Journal of Econometrics*, 2023, *235* (2), 2218–2244.
- Rubin, Jared**, “Institutions, the rise of commerce and the persistence of laws: Interest restrictions in Islam and Christianity,” *The Economic Journal*, 2011, *121* (557), 1310–1339.
- Sant’Anna, Pedro HC and Jun Zhao**, “Doubly robust difference-in-differences estimators,” *Journal of econometrics*, 2020, *219* (1), 101–122.
- Schulz, Jonathan F., Duman Bahrami-Rad, Jonathan P. Beauchamp, and Joseph Henrich**, “The Church, intensive kinship, and global psychological variation,” *Science*, 2019, *366* (6466), eaau5141.
- Sequeira, Sandra, Nathan Nunn, and Nancy Qian**, “Immigrants and the Making of America,” *The Review of Economic Studies*, 2020, *87* (1), 382–419.
- Shaanxi Provincial Bureau of Statistics**, *Historical Statistical Data Compilation for the Prefectures, Cities, and Counties of Shaanxi Province, 1949–1990*, Beijing: China Statistics Press, 1991. In Chinese.
- Shelley, Bruce**, *History of the Christian Church*, Peking University Press, 2004.

- Song, Zheng, Kjetil Storesletten, and Fabrizio Zilibotti**, “Growing like china,” *American economic review*, 2011, 101 (1), 196–233.
- Spence, Jonathan D.**, *The Search for Modern China*, New York: W. W. Norton & Company, 1990.
- Spufford, Peter**, *Money and Its Use in Medieval Europe*, Cambridge: Cambridge University Press, 1988.
- Stauffer, Milton Theobald**, *The Christian occupation of China: a general survey of the numerical strength and geographical distribution of the Christian forces in China made by the Special Comm. on Survey and Occupation, China Continuation Comm., 1918-1921*, China continuation committee, 1922.
- Sun, Liyang and Sarah Abraham**, “Estimating dynamic treatment effects in event studies with heterogeneous treatment effects,” *Journal of econometrics*, 2021, 225 (2), 175–199.
- Sun, Yuchen, Wanda Wang, and Yuchen Xu**, “Bonds of Love: Patriotism and the Rise of Modern Banks,” 2024.
- Wang, Chengcheng, Jiachen Li, Riccardo Di Cato, and Yuan Xu**, “Returns to Overseas Education among Chinese Republican-Era Elites,” 2026. Unpublished manuscript.
- Wang, Lixin**, *American Missionaries and Modernization of China in the Late Qing Dynasty*, Tianjin People’s Publishing House, 1997.
- Wantchekon, Leonard, Marko Klašnja, and Natalija Novta**, “Education and human capital externalities: evidence from colonial Benin,” *The Quarterly Journal of Economics*, 2015, 130 (2), 703–757.
- Weber, Max**, *The Protestant Ethic and the Spirit of Capitalism*, London, UK: Routledge Classics, 2001. Originally published in German as *Die protestantische Ethik und der ‘Geist’ des Kapitalismus*, *Archiv für Sozialwissenschaft und Sozialpolitik*, vol. 20 (1904/1905), pp. 1–54 and vol. 21, pp. 1–110; reprinted in *Gesammelte Aufsätze zur Religionssoziologie*, pp. 17–206, 1920.
- Woodberry, Robert D.**, “The missionary roots of liberal democracy,” *American political science review*, 2012, 106 (2), 244–274.
- World Bank**, “The World Bank In China,” 2024. Accessed: 27-Feb-2025.

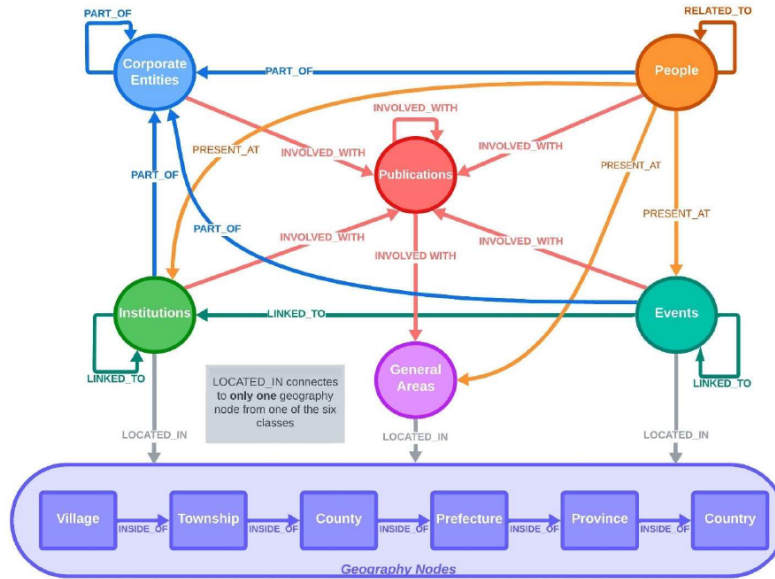
- Xie, Yu and Jingwei Hu**, “An Introduction to the China Family Panel Studies (CFPS),” *Chinese Sociological Review*, 2014, 47 (1), 3–29.
- Xu, Huayu**, “The long-term health and economic consequences of improved property rights,” *Journal of Public Economics*, 2021, 201, 104492.
- Xu, Youchun**, *Minguo Renwu Daxidian [Encyclopedia of Republican-Era Personalities]*, revised edition ed., Shijiazhuang: Hebei People’s Publishing House, 2007. In Chinese.
- Zhao, Jinsong, Peilin He, and Chin-Hsien Yu**, “Transportation Revolution and Banking Development: Evidence from Prewar China,” 2023.
- Zhao, Yiling and Ningning Ma**, “Catholicism in early 20th century China: A re-examination,” *Available at SSRN 5211557*, 2025.

Appendix

Appendix A: Diffusion of Missionary Institutions, Banks, and Enterprises

Missionary Data Structure and Institution Types

Figure A1: Data Structure of CHCD



Notes: The CHCD includes nodes for people, corporate entities, institutions, publications, general areas, and events, along with six types of geographic nodes. This schema illustrates possible relationships among these nodes. Person nodes represent human beings. Corporate-entity nodes represent organizations that do not have a direct geographic footprint. Institution nodes include churches, hospitals, and schools. Publication nodes represent textual documents.

Table A1: Category of Religious Institutions

Institution Category	Freq.	Percent
Ecclesial	4,880	71.18%
Education	1,353	19.73%
Medical	517	7.54%
Other	52	0.76%
Rescue	23	0.34%
Press	16	0.23%
Business	12	0.18%
Scientific	3	0.04%
Total	6,856	100.00%

Notes: The table counts each baseline CHCD v3.0.0 institution once after matching to 1911 prefectures and adds 971 new directory-based CHCD records used in the supplementary descriptive measures, for 6,856 institutions in total. Press coverage combines 16 CHCD press institutions with 677 located CHCD publication records and 16 unique print entities in the reviewed McIntosh data, including 7 mission presses across 10 prefectures. These source counts are complementary and are not deduplicated one-for-one.

Table A2: CMS/Stauffer and CHCD Missionary Coverage, 1920

	Count in 1920	Prefectures with presence	Correlation with CMS/Stauffer count
CMS/Stauffer: Protestant missionaries (1920)	10,715	227	–
CHCD institutions: Protestant	1,860	211	0.737
CHCD institutions: any denomination	1,958	212	0.705
CHCD people: Protestant	3,715	183	0.623
CHCD people: any denomination	4,147	184	0.581

Notes: The comparison is at the 1911 prefecture level and uses the 245 prefectures present in both the CMS/Stauffer source and the main CHCD panel. CMS/Stauffer reports Protestant missionary persons in 1920 only; accordingly, the CHCD “any denomination” rows are broader CHCD measures correlated with the Protestant CMS/Stauffer count, not an all-denomination external benchmark. CHCD institution and person rows count stocks active in 1920. The correlation column reports Pearson correlations between the CMS/Stauffer Protestant missionary count and the corresponding CHCD prefecture-level count. The CMS/Stauffer headcount may be somewhat understated or imprecise as a 1920 snapshot because the survey identifies missionaries who were on the ground without always recording a fully precise reference date. For our purposes, the more important comparison is geographic coverage, which is very similar across the sources: CMS/Stauffer records presence in 227 prefectures, compared with 211–212 for CHCD institutions and 183–184 for CHCD people. The presence column counts prefectures with any recorded measure, using zeros for prefectures without a corresponding CHCD record. Supplementary directory records are not added because their denominational tradition is not coded.

Summary Statistics and Historical Diffusion

Table A3: Summary Statistics: Historical DiD Outcomes

Variable	Obs.	Mean	SD	P5	Median	P95
<i>Panel A. Headline outcomes</i>						
Modern banks (stock)	19,760	0.76	6.16	0.00	0.00	3.00
New modern banks	19,760	0.08	0.67	0.00	0.00	0.00
Industrial enterprises (stock)	15,314	4.78	40.10	0.00	0.00	17.00
<i>Panel B. Ownership-specific outcomes</i>						
Private modern banks (stock)	19,760	0.33	4.14	0.00	0.00	0.00
Government modern banks (stock)	19,760	0.29	1.35	0.00	0.00	2.00
Foreign modern banks (stock)	19,760	0.10	1.16	0.00	0.00	0.00
Foreign enterprises (stock)	15,314	0.05	0.48	0.00	0.00	0.00
Government enterprises (stock)	15,314	0.33	1.40	0.00	0.00	2.00
<i>Panel C. Other historical outcomes and infrastructure</i>						
Trade-oriented enterprises (stock)	15,314	0.87	8.33	0.00	0.00	3.00
Domestic-oriented enterprises (stock)	15,314	0.48	5.11	0.00	0.00	1.00
<i>Piaohao</i> branches (stock)	18,320	0.91	3.22	0.00	0.00	7.00
Railway lines (stock)	19,760	0.27	1.34	0.00	0.00	2.00
Government industrial investment (thousands)	18,880	21.65	249.77	0.00	0.00	0.07

Notes: The unit is a 1911-boundary prefecture-year. Statistics use the exact historical panel behind the headline DiD results. The modern-bank sample covers 1857–1936 (247 prefectures and 19,760 prefecture-years). Enterprise variables are observed from 1875–1936, so their maximum sample is 15,314 prefecture-years. Stocks are cumulative within prefecture. Government industrial investment is shown in thousands of nominal historical currency units. Observation counts vary only when the underlying source does not cover the full analysis window.

Table A4: Summary Statistics: Missionary Treatment, Mechanisms, and Controls

Variable	Obs.	Mean	SD	P5	Median	P95
<i>Panel A. Prefecture-year treatment and mechanism measures</i>						
Missionary institutions (stock)	19,760	3.58	8.34	0.00	0.00	17.00
Mission presence	19,760	0.46	0.50	0.00	0.00	1.00
Missionary people	18,800	5.98	24.12	0.00	0.00	25.00
Mission education institutions (stock)	19,760	0.28	1.47	0.00	0.00	2.00
Mission schools (stock)	19,760	0.23	1.26	0.00	0.00	1.00
Mission higher-education institutions (stock)	19,760	0.03	0.22	0.00	0.00	0.00
Mission publication activity	19,760	0.04	0.20	0.00	0.00	0.00
Protestant mission institutions (stock)	19,760	3.21	7.88	0.00	0.00	15.00
Catholic mission institutions (stock)	19,553	0.37	2.23	0.00	0.00	1.00
Christianity-related bankers / all bankers	19,760	0.030	0.100	0.000	0.000	0.237
Church-school-educated bankers (stock)	19,760	4.30	19.26	0.00	0.00	25.00
<i>Panel B. Prefecture-level treatment and heterogeneity measures</i>						
Ever received a mission by 1936	247	0.94	0.24	0.00	1.00	1.00
Year of first mission (treated prefectures)	232	1,898	17	1,857	1,902	1,926
Ever had mission education	247	0.54	0.50	0.00	1.00	1.00
Ever had mission publication activity	247	0.22	0.42	0.00	0.00	1.00
<i>Panel C. Predetermined controls used in balance tests</i>						
Population, 1880 (thousands)	247	1,408.31	1,273.26	143.90	1,018.48	4,008.94
Provincial capital, 1880	231	0.16	0.36	0.00	0.00	1.00
Main river	247	0.63	0.48	0.00	1.00	1.00

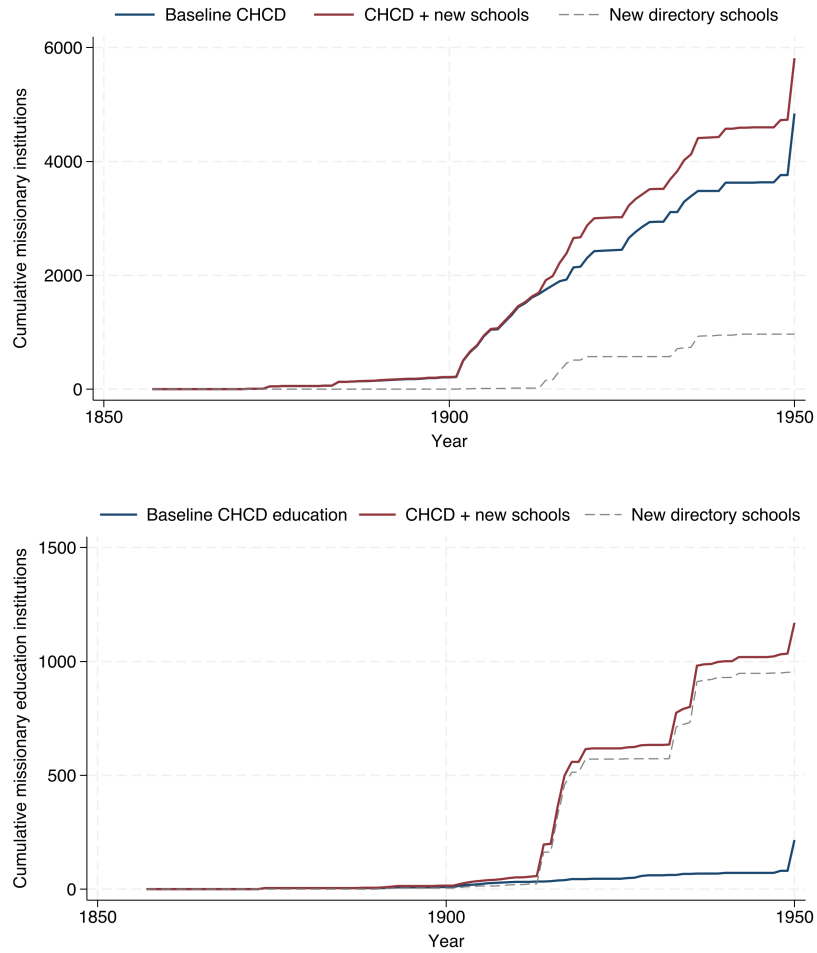
Notes: Panel A uses the exact 1857–1936 prefecture-year panel behind the headline modern-bank DiD. CHCD institution stocks define treatment timing; education adds unmatched directory schools, publication activity defines the school-by-publication split, and the banker share is set to zero when no banker is recorded, matching the corresponding figures. Panels B and C report one observation per 1911-boundary prefecture. First-treatment year is reported only for treated prefectures. Population is in thousands; provincial-capital status is measured in 1880 and available for 231 prefectures; main-river access is time invariant.

Table A5: Summary Statistics: Main Long-Run IV Samples

Variable	Obs.	Mean	SD	P5	Median	P95
<i>Panel A. Historical-bank IV cross-section (1936)</i>						
Missionary institutions, 1850–1936	247	10.83	10.75	0.00	8.00	32.00
Predicted missionary exposure	247	0.547	1.348	0.000	0.000	3.740
Railway connection duration (years)	247	24.09	39.54	0.00	0.00	97.00
Cumulative railway density	247	0.0018	0.0053	0.0000	0.0000	0.0123
Modern banks, 1936	247	5.85	13.59	0.00	1.00	26.00
Standardized modern banks, 1936	247	-0.000	1.002	-0.431	-0.358	1.486
<i>Panel B. Long-run bank and GDP IV cross-section</i>						
Missionary institutions, 1850–1936	231	11.37	10.88	0.00	8.00	34.00
Predicted missionary exposure	231	0.585	1.387	0.000	0.000	4.147
Railway connection duration (years)	231	25.76	40.36	0.00	0.00	98.00
Cumulative railway density	231	0.0020	0.0055	0.0000	0.0000	0.0131
Total banks, 2020	231	4,657.19	2,371.87	1,348.00	4,192.00	9,059.00
Standardized total banks, 2020	231	0.360	0.976	-1.001	0.169	2.171
Real GDP, 2019 (millions of 2017 USD)	231	267,699	172,929	66,809	231,541	535,774

Notes: The unit is a 1911-boundary prefecture. Each panel is restricted to the complete-case sample used by the corresponding headline IV regressions, including province fixed effects, missionary exposure, predicted missionary exposure, railway duration, and railway density. Panel A contains 247 prefectures. Panel B contains 231 prefectures and matches the 2020 bank and 2019 GDP headline cross-sections stored in the main IV analysis dataset. Cumulative railway density is cumulative railway stock divided by prefecture area.

Figure A2: Evolution of Missionary Institutions Over Time

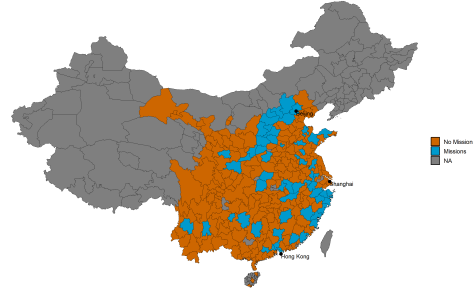
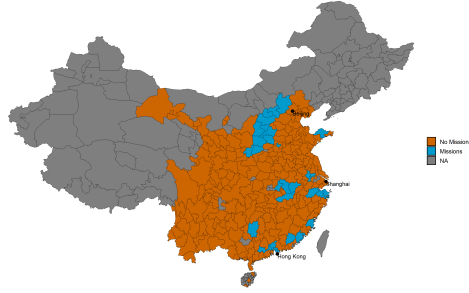


Notes: The first panel plots cumulative missionary institutions. The second panel plots cumulative missionary education institutions. Baseline counts use the CHCD v3.0.0 institution panel; the added series includes new directory-based school records not duplicated in the baseline panel.

Figure A3: Expansion of Missionaries

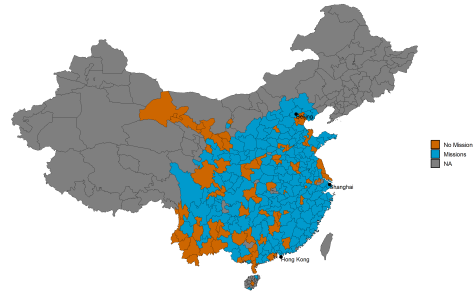
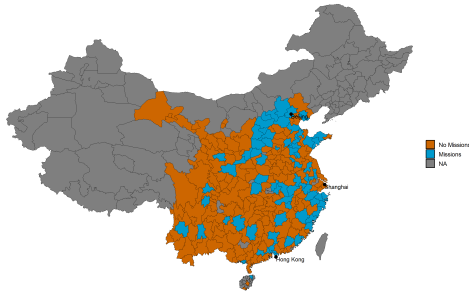
(a) 1875

(b) 1885



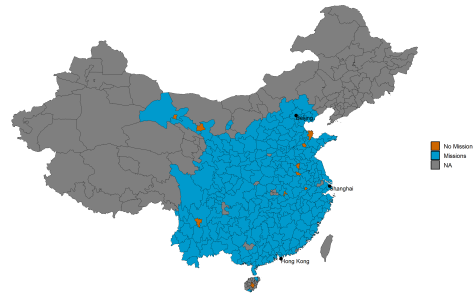
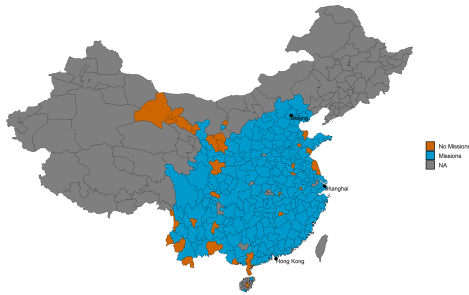
(c) 1900

(d) 1905



(e) 1926

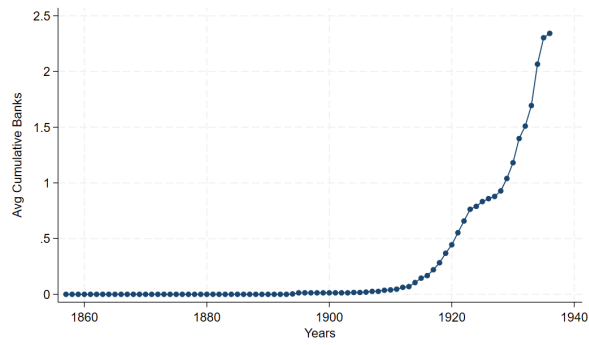
(f) 1950



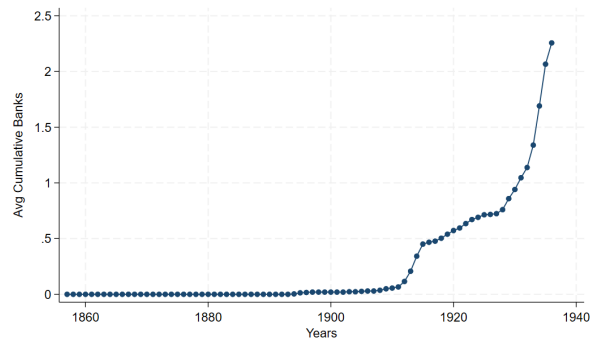
Notes: These maps show the expansion of missionaries over time for selected periods. Prefectures with missionary presence are shown in blue, those without missionaries in orange, and prefectures not included in the main analysis are in gray. Panel A shows prefectures with missionaries by 1875, Panel B by 1885, Panel C by 1900, Panel D by 1905, and subsequent panels continue chronologically. Data on missions are from the CHCD, and prefecture boundaries are from CHGIS 1911.

Figure A4: Evolution of Different Modern Banks Over Time

(a) Private Modern Banks



(b) Government Modern Banks



(c) Foreign Modern Banks

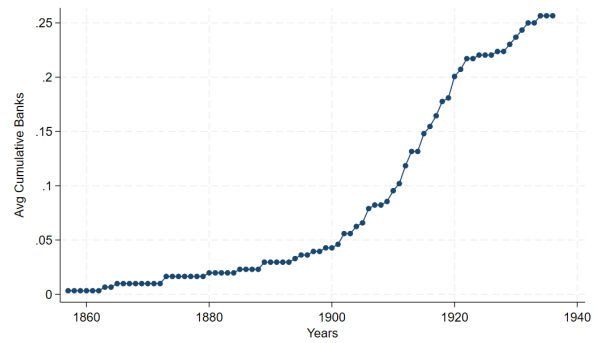
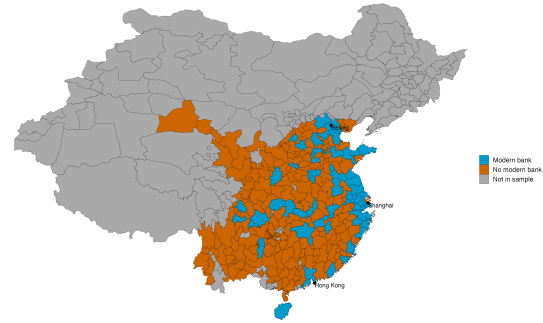
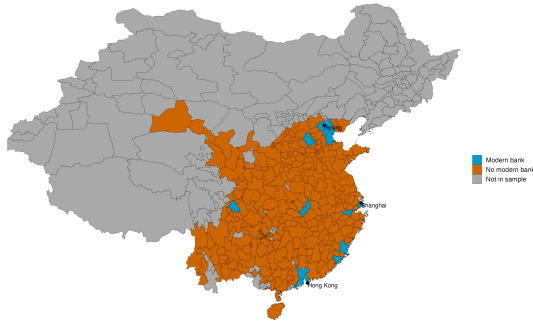


Figure A5: Expansion of Modern Banks

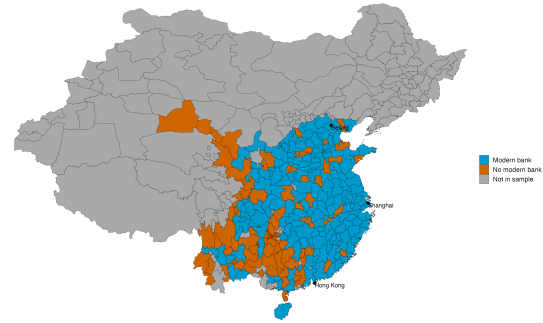
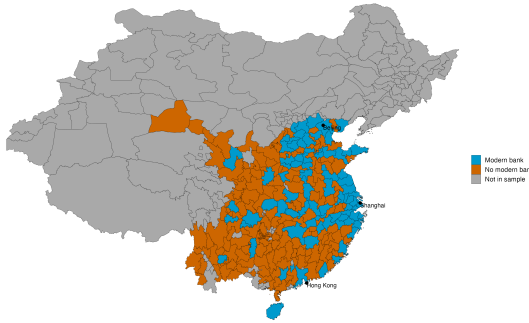
(a) By 1905

(b) By 1915



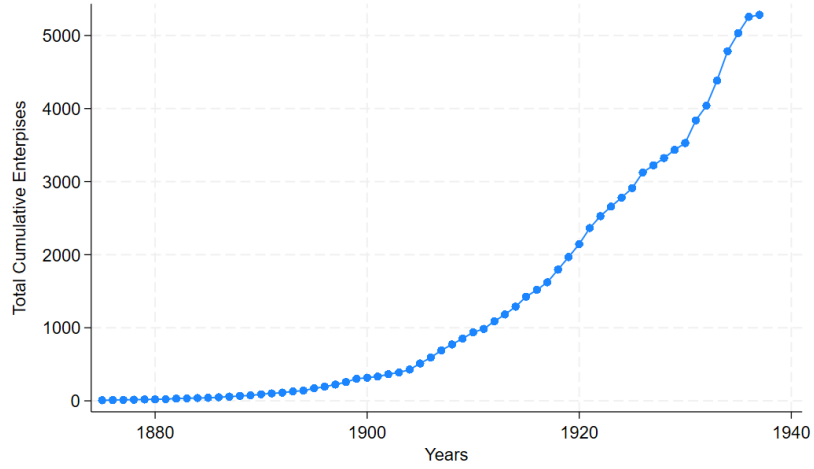
(c) By 1926

(d) By 1936



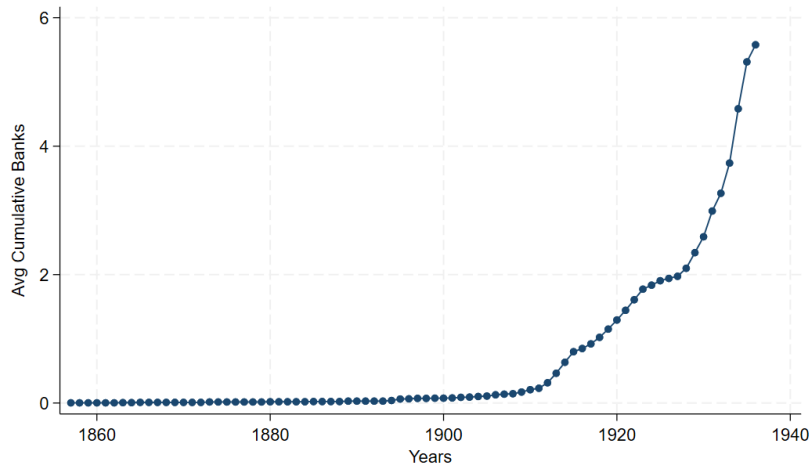
Notes: These maps show the expansion of modern banks in the current prefecture-year panel. Prefectures with at least one modern bank are shown in blue, sample prefectures without modern banks are shown in orange, and prefectures outside the analysis sample are shown in gray. Data on modern banks are from the Bank of China Economic Research Office, the *National Banking Yearbook*, and [Huang \(1995\)](#). Prefecture boundaries are from CHGIS 1911.

Figure A6: Evolution of Enterprises Over Time



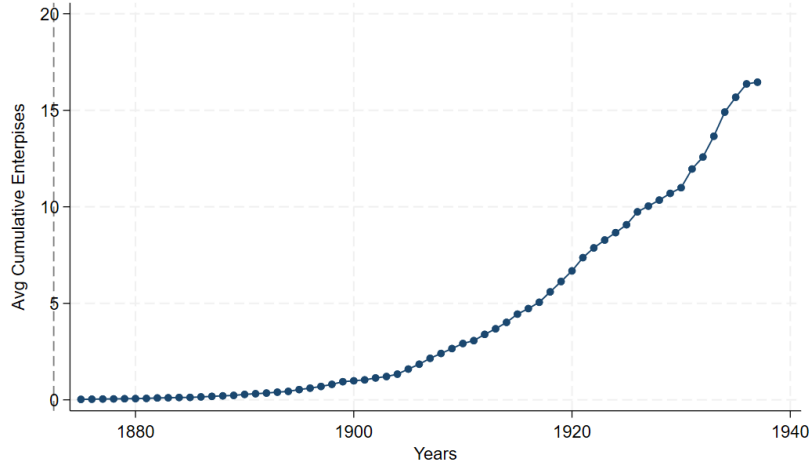
Notes: The figure plots enterprises recorded in [Du \(1991, 2019\)](#). These sources cover domestic-capital and old Chinese government enterprises between 1840 and 1937.

Figure A7: Evolution of Average Total Modern Banks Over Time



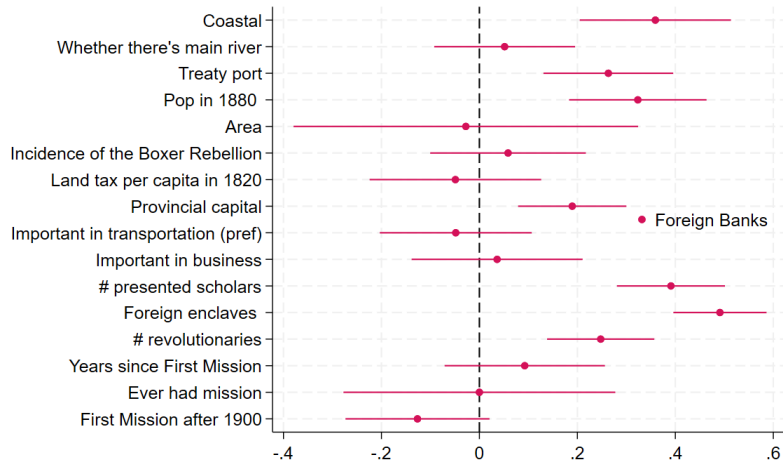
Notes: The figure plots average total modern banks over time. Sources: Bank of China Economic Research Office and the *National Banking Yearbook*.

Figure A8: Evolution of Average Enterprises Over Time



Notes: The figure plots average enterprises over time using records from [Du \(1991, 2019\)](#). These sources cover domestic-capital and old Chinese government enterprises between 1840 and 1937.

Figure A9: Predictors of Receiving Foreign Modern Banks First



Notes: The figure reports separate one-by-one regressions for 253 observations. All variables are standardized, and bars show 95% confidence intervals.

Table A6: Three Types of Financial Institutions

Feature	Piaohao	Qianzhuang	Bank
Main Function	Long-distance remittances	Short-term local lending	Deposit and lending
Clients	Governments, Large merchants	Small businesses, artisans	All economic agents
Branch Networks	Many branches	No branches, locally operated	Many branches
Lending Focus	Limited lending, high-trust clients	Small, short-term loans	Large mortgage loan

Notes: The table compares the differences among the three main types of financial institutions in China between 1840 and 1936, focusing on their primary functions, clientele, branch networks, and lending focus. Notably, only modern banks operated as contemporary enterprises with deposit and loan services as their core business and adopted a limited liability organizational structure.

Appendix B: Data Appendix

Historical Biographies

Banker biographies. We created a person-level dataset from *Jindai Zhongguo Yinhangye Jigou Renming Dacidian* (Jiang, 2014), a biographical dictionary of people in modern Chinese banking. The file contains 1,041 bankers who held positions such as director, founder, sponsor, or manager at Chinese-funded banks. We transcribed narrative entries and coded names, life dates, native place, education, Christian-school or other Christian connections, comprador experience, banking positions, and work across Chinese and foreign financial institutions. These fields make it possible to examine not only where banks appeared, but also the human-capital and intermediary networks of the people who operated them. Figure B2 illustrates the underlying narrative-to-field coding.

The dictionary is a selected biographical source rather than a census of all bank employees. It is most informative about people in senior or otherwise documented positions, and missing information in a biography is not interpreted as evidence that an attribute was absent. We therefore use these measures descriptively and for mechanism evidence, rather than treating them as a representative individual-level sample.

Republican-era elite database. The Republican China Elite Database was assembled by Wang et al. (2026) from the revised edition of Xu Youchun’s *Minguo Renwu Dacidian* (Xu, 2007). The source volume covers prominent individuals active between the founding of the Chinese Revolutionary Alliance in August 1905 and September 30, 1949, although individual biographies can extend beyond those dates. Its inclusion criteria span senior political and military figures; prominent businesspeople, bankers, and economists; scientists, engineers, and medical professionals; leading educators, writers, artists, journalists, and lawyers; and prominent religious figures. The resulting individual-level file contains 16,445 biographies and should therefore be interpreted as a selected record of prominent people, not a representative sample of the Chinese population.

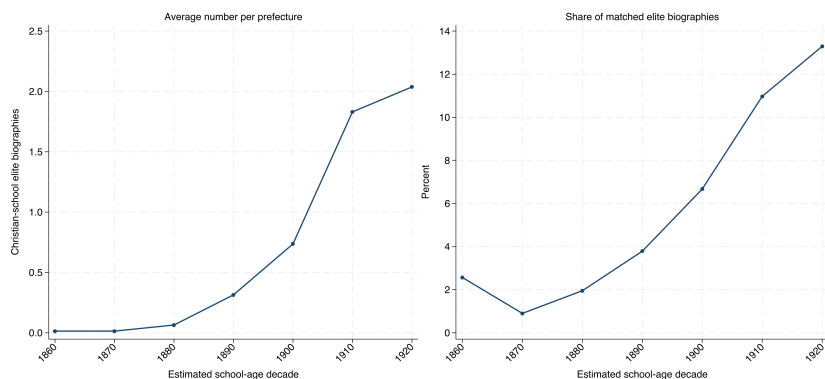
The source PDF was converted to machine-readable text using optical character recognition, manually cleaned and structured, and then parsed with DeepSeek-R1 into person-level fields. The extracted information includes native and childhood place, birth and death years, education and discipline, foreign study and destination, career history and primary position, regime affiliation, publications, and other biographical markers. We harmonize native-place strings to Qing-era province names and 1911 prefecture boundaries. Of the 16,445 biographies, 14,763 match to one of 236 prefectures in the historical mission panel; 14,113 of those

also report a birth year.

For the education proxy, we search the raw education narrative for a dictionary of major Christian institutions and church-related terms. The dictionary includes, among others, St. John's, Ginling, Yenching, Aurora, Fu Jen, West China Union, Lingnan, Cheeloo, Soochow, Hangchow, and the University of Shanghai, as well as generic references to church or Christian schooling. This procedure identifies 1,282 Christian-school-background biographies among the 14,763 geographically matched elites. It measures an education-background connection to Christian institutions, not the individual's religious belief. Because the biographies usually omit the exact date and place of attendance, the event-study proxy assigns each flagged person to the native prefecture at birth year plus 15. This construction assumes that native place is informative about school-age location and uses age 15 as a common marker for possible secondary-school exposure; mobility or attendance at a school elsewhere will introduce measurement error.

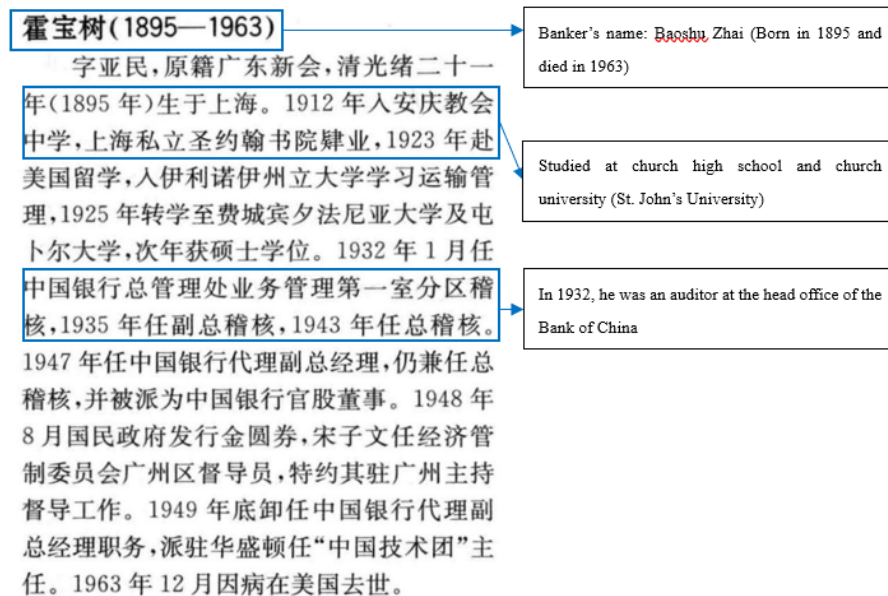
The foreign-study indicator comes from the person-level biographical coding. The associated year field is free text and sometimes contains ranges or multiple dates, so we extract the first four-digit year between 1800 and 1999 as the departure-year proxy. The event-study outcome then cumulates these departures within the native prefecture.

Figure B1: Christian-School Background Among Republican-Era Elites by School-Age Cohort



Notes: The sample contains biographies matched to 1911 prefectures with a non-missing birth year. School-age year is defined as birth year plus 15. Panel A reports, for each full decade from 1860 through 1920, the number of biographies coded with a Christian-school background divided by all 236 matched prefectures, including prefectures with zero such biographies. Panel B reports these biographies as a percentage of all matched elite biographies in the same school-age decade. The Christian-school flag is based on the education-field dictionary described in the text. The figure is descriptive: changes can reflect cohort composition and source coverage as well as changes in education, and the selected biographies are not representative of the population.

Figure B2: Illustration of Banker-Biography Digitization



Notes: This image illustrates how the banker-biography data were digitized: passages from a source biography are linked to coded fields such as name, education, and banking positions. It is an example of the transcription and coding workflow, not a complete observation or a separate outcome measure.

Historical Financial Records

Modern-bank balance sheets. We digitized two sources that differ in period, unit of observation, and accounting detail. Republican Ministry of Agriculture and Commerce tables cover the 1910s and list head offices and branches separately ([Ministry of Agriculture and Commerce, 1912–1924](#)). We transcribed the reported accounts, geocoded each office, assigned it to a 1911 prefecture, and aggregated branch-level deposits, capital, paid-in capital, banknote issuance, and surplus reserves. Coverage is not complete in every year, so these records are used primarily as supporting evidence rather than as a balanced panel.

The *National Banking Yearbook* provides bank-level balance sheets for 1932–1936 ([Bank of China Economic Research Office, 1937](#)). We digitized deposits, loans, assets, income, net income, cash, securities, paid-in capital, surplus reserves, and related accounting ratios. Reporting varies across banks and items, and the source is not a complete opening-and-closure panel. Specifications using these accounts therefore report the available outcome-specific sample and do not treat unreported values as zeros.

***Piaohao* and *Qianzhuang*.** The *Piaohao* data combine the branch histories assembled by [Huang \(2002\)](#) and [Liu \(2018\)](#). The underlying materials draw on museum holdings, archival chamber-of-commerce files, recovered historical records, founders’ descendants, private collections, and original Shanxi merchant documents.⁴¹ We retain the institution name, founder-origin affiliation, location, head-office or branch status, and opening, suspension, and closure year-month. Because branches have their own dated entry and exit histories, these data permit the stock, openings, and closures of *Piaohao* to be reconstructed with substantially less survivor bias than a single directory snapshot.

We separately transcribed and geocoded *Qianzhuang* entries from the *National Banking Yearbook*. The source records establishment dates, addresses, and the names and positions of core staff. The empirical series covers eight major provinces rather than all of China and lacks the complete branch-level closure histories available for *Piaohao*. We therefore use it to study financial interaction within its observed geography, not to infer a national census of local money houses.

⁴¹The sources described by [Huang \(2002\)](#) include materials from the Chinese Piaohao Museum in Pingyao, files of the Fengtian Chamber of Commerce in the Liaoning Provincial Archives, historical materials recovered from Japan, and collections held by founders’ descendants and private collectors. [Liu \(2018\)](#) collates original Shanxi merchant documents from multiple collections.

Other Historical Outcomes and Controls

Enterprises and trade. The domestic-enterprise data record head offices of registered Chinese enterprises from [Du \(1991, 2019\)](#). The source applies capitalization thresholds: more than 10,000 silver dollars for shipping, industrial, and mining enterprises and more than 50,000 silver dollars for financial enterprises. It therefore captures relatively formal and capitalized firms while underrepresenting smaller businesses. We geocode head offices to 1911 prefectures and complement the source with foreign-enterprise directories compiled by [Huang \(1995\)](#). In the combined historical enterprise file, approximately 70% of establishments are private Chinese enterprises, 5% are government enterprises, and 24% are foreign enterprises.

The trade data come from the 1934–1936 postal census compiled in [Ministry of Communications \(1937\)](#). The Republican government entrusted local postal offices with collecting information at the county and township levels on product types, production volumes, prices, and sales destinations. We digitize domestic and international export values and the number and composition of commodities traded, then aggregate the underlying local records to 1911 prefectures. These outcomes measure recorded trade flows and product coverage, not all informal exchange.

Railways, geography, and clans. We reconstruct railway access from station locations and opening dates in [Ma et al. \(1983\)](#). Stations are geocoded and assigned to 1911 prefectures, producing time-varying indicators, railway duration, counts, and exposure measures. Baseline geographic and economic controls from [Bai and Jia \(2016\)](#) include historical population, coastal and major-river access, rice suitability, provincial-capital status, and treaty-port status. Selected balance and mechanism exercises add genealogy-book counts from [Dong and Zhang \(2024\)](#) as a proxy for clan intensity. These variables are treated as predetermined controls; potentially endogenous historical market outcomes are introduced only in explicitly labeled mechanism or robustness exercises.

Long-Run National Outcomes and Other Measures

Financial institutions, GDP, and population. For the post-1949 bank series, we standardize the administrative names in the government license registry, assign each modern county or district to the 1911 prefecture containing its centroid, and aggregate institution openings and closures to historical-prefecture years. This preserves a consistent historical geography as modern administrative boundaries change. As described in the main data section, the registry covers a broader modern regulatory category than the Republican-era

modern-bank records, so levels across the two sources are not interpreted as directly comparable.

The GDP data of [Chen et al. \(2022a\)](#) provide 1-km gridded estimates for 1992–2019 based on nighttime lights. We intersect and sum the grid cells directly within accepted CHGIS 1911 prefecture polygons, avoiding an intermediate modern-prefecture crosswalk and the measurement error it would introduce. Population controls and denominators combine historical population with county-level totals constructed from the National Bureau of Statistics’ 2000 Population Census and 2015 one-percent population sample survey. Modern counties are assigned to 1911 prefectures by centroid so the population scale matches the bank geography.

Industrial enterprises and foreign investment. The Annual Survey of Industrial Production is a survey of industrial enterprises above the applicable reporting threshold, not the universe of registered firms. We use the number of surveyed enterprises, rather than their sales, employment, output, or borrowing, and distinguish private firms using the paid-in-capital ownership classification. Geocoded firm locations are assigned directly to accepted 1911 prefecture polygons; records without a usable direct match use a modern county or district centroid, followed by an area-weighted city crosswalk as a final fallback. The private-enterprise series omits 2009 because the ownership field mechanically assigns all observations to the same category in that year; the all-enterprise count remains usable. Foreign direct investment in 2017 comes from the *China City Statistical Yearbook* and is mapped to the same historical prefecture geography.

Biographies and household surveys. The Chinese Engineers Relational Database structures education, employment, and location histories for engineers in Republican China (1912–1949) and early postwar Taiwan (1945–1974) ([Pelzer et al., 2025](#)). We use it to study geographic origins, foreign education, and later location while retaining its source-defined coverage of documented engineers. The 2000 Population Census provides educational attainment. The China Family Panel Studies provide panel measures of religious affiliation, values, and household financial behavior ([Xie and Hu, 2014](#)), while the 2007 [Spiritual Life Study of Chinese Residents](#) provides complementary cross-sectional measures of religiosity and beliefs. Survey locations are linked to 1911 counties or prefectures at the finest disclosed geography; the resulting regressions retain the survey-specific sampling and geographic limitations described with the mechanism tables.

Shaanxi Provincial Data

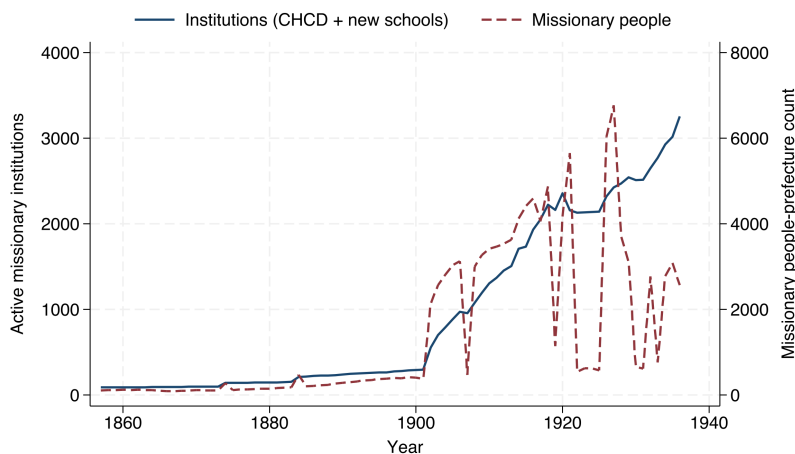
The county panel is transcribed from the Shaanxi Provincial Bureau of Statistics' *Historical Statistical Data Compilation for the Prefectures, Cities, and Counties of Shaanxi Province, 1949–1990* (Shaanxi Provincial Bureau of Statistics, 1991). It reports benchmark years before reform—1949, 1952, 1957, 1962, 1965, 1970, 1975, and 1978—and annual observations from 1978 through 1990. We use its county-level measures of population, students, fiscal revenue and expenditure, fixed-asset investment, electricity, health infrastructure, industrial and township enterprises, retail sales, savings, and production.

We exclude provincial, prefectural, and broad city aggregates, harmonize the remaining county, district, and county-level-city names to CHGIS 1911 counties using reviewed Chinese-name aliases, and aggregate multiple successor units assigned to the same historical county-year. The resulting panel covers 88 historical counties in the pre-1978 benchmark years and 89 from 1978 onward. Per-capita outcomes use population reported in the same source. Mission exposure is defined by whether a CHCD institution was recorded in the historical county before 1950; the railway and predicted-exposure measures are assigned to the same 1911 county geography. For this exercise, we also conducted a focused recovery of Shaanxi CHCD geography: unresolved province- or prefecture-level place records were reviewed using institution names, historical place-name variants, directory and gazetteer evidence, and the full CHGIS 1911 county reference, allowing additional institutions to be assigned to counties when the evidence was sufficiently specific. A national county-level design is not currently feasible without substantially greater record-by-record recovery, because many CHCD locations elsewhere are identified only at the prefecture or broader level; restricting the national analysis to county-resolved records would discard considerable information and produce a selected sample. Because the source is confined to one province, these estimates are complementary to the national prefecture analysis.

For the religion series, we separately digitize dated references to Christian institutions, activity, closures, persecution, and Cultural Revolution disruption from Shaanxi county and city gazetteers. We aggregate these source references into county-period mention shares through 1998. They measure the content recorded in local histories, not a complete annual census of religious activity.

Supplementary Missionary Data

Figure B3: Missionary People and Institutions Over Time



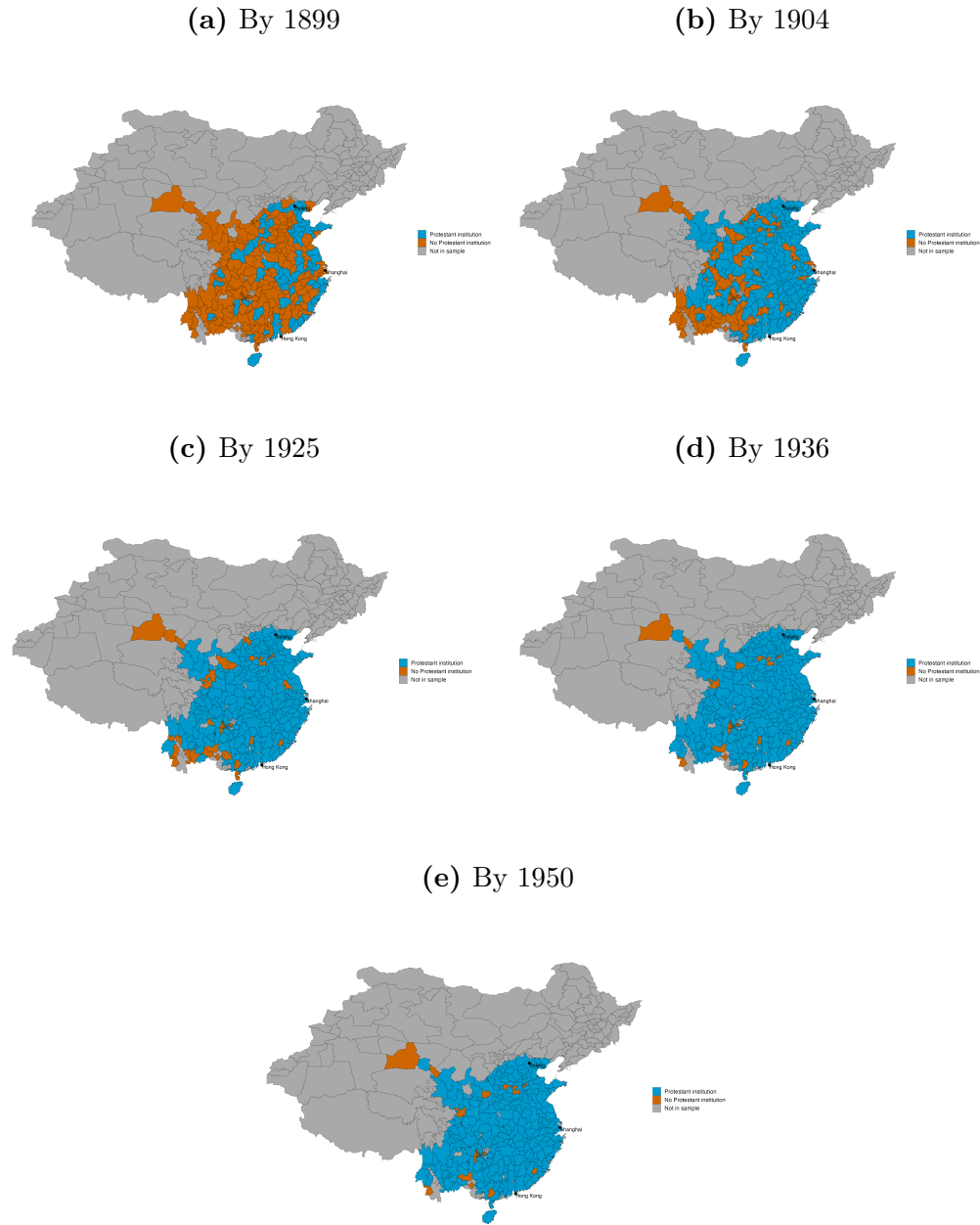
Notes: The figure compares active missionary institutions and missionary people in the CHCD prefecture-year panel through 1936. The institution series includes new directory-based CHCD school records not duplicated in the baseline v3.0.0 panel. The figure is descriptive; the main treatment is based on institutional arrival.

Table B1: Distribution of Protestant and Catholic Mission Institutions

Institution Tradition	Pre-1857		1857–1899		1900–1904		1905–1925		1926–1936		1937–1950		Undated		Total	
	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.
Protestant	4	4.1	204	99.5	548	100.0	1,664	99.9	1,024	100.0	1,292	99.8	265	36.5	5,001	89.9
Catholic	93	95.9	1	0.5	0	0.0	1	0.1	0	0.0	2	0.2	462	63.5	559	10.1

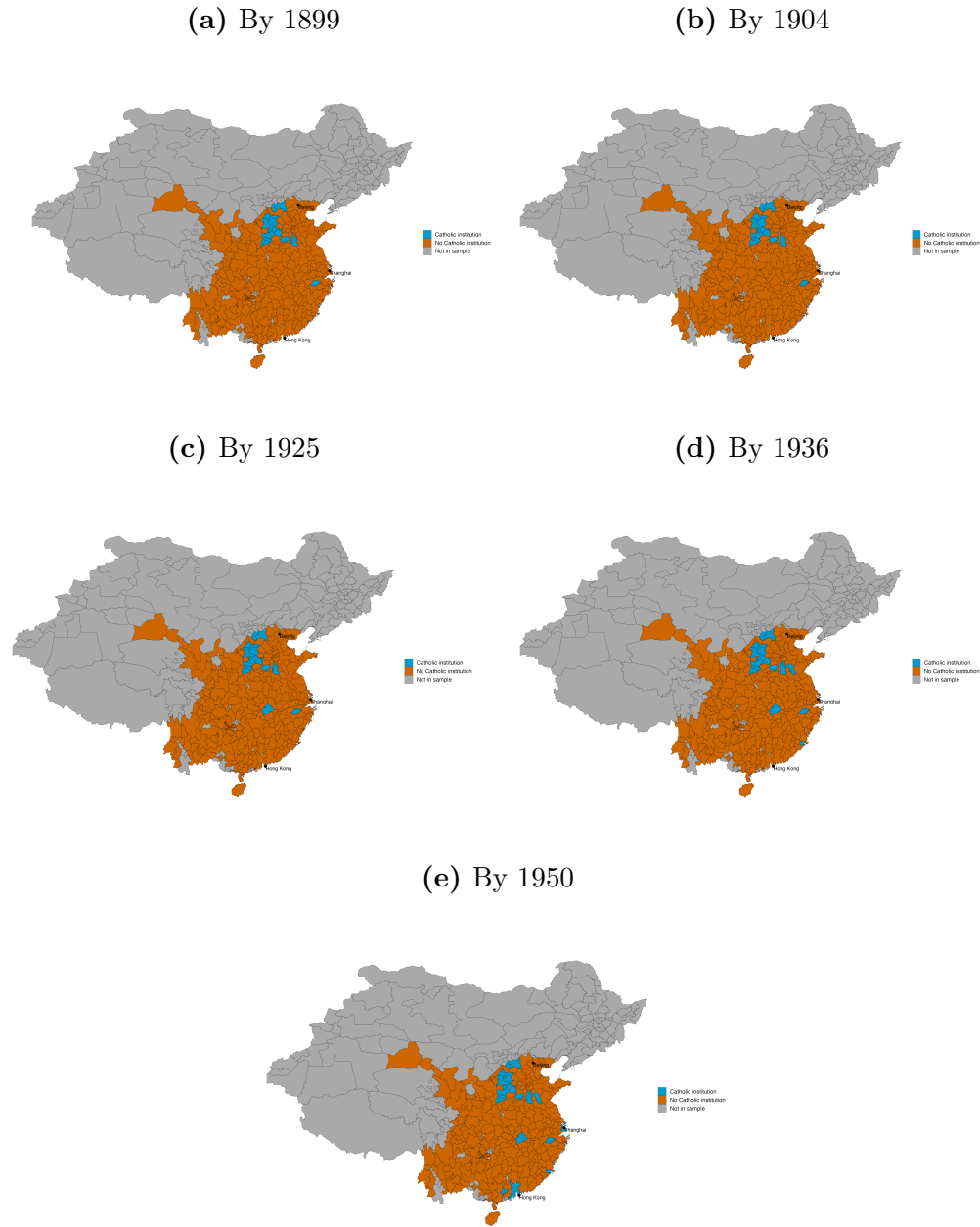
Notes: The table counts unique CHCD v3.0.0 mission institutions with prefecture matches, using the institution tradition field and the cleaned start-year measure used to construct treatment timing. Percentages are within columns. The “Undated” column reports institutions with known tradition and location but no usable cleaned start year. Orthodox, non-religious, unknown, and missing traditions are excluded from the two-way Protestant–Catholic split. Source: CHCD v3.0.0.

Figure B4: Expansion of Protestant Mission Institutions



Notes: These maps show the cumulative expansion of Protestant mission institutions over time. Prefectures with at least one Protestant institution are shown in blue, sample prefectures without a Protestant institution are shown in orange, and prefectures outside the analysis sample are shown in gray. The split uses the CHCD institution tradition field. Prefecture boundaries are from CHGIS 1911.

Figure B5: Expansion of Catholic Mission Institutions



Notes: These maps show the cumulative expansion of Catholic mission institutions over time. Prefectures with at least one Catholic institution are shown in blue, sample prefectures without a Catholic institution are shown in orange, and prefectures outside the analysis sample are shown in gray. The split uses the CHCD institution tradition field. Prefecture boundaries are from CHGIS 1911.

Table B2: Distribution of Missionary Subdenominations

Religious Family	Pre-1857		1857–1899		1900–1904		1905–1925		1926–1936		1937–1950		Total	
	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.
Interdenominational	4	33.3	63	12.5	368	12.4	1,710	17.9	1,047	17.0	0	0.0	3,192	16.4
Presbyterian	0	0.0	116	23.1	500	16.8	1,323	13.8	838	13.6	0	0.0	2,777	14.3
Methodist	1	8.3	82	16.3	358	12.0	1,214	12.7	678	11.0	2	1.0	2,335	12.0
Anglican	0	0.0	67	13.3	385	13.0	872	9.1	506	8.2	0	0.0	1,830	9.4
Lutheran	0	0.0	10	2.0	167	5.6	806	8.4	626	10.1	0	0.0	1,609	8.3
Baptist	1	8.3	23	4.6	519	17.5	630	6.6	356	5.8	0	0.0	1,529	7.9
Congregational	1	8.3	91	18.1	267	9.0	751	7.9	353	5.7	1	0.5	1,464	7.5
Other Protestant	3	25.0	24	4.8	396	13.3	2,111	22.1	1,414	22.9	10	5.1	3,958	20.4
Maryknoll	0	0.0	0	0.0	0	0.0	12	0.1	30	0.5	42	21.5	84	0.4
Passionist	0	0.0	0	0.0	0	0.0	0	0.0	56	0.9	21	10.8	77	0.4
Other Catholic	2	16.7	27	5.4	12	0.4	131	1.4	266	4.3	119	61.0	557	2.9

Notes: The table counts unique missionary people in the most recent CHCD person-prefecture file. It uses people data because detailed religious-family information is recorded at the person level. Smaller Protestant families are grouped as other Protestant, and smaller or missing Catholic families are grouped as other Catholic. Source: CHCD.

Appendix C: Balance & Identification

Figure C1: Distribution of Treatment Variable G_i

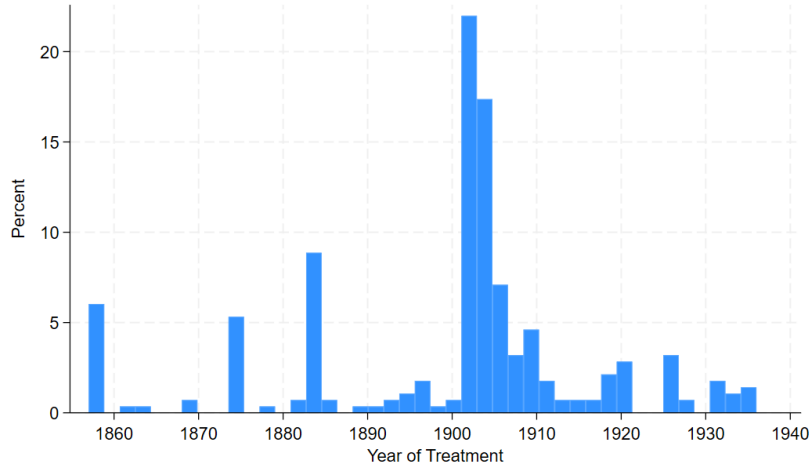
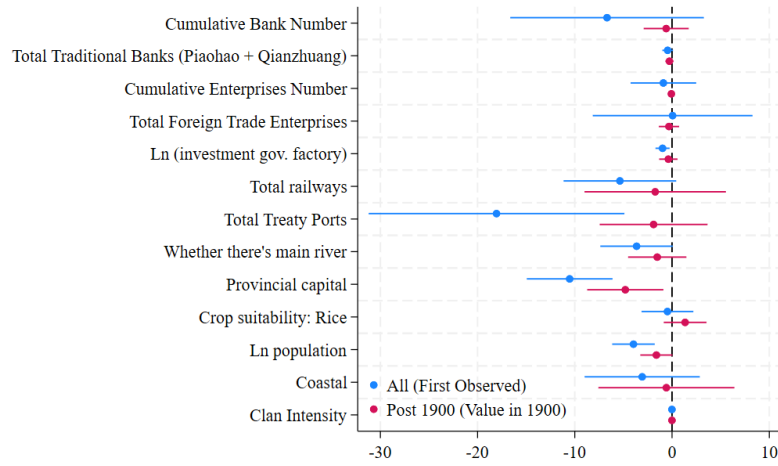
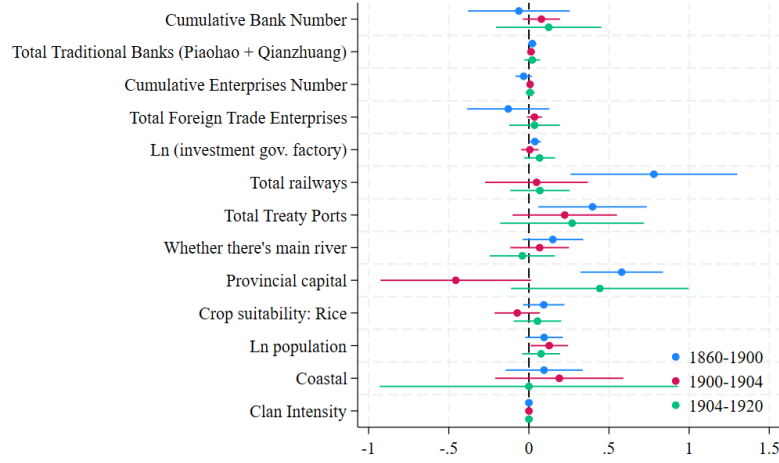


Figure C2: Predictors of Mission Establishment Year: One-by-One Regressions



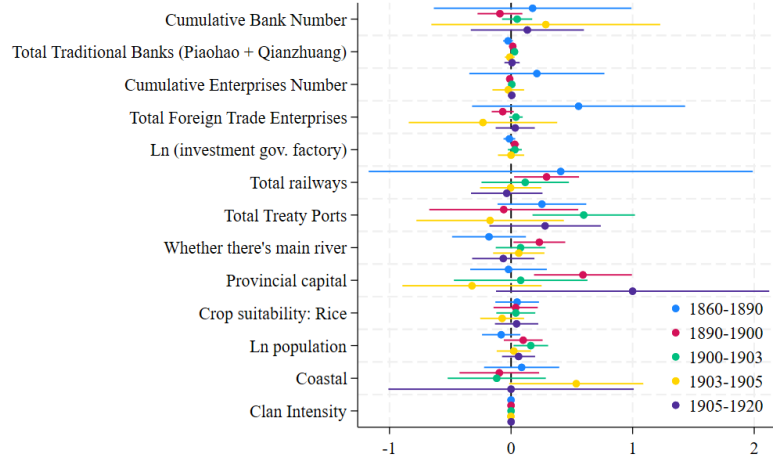
Notes: Predetermined covariates are separately regressed on missionary establishment year. The figure plots the coefficients corresponding to columns 1 and 2 of Table 2. All regressions include province fixed effects. Covariates are in raw units.

Figure C3: Predictors of Mission Establishment by Broad Treatment Cohort: One-by-One Regressions



Notes: Univariate regressions of the treated-cohort dummy on each covariate. This figure reports adjacent broad-cohort comparisons as a complementary local-timing diagnostic. All regressions include province fixed effects. Covariates are in raw units.

Figure C4: Predictors of Mission Establishment by Narrow Treatment Cohort: One-by-One Regressions



Notes: Univariate regressions of the treated-cohort dummy on each covariate. This figure uses a coarser five-bin cohort partition. All regressions include province fixed effects. Covariates are in raw units.

Appendix D: Historical DiD Results and Robustness

D.1 Main Estimates, Pre-Period Variation, and Outcome Transformations

Table D1: Pre- and Post-Treatment Estimates for Main Historical Outcomes

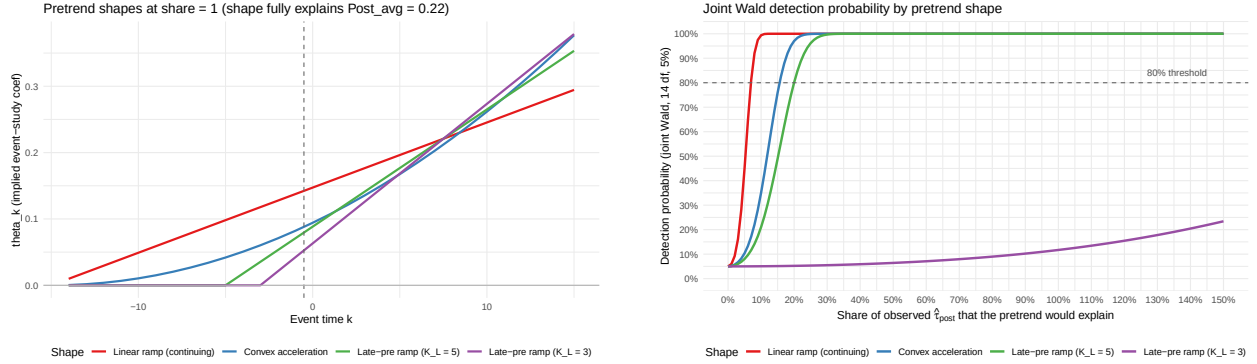
VARIABLES	(1) Modern banks	(2) Enterprises	(3) Log banks	(4) Bank growth	(5) New banks
Pre-treatment ATT	0.024 (0.016)	0.060* (0.036)	0.013 (0.009)	0.007 (0.004)	0.011 (0.007)
Post-treatment ATT	0.221*** (0.071)	1.292** (0.610)	0.075*** (0.021)	0.019*** (0.005)	0.059** (0.024)
Prefecture FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Period	1857–1936	1875–1936	1857–1936	1857–1936	1857–1936
Observations	18,560	13,268	18,560	18,328	18,560
Prefectures	232	214	232	232	232
Years	80	62	80	79	80
Mean Dep. Var.	0.761	4.072	0.164	0.014	0.082
S.D. Dep. Var.	6.320	41.477	0.540	0.090	0.680

Notes: The first two rows report average pre-treatment and post-treatment ATT estimates from the event-study aggregation over the event-time window $[-15, 15]$. Standard errors are in parentheses. Column 1 corresponds to Figure 5; column 2 corresponds to Figure 6. Columns 3–5 report alternative modern-bank outcomes: $\log(\text{total modern banks} + 1)$, one-year growth in this log outcome, and new modern banks. The post-treatment estimates imply about 31% more modern banks and 27% more enterprises using the outcome means reported in Figures 5 and 6. The log specification implies a 7.8% increase in total modern banks plus one; the growth specification implies a 1.9 percentage-point increase in annual log growth; and new-bank flows rise by about 73% relative to their mean. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Pre-period variation. A natural concern is that modern banks are rare in the early part of the sample, so flat pre-treatment coefficients may partly reflect limited pre-period variation. We therefore complement the event-study pre-trend tests with a design-based minimum detectable effect exercise. The exercise uses the estimated pre-period coefficients and covariance matrix from the baseline modern-bank event study. It then asks how large different hidden pre-trend shapes would need to be for the design to detect them. The design is informative against several smooth pre-trend shapes, although a pre-trend that activates only in the final few pre-treatment years is harder to detect. This exercise is intended as a diagnostic for the baseline design, not as a separate identification assumption.

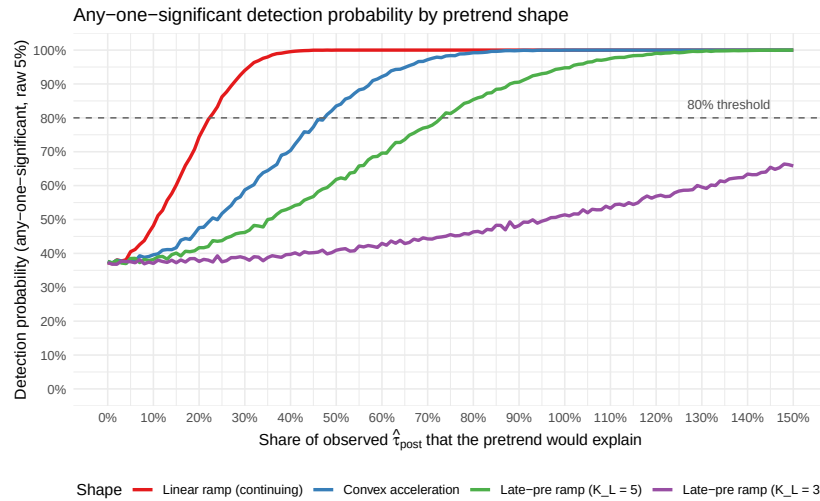
Alternative transformations. We also re-estimate the historical DiD using alternative transformations of the modern-bank outcome. These specifications address the concern

Figure D1: Design-Based Power Against Hidden Pre-Trends



Notes: The left panel shows candidate hidden pre-trend shapes, scaled relative to the observed post-treatment effect. The right panel reports how often the joint Wald test over pre-treatment coefficients would reject for each candidate shape. The exercise uses the estimated pre-period coefficients and covariance matrix from the baseline modern-bank event study.

Figure D2: Alternative Pre-Trend Detection Rule



Notes: The figure reports rejection probabilities under an alternative rule based on whether any individual pre-treatment coefficient is statistically significant. This rule is more mechanical than the joint Wald test but gives a useful check that the pre-period is not completely uninformative.

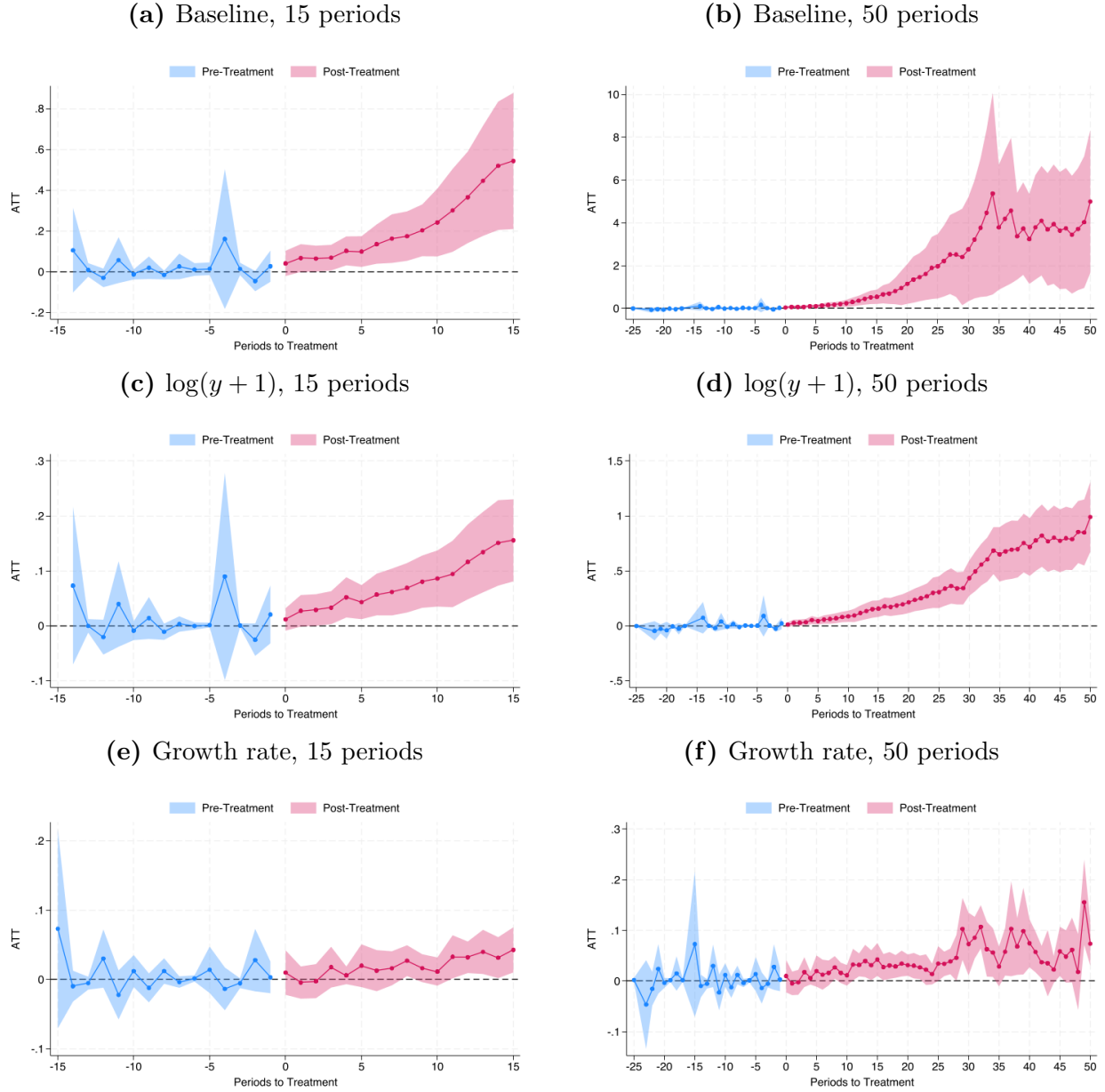
that the baseline level outcome may be affected by zeros or limited early variation. Table D2 reports the post-treatment ATT for each specification, and Figures D3–D5 plot the corresponding event studies.

Table D2: Post-Treatment ATT Under Alternative Modern-Bank Outcomes

Specification	ATT	SE	p -value
Baseline: total modern banks	0.221	0.071	0.002
$\log(y + 1)$	0.075	0.021	< 0.001
Growth rate	0.019	0.005	< 0.001
Extensive margin ($y > 0$)	0.110	0.046	0.017
New modern banks	0.059	0.024	0.014
Province-conditioned levels	0.221	0.071	0.002
Drop pre-1900, levels	0.258	0.092	0.005
Drop pre-1900, $\log(y + 1)$	0.084	0.026	0.001

Notes: The table reports post-treatment ATT estimates averaged over event times 0 through 15. The baseline reproduces the main effect reported in Figure 5. The remaining rows use alternative transformations or definitions of the modern-bank outcome.

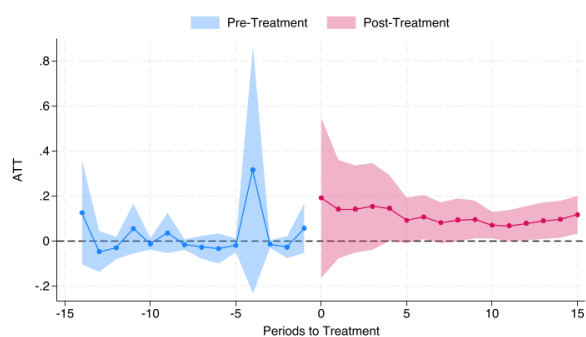
Figure D3: Alternative Outcome Transformations: Baseline, Logs, and Growth



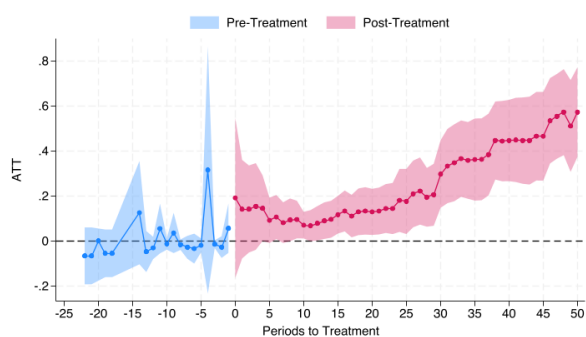
Notes: The figure reports event studies for alternative transformations of the modern-bank outcome. Panels use the same treatment definition as the baseline specification.

Figure D4: Alternative Outcome Transformations: Extensive Margin and New Banks

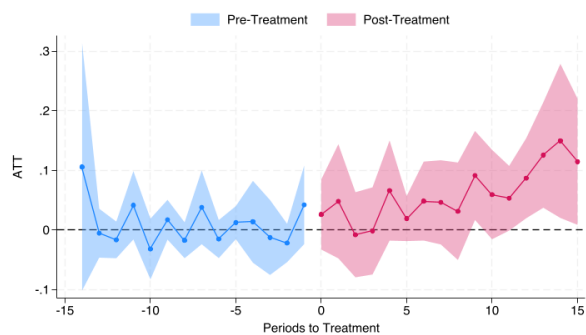
(a) Extensive margin, 15 periods



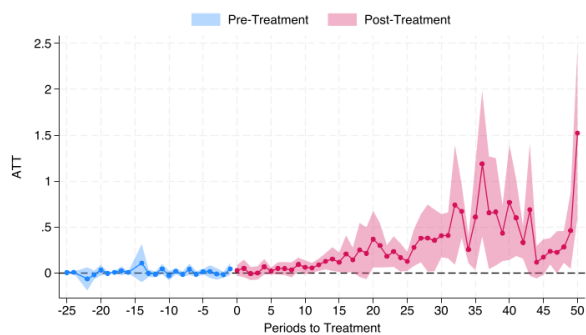
(b) Extensive margin, 50 periods



(c) New banks, 15 periods



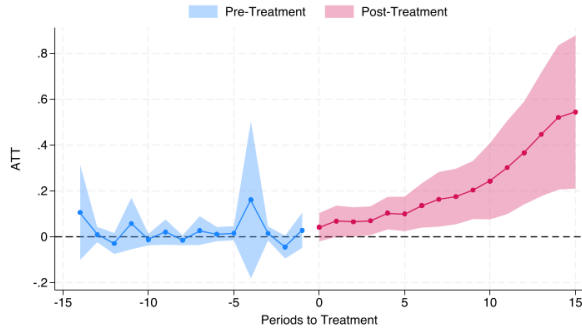
(d) New banks, 50 periods



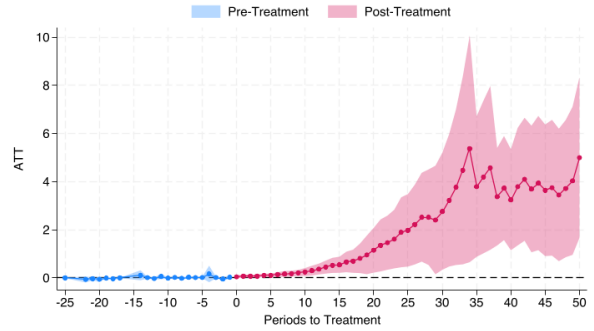
Notes: The figure reports additional outcome transformations. The post-treatment averages corresponding to these figures are reported in Table D2.

Figure D5: Alternative Outcome Transformations: Province-Conditioned and Restricted Samples

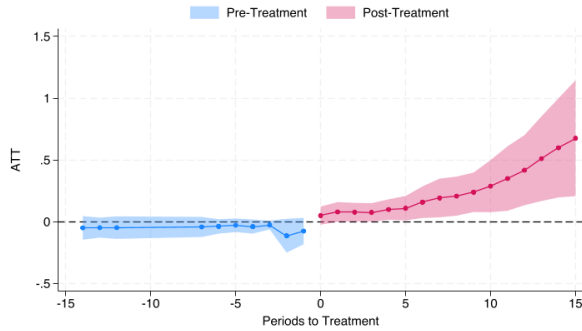
(a) Province-conditioned, 15 periods



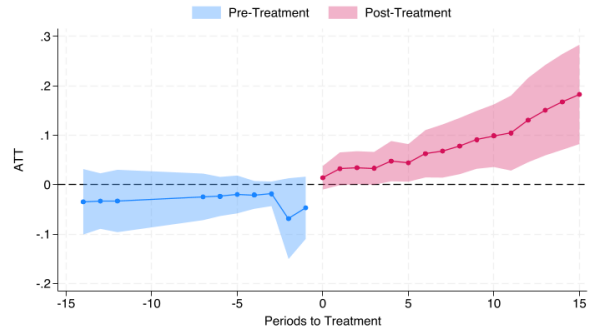
(b) Province-conditioned, 50 periods



(c) Drop pre-1900, levels



(d) Drop pre-1900, $\log(y + 1)$

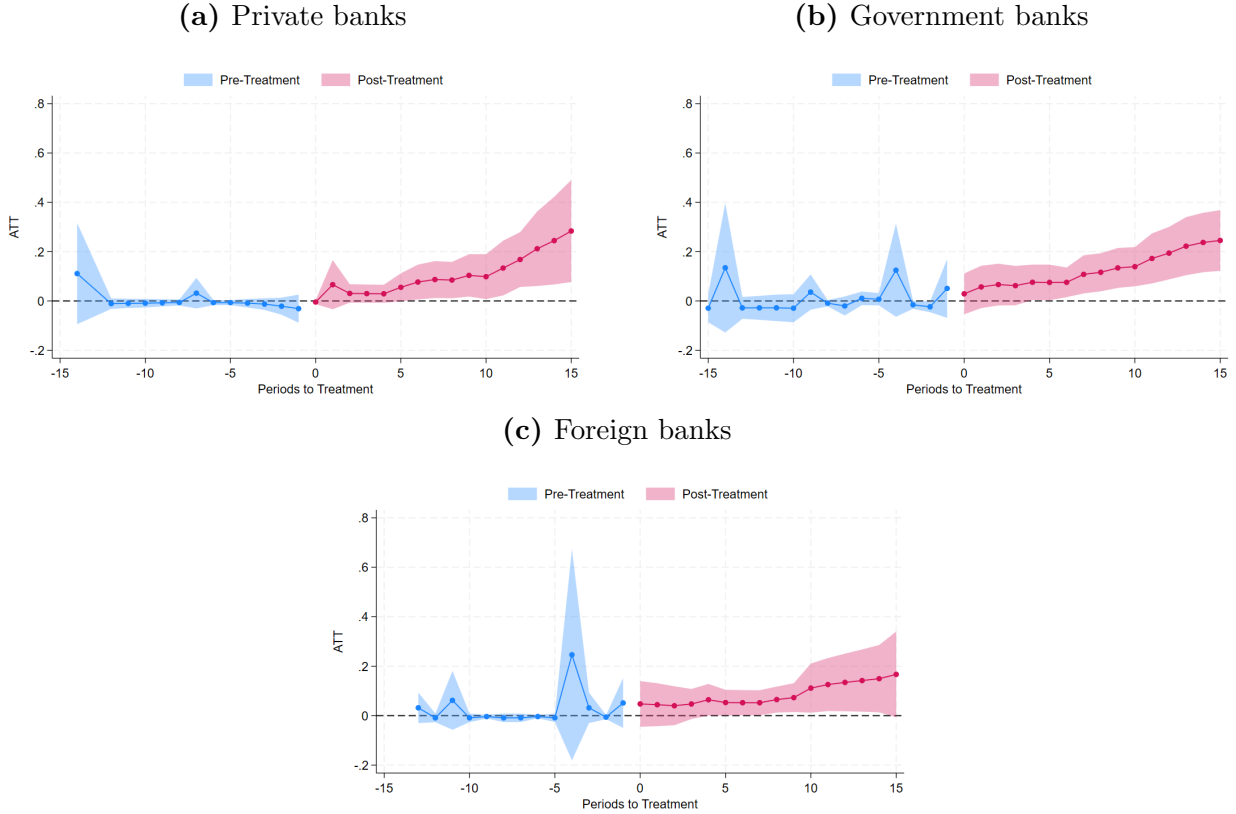


Notes: The figure reports province-conditioned specifications and sample restrictions. The post-treatment averages corresponding to these figures are reported in Table D2.

D.2 Propagation Across the Banking Sector

The baseline outcome counts all modern banks. We next show that the result is not driven by a single bank type or by a purely extensive-margin count. Figure D6 reports event studies separately for private, government, and foreign banks. All three outcomes move in the same direction after missionary arrival, suggesting that missionary exposure affected the broader modern-banking sector rather than only one ownership category.

Figure D6: Missionary Presence and Modern Banks by Ownership Type



Notes: Each panel reports the event-study estimates from the historical DiD design using the indicated bank type as the outcome. The estimation window is 15 periods before and after missionary arrival.

We also use balance-sheet data to study whether missionary exposure is associated with the scale of banking activity. These exercises are supporting evidence rather than the baseline DiD specification, but they point in the same direction: places with greater missionary exposure have larger deposits, loans, assets, and related measures of bank size. The prefecture-level table uses the branch/location balance-sheet records from the 1910s, aggregated to 1911 prefectures, and relates them to the predicted missionary exposure constructed from the railway-flow instrument in Section 4.2. The bank-level table uses the 1932–1936 bank balance sheets and constructs each bank’s missionary exposure from its branch network

across prefectures.

Table D3: Missionary Exposure and Bank Balance Sheets: Prefecture Level

Variables	(1) Ln Deposits	(2) Ln Banknotes	(3) Ln Surplus Reserves	(4) Ln Capital Paid	(5) Ln Capital	(6) Total Banks (1936)
Predicted Tot Missions	0.448** (0.175)	0.498*** (0.183)	0.396*** (0.151)	0.379** (0.166)	0.368** (0.180)	1.634* (0.959)
Railway Duration	-0.0268 (0.0266)	-0.0396 (0.0282)	-0.0311 (0.0227)	-0.0261 (0.0251)	-0.0245 (0.0270)	-0.179 (0.130)
Railway Density	323.8*** (109.6)	259.9** (124.2)	265.9** (113.8)	259.3** (108.7)	269.2** (113.2)	1,438** (597.7)
Obs.	247	247	247	247	247	247
Mean Dep. Var.	2.271	2.062	0.848	1.458	1.520	3.622

Notes: The table reports prefecture-level balance-sheet regressions using the 1910s branch/location records. Outcomes are measured using historical bank balance-sheet data described in the data section. Predicted total missions is constructed using the railway-flow instrument described in Section 4.2. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table D4: Missionary Exposure and Bank Balance Sheets: Bank Level

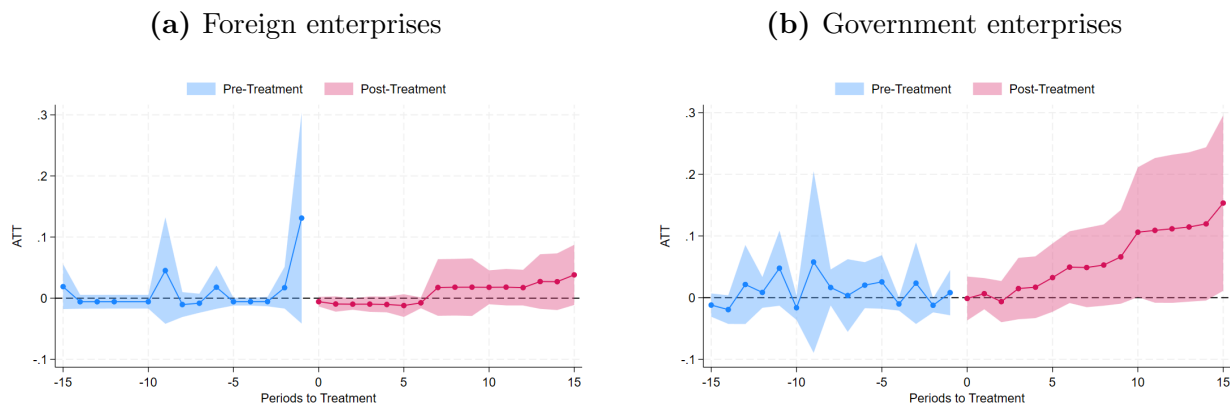
Variables	(1) Ln Deposits	(2) Ln Loans	(3) Ln Assets	(4) Ln Income	(5) Ln Security	(6) Ln Surplus Reserves	(7) Tot Branches	(8) Ln Net Income
Mission Exposure (STD)	0.954*** (0.125)	0.858*** (0.107)	0.867*** (0.117)	0.689*** (0.106)	0.844*** (0.143)	1.262*** (0.141)	0.676 (0.799)	0.585*** (0.120)
Exposure to Population (STD)	0.488*** (0.0867)	0.472*** (0.0784)	0.479*** (0.0824)	0.396*** (0.0782)	0.457*** (0.120)	0.268*** (0.0849)	17.47*** (0.776)	0.329*** (0.0669)
Obs.	155	156	157	156	136	144	160	153

Notes: The table reports bank-level regressions using 1932–1936 historical balance-sheet data. Mission exposure is standardized and is based on the historical missionary exposure of the prefectures in which each bank operated branches. Population exposure is constructed analogously using historical population. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

D.3 Downstream Market Outcomes

The main text reports the effect of missionary presence on total enterprises. This appendix provides additional evidence on downstream market activity. First, we separate enterprises by ownership type. The positive enterprise result is mainly driven by private enterprises, which account for most of the enterprise panel. Government and foreign enterprises move in the same direction but are less precisely estimated.

Figure D7: Missionary Presence and Enterprises by Ownership Type



Notes: The figure reports event studies for foreign and government enterprises. The total enterprise result is reported in the main text. Private enterprises account for most of the enterprise panel.

Second, we use trade data from 1934–1936 to examine whether missionary exposure is associated with broader market exchange. We treat this as downstream evidence, not as a mechanism. The trade data come from a government census of trade product types, production volumes, prices, and sales areas. In 1934, local postal offices were entrusted with collecting the census at the county and township levels, and the data collection was completed in 1936 ([Ministry of Communications, 1937](#)). The data report domestic and international export activity, including trade values and commodity types.

We use these data to classify enterprise industries as trade-oriented or domestic-oriented. The idea is that treaty ports were the main departure points for overseas-bound goods, so products disproportionately exported from treaty ports are more likely to be linked to international trade. We compare export composition across treaty and non-treaty ports, compute the relative importance of each product category, and identify products with the largest positive and negative differences in treaty-port export shares. Products with high treaty-port export intensity are classified as “trade-likely,” while products with low treaty-port export intensity are classified as “domestic-likely.” We then map these products to enterprise industries in the historical enterprise panel. This gives us two downstream outcomes: trade-likely enterprises and domestic-likely enterprises.

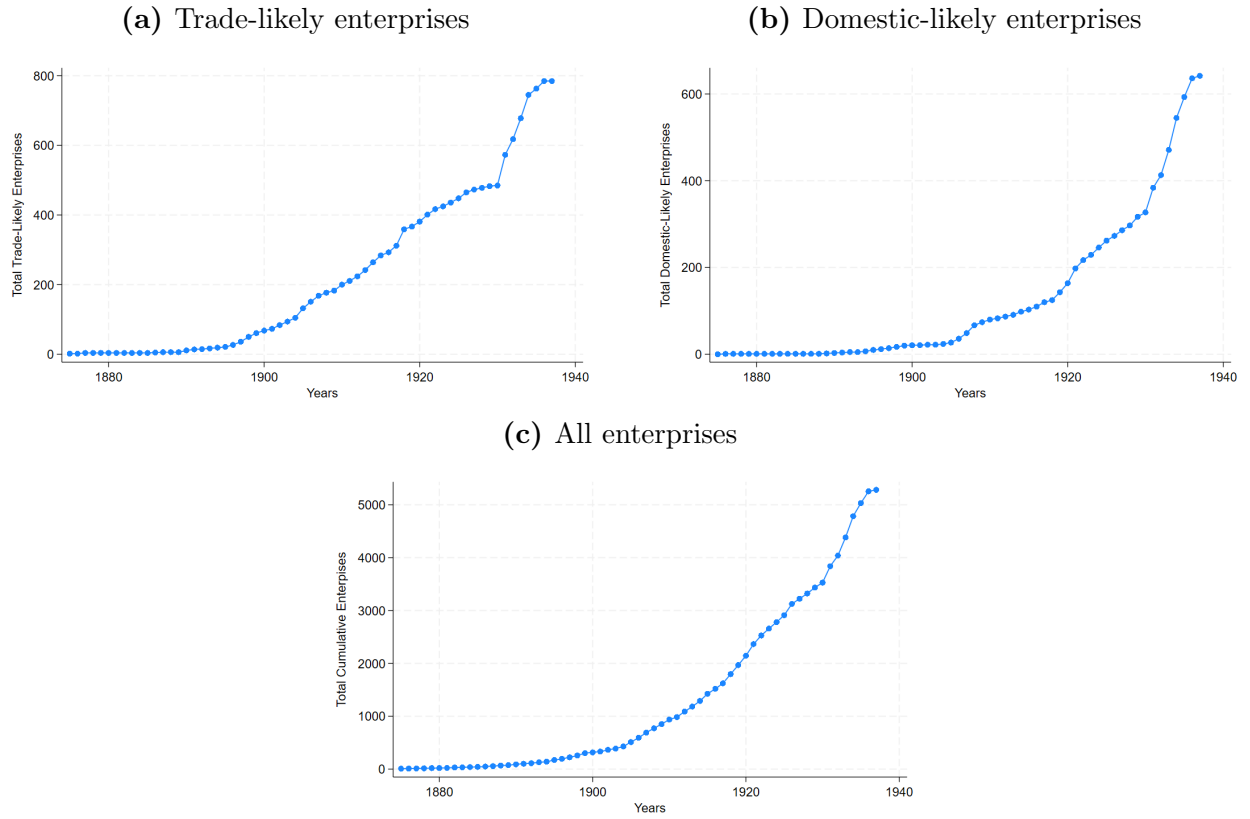
Table D5 reports the resulting classification. Figure D8 plots the aggregate evolution of these enterprise categories, and Figure D9 estimates the historical DiD separately for trade-likely and domestic-likely enterprises. The trade-likely response appears only after a delay, mirroring the timing of the total-enterprise result in the main text. Table D6 also reports the relationship between missionary exposure and export transactions in 1936. These exercises show that missionary exposure is associated with stronger trade-related activity, consistent with modern banks supporting broader market integration.

Table D5: Trade-Likely and Domestic-Likely Enterprise Categories

Trade-Likely Enterprises	Domestic-Likely Enterprises
Silk	Brick
Tea	Cement
Tobacco	Construction
Seafood	Building
Fish	Real Estate
Herbal Products	Services
Shipping	Textile
Foreign Trade	Animal Products
Transshipment	Agriculture
Beverages	Farming
Porcelain	Clothing
Cigarette	Entertainment
	Rice

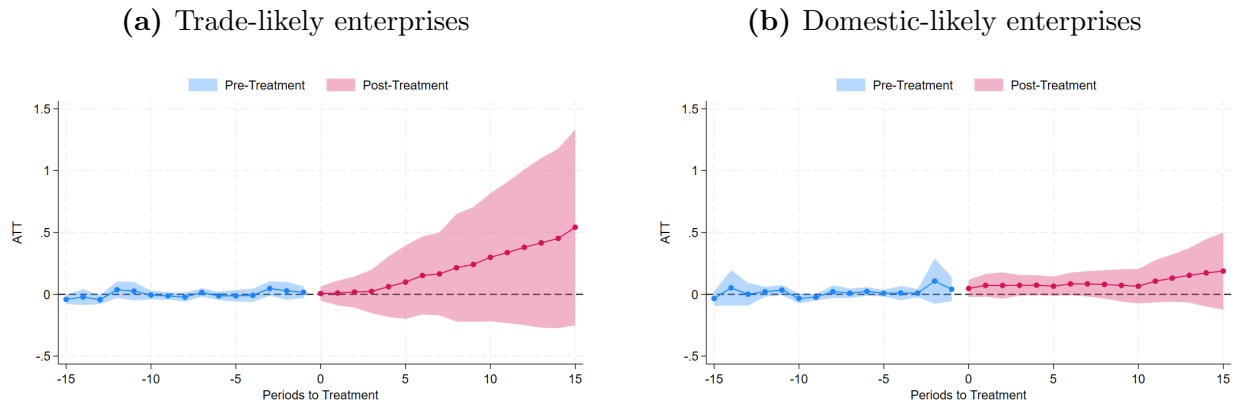
Notes: Product groups are classified using differences in export composition between treaty ports and non-treaty ports in the 1930s trade data. We then map these products to enterprise industries.

Figure D8: Evolution of Trade-Likely and Domestic-Likely Enterprises



Notes: The figure plots aggregate enterprise categories over time. Data on enterprises are from Du (1991, 2019).

Figure D9: Missionary Presence and Trade-Likely Enterprises



Notes: Panel A uses trade-likely enterprises as the outcome, and Panel B uses domestic-likely enterprises. Trade-likely and domestic-likely categories are defined using the export data described in Table D5. The delayed response in Panel A is consistent with interpreting trade as a downstream market outcome.

Table D6: Missionary Exposure and Export Transactions in 1936

Variables	IV	OLS
	Export Transactions	Export Transactions
Tot Miss Inst	0.865* (0.496)	0.289*** (0.0893)
RR Duration	0.00212 (0.146)	0.185** (0.0719)
RR Density	-595.0 (637.4)	-617.4 (491.6)
Province FE	Yes	Yes
Observations	231	231
Adj. R-squared	-0.214	0.385
Mean Dep. Var.	16.07	16.07
S.D. Dep. Var.	33.66	33.66
K-P F-Stat	11.68	NA

Notes: The dependent variable is the number of export transactions in 1936. The IV specification instruments total missionary institutions with the railway-flow instrument described in Section 4.2. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

D.4 Traditional Finance and Financial Interaction

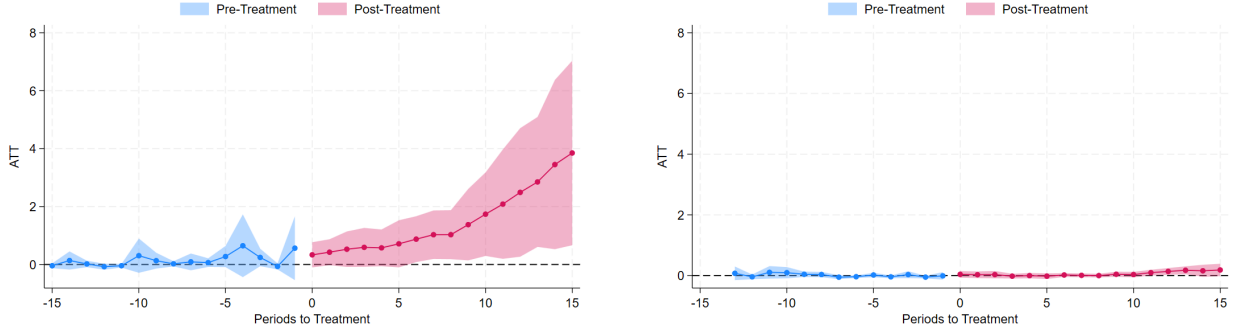
This section studies how missionary exposure interacted with traditional financial institutions. The main text emphasizes the transition from traditional finance to modern banking. We therefore begin with *Piaohao*, where the substitution pattern is clearest.

Modern banks in places with preexisting *Piaohao*. We first split the sample by whether a prefecture had *Piaohao* before missionary arrival. Prefectures with preexisting *Piaohao* experienced a sharper rise in modern banks following missionary entry, suggesting that modern banks expanded where financial demand was already present.

Figure D10: Modern Banks in Places With and Without Preexisting *Piaohao*

(a) Prefectures with preexisting *Piaohao*

(b) Prefectures without preexisting *Piaohao*



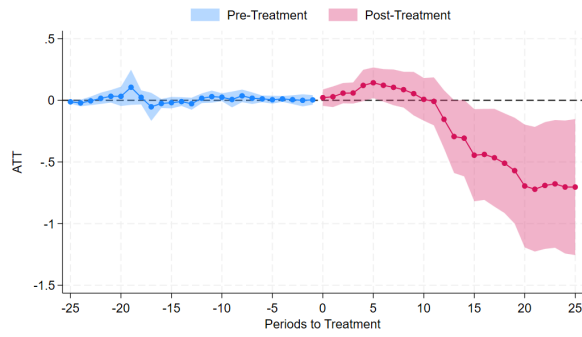
Notes: The outcome is the number of modern banks. Panel A includes prefectures with *Piaohao* before missionary arrival and never-treated prefectures. Panel B includes prefectures without *Piaohao* before missionary arrival and never-treated prefectures.

***Piaohao*.** We then use *Piaohao* as the outcome. The event study shows that *Piaohao* initially rise but later decline after missionary arrival, consistent with modern banks eventually replacing older remittance institutions. The lack of a clear *Piaohao* pre-trend is also useful because these institutions were already active before the modern-bank wave. Unlike the modern-bank outcome, the pre-period has meaningful variation, so the flat pre-period is less likely to be only a mechanical consequence of early zeros.

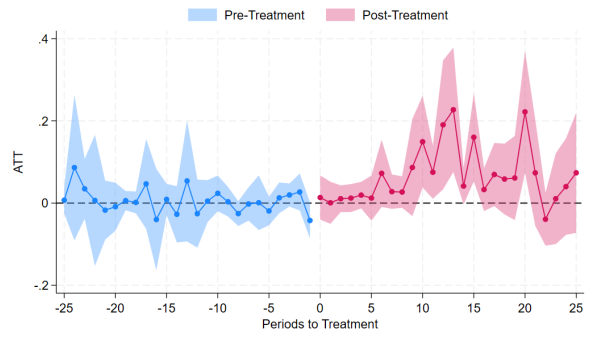
***Qianzhuang*.** The pattern for *Qianzhuang* is different. Because *Qianzhuang* were local money shops with dense merchant networks, they could complement modern banks that initially lacked local trust and information. We therefore treat the *Qianzhuang* results as evidence on financial interaction rather than simple substitution. Taken together, the traditional-finance results suggest that missionary exposure did not mechanically erase all

Figure D11: Missionary Presence and *Piaohao*

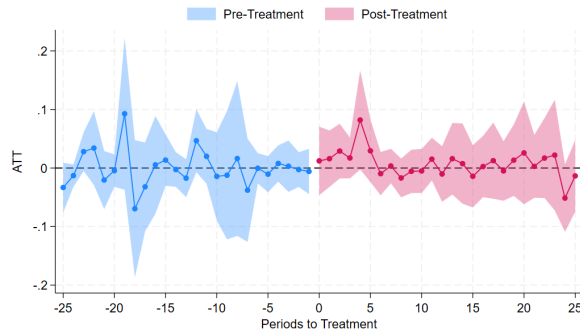
(a) Total *Piaohao*



(b) New closings



(c) New openings

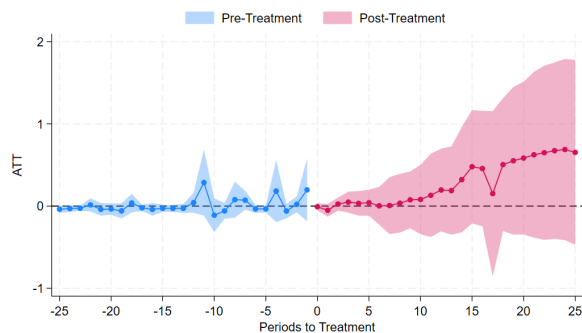


Notes: The figure reports event studies using total *Piaohao*, new *Piaohao* closings, and new *Piaohao* openings as outcomes. The absence of a clear pre-trend is informative because *Piaohao* have substantial pre-period variation and branch-level opening and closing dates.

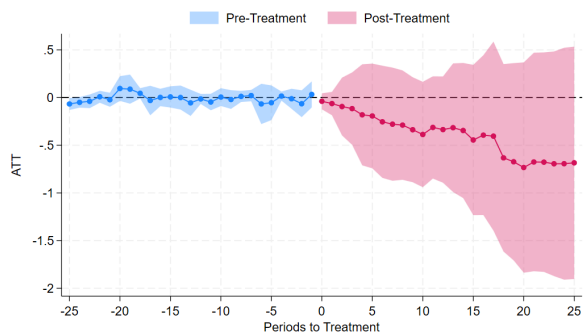
older institutions; the substitution pattern is clearest for *Piaohao*, while *Qianzhuang* appear more complementary.

Figure D12: Missionary Presence and *Qianzhuang*

(a) Total *Qianzhuang*, eight provinces



(b) Total *Piaohao*, same eight provinces

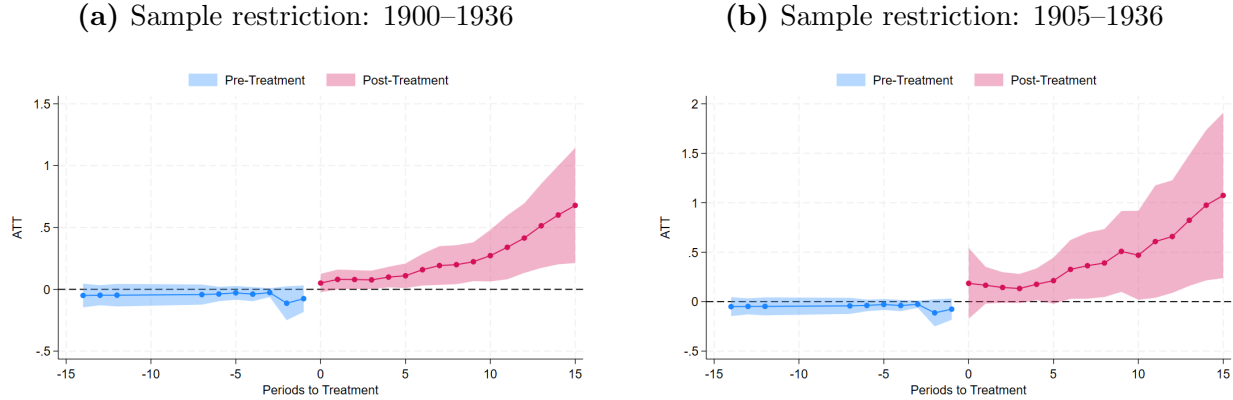


Notes: Panel A uses total *Qianzhuang* in eight major provinces as the outcome. Panel B uses total *Piaohao* in the same eight provinces for comparison.

D.5 Identification Robustness

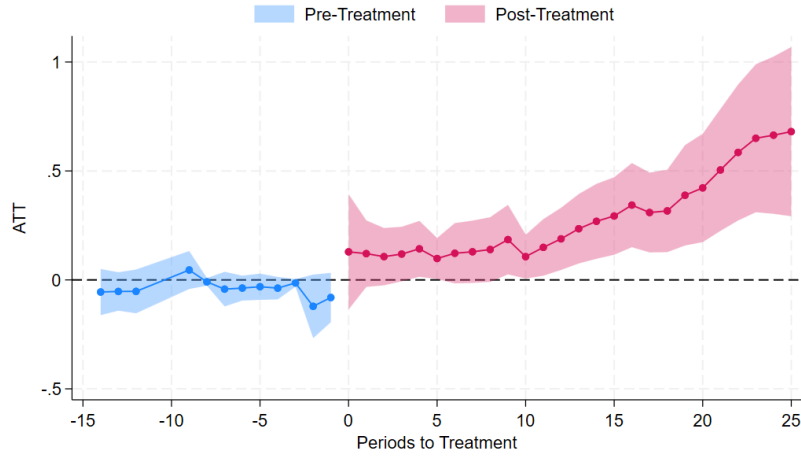
This section reports additional checks for the historical DiD design. First, we restrict the sample to the post-1900 and post-1905 periods, when selection into missionary entry is weaker and the modern-bank outcome has more pre-period variation. Second, we exclude treaty ports to address the concern that the result is driven by locations with early foreign presence. Third, we use longer event windows, balanced pre/post windows, and group-specific ATTs. Fourth, we add railway controls and examine pre-trends in other predetermined outcomes. Across these exercises, the qualitative pattern remains similar to the baseline result: the effect is not driven by the earliest part of the sample, treaty-port prefectures, one particular cohort, or one event-window choice.

Figure D13: Missionary Presence and Modern Banks: Post-1900 and Post-1905 Samples



Notes: The figure reports the baseline event study after restricting the estimation sample to later periods. The dependent variable is the total number of modern banks. The positive post-treatment pattern remains when the early years with sparse modern-bank activity are excluded.

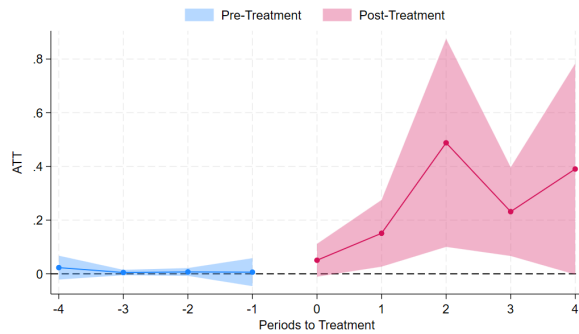
Figure D14: Missionary Presence and Modern Banks: Excluding Treaty Ports



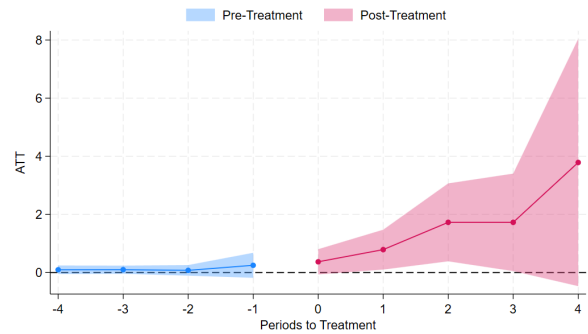
Notes: The figure reports the baseline event study after excluding treaty-port prefectures. The dependent variable is the total number of modern banks. The persistence of the effect outside treaty ports suggests that the result is not only a treaty-port foreign-presence effect.

Figure D15: Pre-Trends in Other Predetermined Outcomes

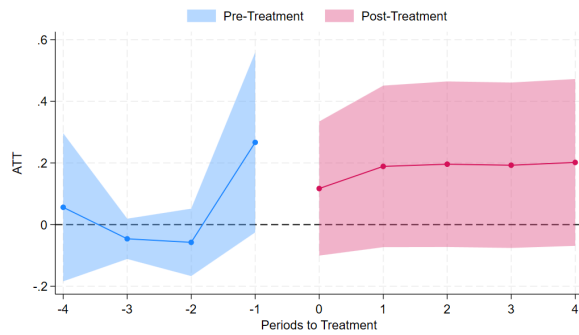
(a) Modern banks



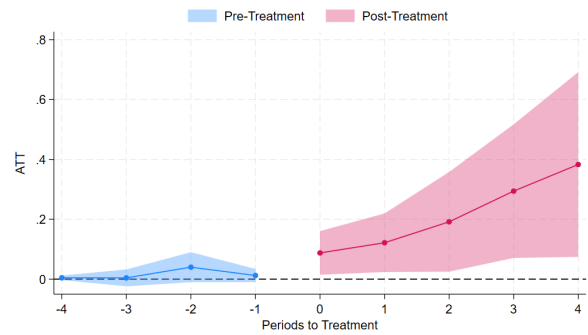
(b) Enterprises



(c) Government investments

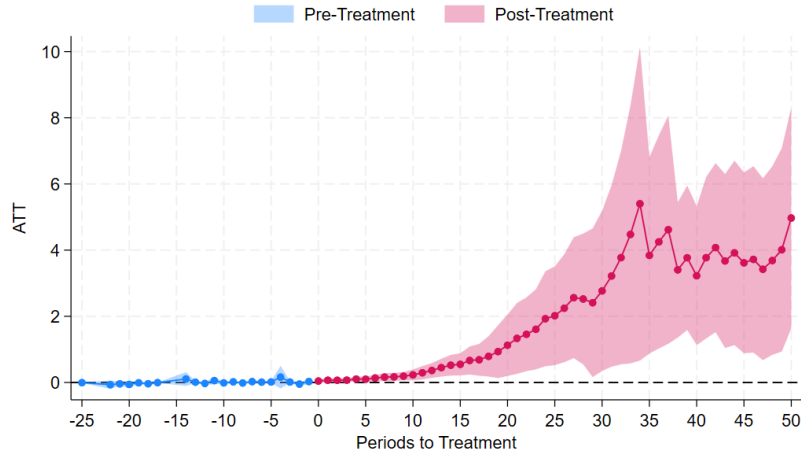


(d) Railways



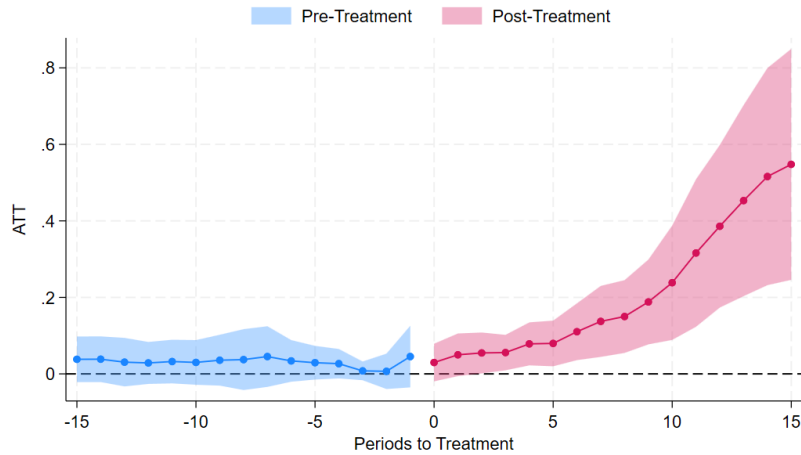
Notes: The figure reports cohort-based pre-trend diagnostics for outcomes and covariates measured before missionary entry. These checks compare treated cohorts before missionary arrival and ask whether they were already diverging in other predetermined outcomes. The panels show no systematic pre-treatment divergence.

Figure D16: Missionary Presence and Modern Banks: Longer Event Window



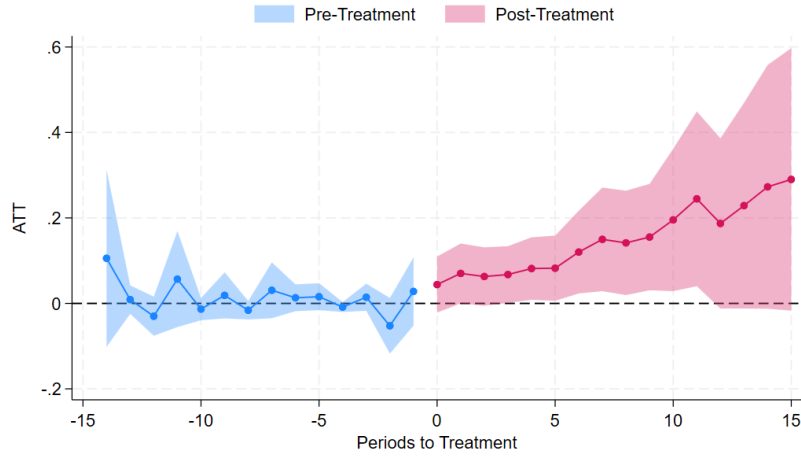
Notes: The figure reports the event study using the longer event window. The dependent variable is the total number of modern banks. The longer window shows a similar post-treatment increase.

Figure D17: Missionary Presence and Modern Banks: Long Gaps in the Pre-Period



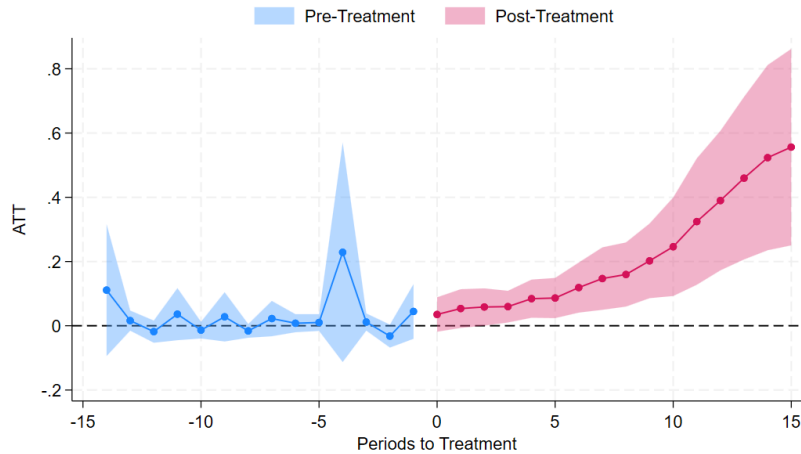
Notes: The figure reports the event study allowing for longer gaps in the pre-period. The exercise checks that the result is not an artifact of how long pre-treatment gaps are handled.

Figure D18: Missionary Presence and Modern Banks: Number of Railways as Control



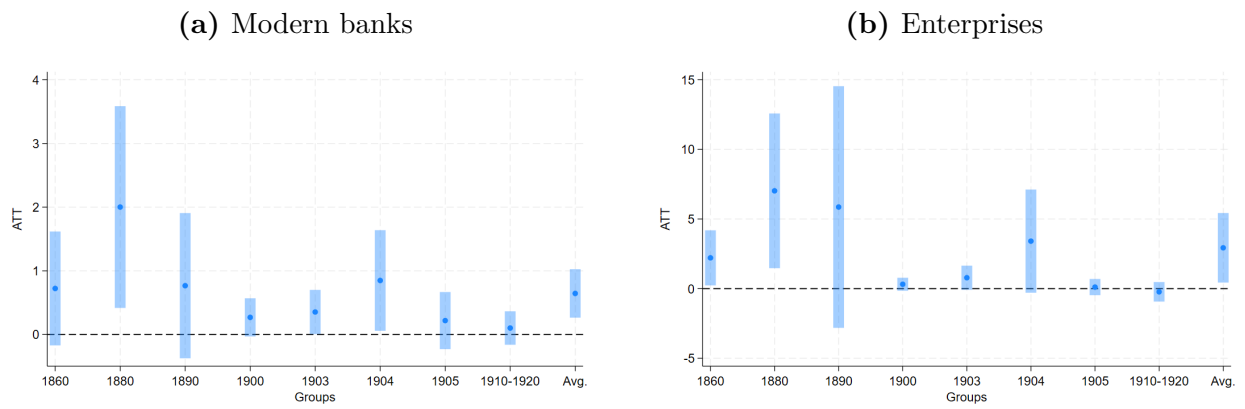
Notes: The figure reports the baseline event study after adding the number of railways as a control. The result is similar, suggesting that railway expansion is not driving the historical DiD pattern.

Figure D19: Missionary Presence and Modern Banks: Balanced Pre/Post Window



Notes: The figure reports the event study using a balanced number of pre- and post-treatment periods. The pattern remains similar after imposing the balanced event window.

Figure D20: ATT by Cohort Group



Notes: The figure reports average treatment effects by treatment-cohort group for modern banks and enterprises. The estimates do not point to the result being driven by a single treatment cohort.

Appendix E: Mechanism Results and Examples

Mechanism 1: Missionaries spread economic ideas through translating and education

.1 Mission Organizations as Intermediaries Between Foreign and Chinese Systems

This appendix provides brief historical background for the claim in Section 2.1 that mission organizations could act as intermediaries between foreign and Chinese systems. The point is not that missionaries alone possessed local knowledge, nor that foreign knowledge automatically generated institutional change. The argument is narrower. Mission organizations often combined access to foreign ideas and networks with the local linguistic and social knowledge needed to translate those ideas into forms that Chinese actors could adapt and use.

Language and contextual learning. Language acquisition was often systematic rather than incidental. Mission societies commonly required newly arrived personnel to devote substantial time to Chinese language study before taking on independent responsibilities. In some cases, missionaries were examined in the vernacular before being allowed to lead local work. Training could also include geography, government, religion, etiquette, and other aspects of local society. In Hakka-speaking areas, for example, mission work involved extended language study with Chinese teachers, regular examinations, and the production of dictionaries, translations, and other vernacular materials. These practices made linguistic and contextual knowledge more reproducible across successive cohorts of workers.

Chinese personnel. Mission organizations also relied heavily on Chinese personnel. This is important because the relevant intermediary was often the organization rather than the foreign missionary taken in isolation. Chinese pastors, teachers, translators, evangelists, printers, doctors, nurses, and other workers helped link mission institutions to local communities. A Protestant count for 1921 reported 6,636 foreign missionaries and 24,732 paid Chinese workers. The Chinese workforce therefore substantially exceeded the foreign missionary presence. This broader Chinese participation is more important for the paper's mechanism than the nationality of the formally recorded missionary individuals in our own data.

Women and social reach. Mission organizations were also unusually female. In our linked person data, 54 percent of named missionary personnel are women. Historical counts show a similar pattern among foreign Protestant missionaries, and they also record a large number of Chinese Bible women. This mattered because female missionaries and Chinese Bible women could work in households, schools, clinics, and women's networks that were less accessible to foreign men. The implication is not that women necessarily possessed special commercial information. It is that the gender composition of mission organizations broadened their social reach and helped diffuse education, literacy, and information through a wider set of local environments.

Relation to commercial intermediaries. This mechanism does not require a sharp contrast between missionaries and compradors. Foreign commercial firms often relied on compradors, who were themselves highly informed intermediaries between Chinese and foreign markets. This is complementary rather than contradictory to the argument here. Mission schools could help produce some of the same bridging skills that later mattered in commerce and finance. Tang Tingshu is a useful example: educated at the mission-founded Robert Morrison School, he became a leading bilingual commercial intermediary and later chief Shanghai comprador for Jardine Matheson. The broader point is that missionary exposure could help create Chinese actors able to move across foreign and local systems.

Implication for the paper. For the purposes of this paper, the value of these historical details is not to show that missionaries directly selected borrowers or replaced relational finance. Rather, they help explain why missionary exposure could generate more than generic education. Through schools, publications, language instruction, and overseas links, mission organizations helped produce high-skilled people, practical knowledge, and transnational relationships. Those assets could be economically useful in periods of openness, lose much of their return under political closure, and become valuable again when China reopened.

Missionary Interview Evidence

As additional qualitative evidence, we reviewed 44 interviews with Christian missionary workers in China held in the Claremont Colleges Special Collections. To summarize the material systematically, we also coded the short archival summary associated with each interview into broad topics. The counts below are based on those summaries, while the selected passages and photographs draw on the underlying interview materials. This exercise is descriptive and does not enter the causal analysis, but it helps discipline the historical

interpretation of the education mechanism.

Table E1: Topics in Missionary Interviews

Topic	% of archival summaries mentioning
Economic contribution to local economy	18%
Banking link: banks, credit, or bookkeeping	9%
Enterprise link: firms or entrepreneurship	7%
Mentions mission schools	62%
Curriculum: modern subjects, science, math, or English	27%
Girls' education	16%

Notes: The table reports preliminary AI-assisted topic coding of short archival summaries associated with 44 interviews. The underlying interviews are from the Claremont Colleges Special Collections.

Table E2: Selected Short Excerpts from Missionary Interviews

Interviewee	Description	Selected excerpt
Glen Fuller	Methodist; set up a business administration school	“Several graduates went into Chinese banks.”
Francis Price Jones	Methodist; professor and translator	“Government had no plan for universal primary education”; “Superior instruction in English”
Katherine Ward	Methodist; educator and community services	“First schools for girls”; “Higher academic standards”

Notes: These excerpts illustrate themes that appear in the interview material: links between mission education and banks, English instruction, and girls' schooling. Excerpts are shortened from the interview summaries.

Figure E1: Mission Education: Classroom and College Study

(a) Classroom lecture scene



(b) Lingnan college library, 1920s



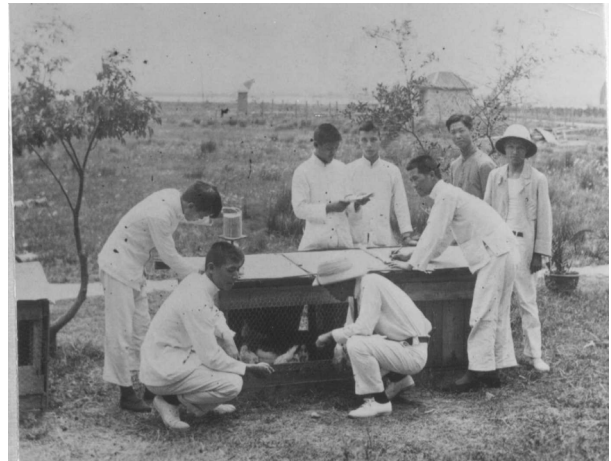
Notes: Panel A is an uncaptioned classroom photograph preserved with the China Missionaries Oral History Project materials at Claremont Graduate School and The Claremont Colleges Library. It illustrates the organized classroom instruction, teacher training, and educational materials described in the interviews. Panel B shows students in the reading room of the college library in Grand Hall at Lingnan University (Canton Christian College) during the 1920s. The library, study space, and student body illustrate the university-level infrastructure that mission institutions built for advanced training. Source for Panel B: Lingnan University Digital Commons, “Library Reading Room in Grand Hall.”

Figure E2: Mission Education: Adult Workers and Practical Training

(a) West China pastors and women workers,
1937



(b) Outdoor practical-demonstration scene



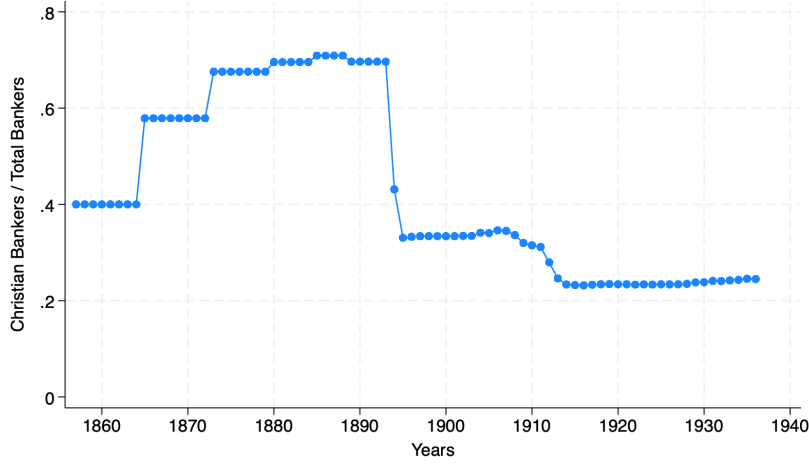
Notes: Panel A shows West China pastors and women workers in 1937. The F. Olin Stockwell manuscript describes the training of pastors, women workers, nurses, and local leaders for village outreach, home programs, and religious-education work. Panel B is an uncaptioned outdoor demonstration photograph preserved with the China Missionaries Oral History Project materials. The practical-training interpretation comes from the Louise Hathaway Stanley manuscript, which describes mission-linked work on seed selection, improved plants and livestock, village fairs, and seminary-based preparation of local leaders. We therefore use the image as an illustration of practical field instruction without assigning it to a specific school or program. Sources: F. Olin Stockwell and Louise Hathaway Stanley manuscripts, China Missionaries Oral History Project, Claremont Graduate School, 1971; The Claremont Colleges Library.

Figure E3: American Presbyterian Mission Press in Shanghai



Notes: The photograph shows the front building and staff of the American Presbyterian Mission Press, reproduced from [McIntosh \(1895\)](#). The press began in Macao in 1844, moved to Ningbo, and relocated to Shanghai in 1860. It combined Chinese typesetting, typesetting, editing, mechanical printing, binding, sales, and distribution under one organization and trained Chinese printers in these activities. Its personnel and skill network later helped seed the Chinese-owned Commercial Press. The Commercial Press's four founders—Xia Ruifang, Bao Xian'en, Bao Xianchang, and Gao Fengchi—came out of the Presbyterian Qingxin school and Huamei Press printing ecosystem, and the firm's official history describes its early operation as deliberately modeled on Huamei Press ([Lu, 2017](#)). This is evidence of skill and organizational diffusion rather than missionary ownership of the Commercial Press.

Figure E4: Evolution of the Ratio of Total Christian Bankers to Total Bankers

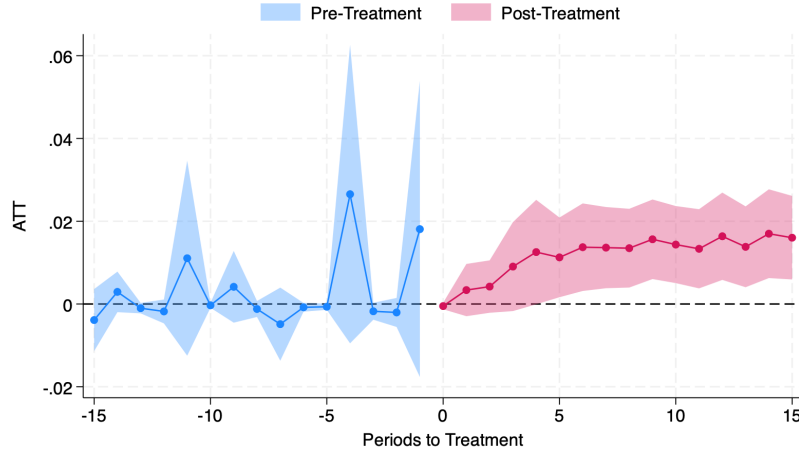


Notes: The figure plots the ratio of Christianity-related bankers to all bankers over time. Changes in the ratio partly reflect increases in the number of non-Christian bankers around the establishment of the first government banks and after the founding of the Republic in 1911.

Heterogeneity Magnitudes and Difference Tests

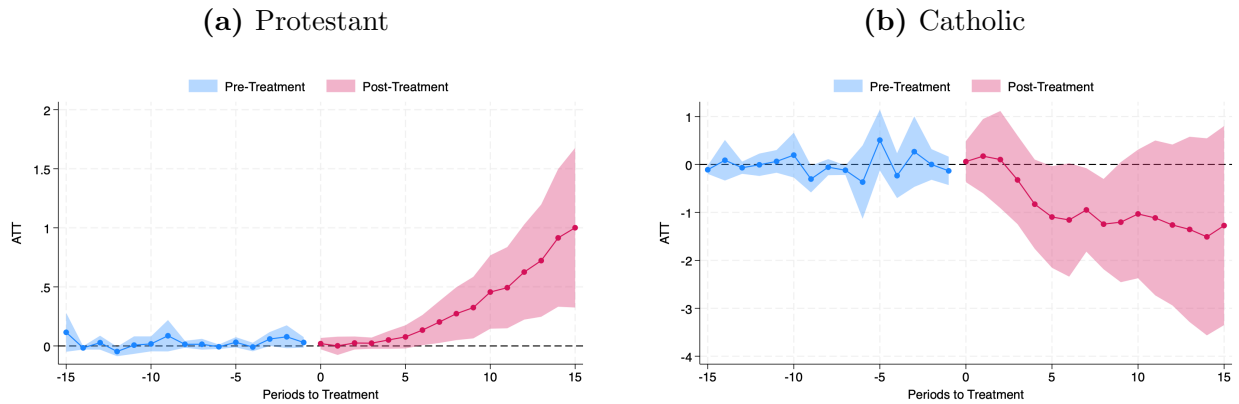
This appendix summarizes the magnitude comparisons behind the heterogeneity discussion in the main text. The table reports average pre- and post-treatment differences between high-exposure and low-exposure groups. The figures then plot coefficient-by-coefficient differences at each event time. These exercises are best read as suggestive heterogeneity checks: they compare magnitudes across groups, but some average pre-period differences are nonzero.

Figure E5: Missionary Presence and the Share of Christian-educated Bankers



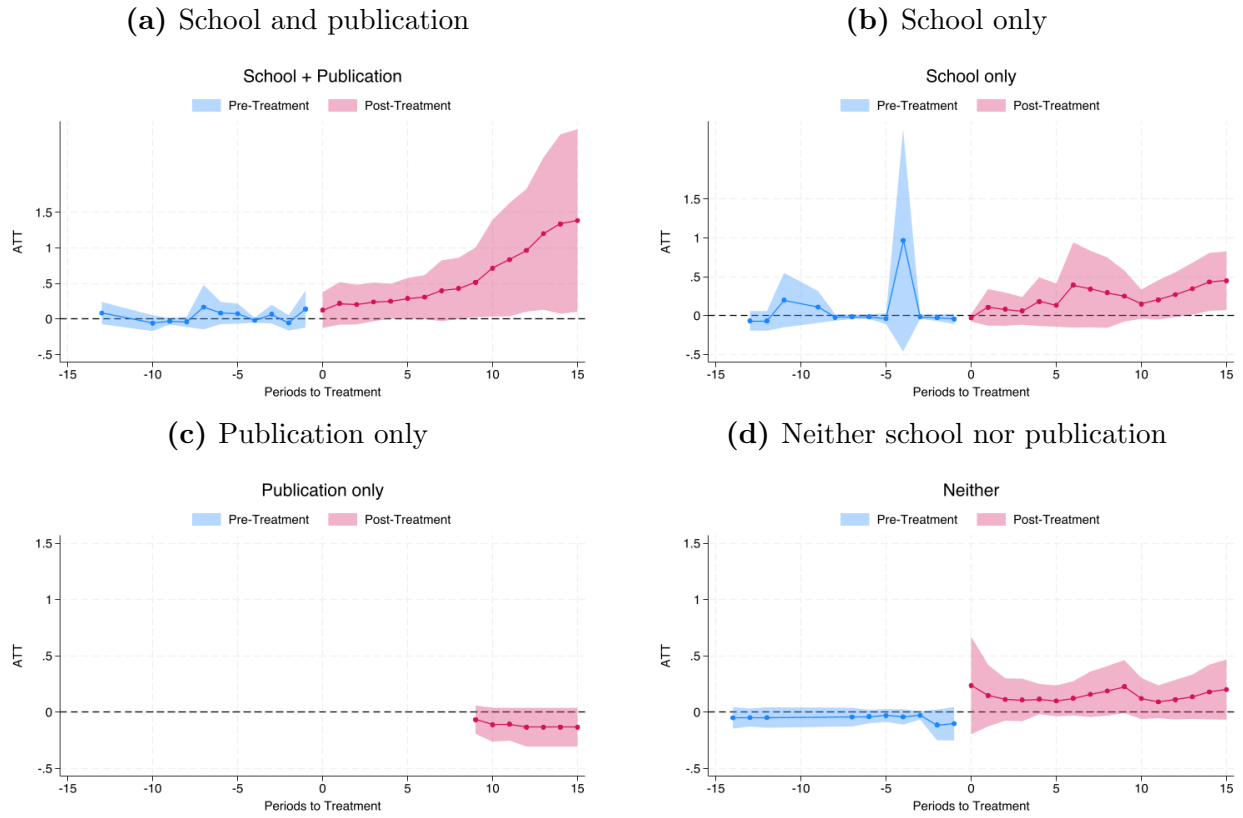
Notes: The figure reports event-study estimates from Equation 2, using the share of Christian-educated bankers as the outcome. The average post-treatment ATT is 0.012. The dependent-variable mean is 0.016 overall, 0.032 in 1920, and 0.066 in 1936.

Figure E6: Missionary Presence on Modern Banks: Protestant and Catholic (with controls)



Notes: The figure reports event-study estimates from Equation 2. Treatment is defined by the arrival of the first Protestant or Catholic mission institution in the prefecture. Each panel controls for time-varying exposure to the other missionary tradition.

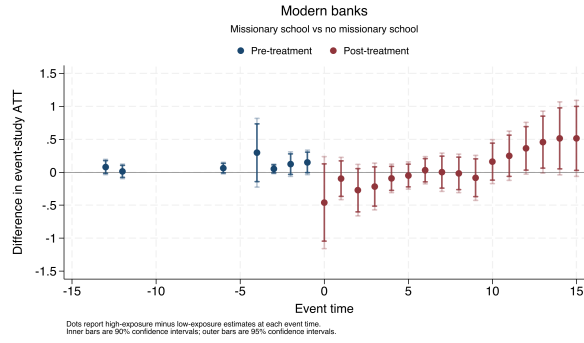
Figure E7: School and Publication Cells



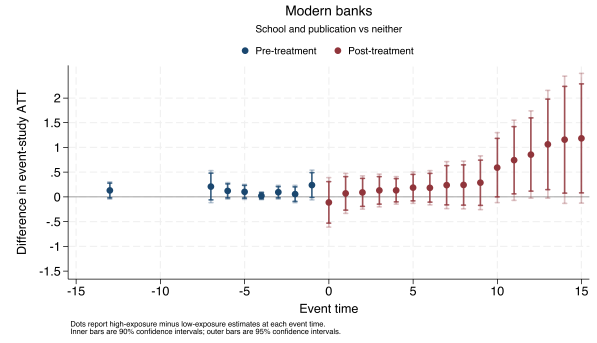
Notes: The figure splits treated prefectures into four cells according to whether missionary exposure included a school and whether missionary publication activity is recorded. Never-treated prefectures are included as controls in each panel.

Figure E8: Coefficient-by-Coefficient Heterogeneity Differences

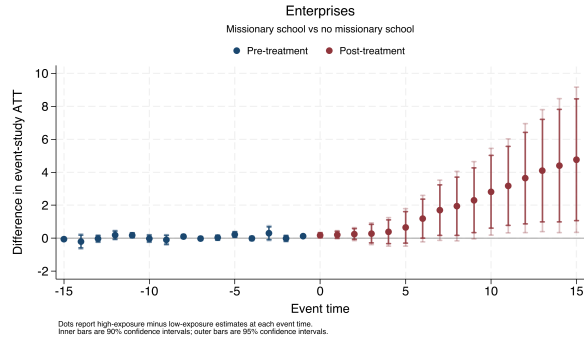
(a) Modern banks: school vs no school



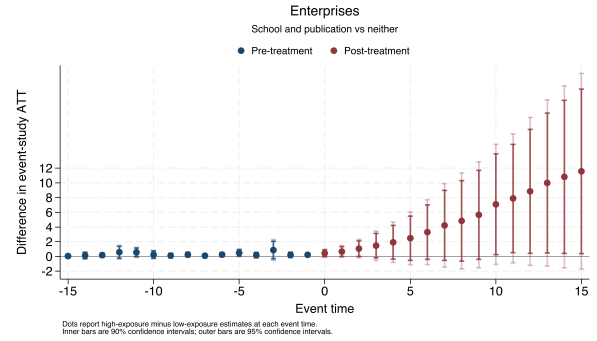
(b) Modern banks: school and publication vs neither



(c) Enterprises: school vs no school



(d) Enterprises: school and publication vs neither



Notes: Each dot is the high-exposure event-study estimate minus the low-exposure estimate at the same event time. Confidence intervals use conservative standard errors computed from the two group-specific standard errors. The bank differences are small early and become more positive at long horizons. The enterprise differences grow more strongly after treatment, consistent with the interpretation that education and school-publication complementarities matter especially for downstream market outcomes.

Table E3: Average Heterogeneity Differences

Outcome	High-exposure group	Low-exposure group	Pre diff.	Post diff.
Modern banks	Missionary school	No missionary school	0.121**	0.063
Modern banks	School and publication	Neither	0.172***	0.441
Enterprises	Missionary school	No missionary school	0.046	1.998**
Enterprises	School and publication	Neither	0.298	5.141
Trade-likely enterprises	Missionary school	No missionary school	-0.320***	0.679*
Domestic-likely enterprises	Missionary school	No missionary school	0.015	0.006

Notes: The table reports high-exposure minus low-exposure differences in average pre- and post-treatment event-study estimates. Standard errors for the differences are computed as the square root of the sum of the two group-specific variances; this is conservative because the groups share the never-treated comparison units. Stars denote two-sided significance using this conservative standard error. The table is intended to summarize magnitudes, so it should be read together with the event-time plots in Figure E8. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Compradors and Foreign-Bank Intermediaries

This section reports descriptive evidence on a possible intermediary-network channel. Compradors were Chinese intermediaries who worked with foreign firms and banks. They could therefore serve as bridges between foreign financial practices and Chinese financial institutions. The evidence in this section is descriptive and does not enter the baseline causal analysis.

Our data contain comprador information at two levels. First, the foreign-enterprise data include a firm-level flag for whether a foreign firm involved a recorded comprador. Second, the banker biography data include an individual-level flag for whether a Chinese banker had a comprador background. The individual-level data are especially useful because they can be linked to the same banker biographies used in the Christian-banker analysis.

Table E4: Comprador Data and Descriptive Patterns

Measure	Value
Cleaned foreign-firm comprador records, relative to raw flags	76.5%
Bankers flagged as comprador-related, Sheet 1	4.1%
Share Christianity-related among comprador-flagged bankers	28%
Share Christianity-related among non-comprador bankers	15%
Comprador-flagged bankers who founded a Chinese bank, <i>Qianzhuang</i> , <i>Yinhao</i> , or <i>Piaohao</i>	63%
Comprador-flagged bankers who worked at both foreign and Chinese banks	72%

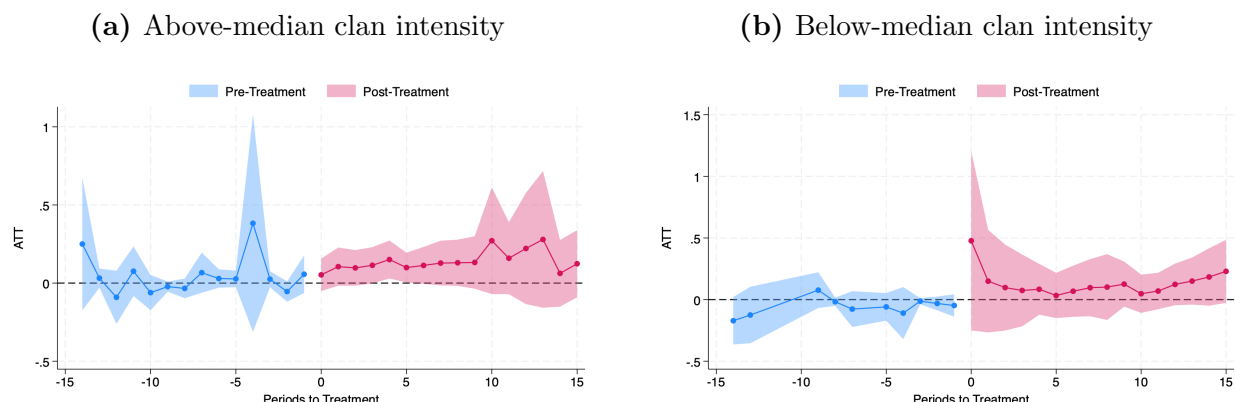
Notes: The table summarizes descriptive evidence from the foreign-enterprise data and the banker biography data. The foreign-firm row reports the cleaned comprador records as a share of raw comprador flags (202 of 264). The banker rows use Sheet 1 of *Bankers CV.xlsx*, which contains 1,041 bankers; 43 are flagged as comprador-related. A separate supplemental sheet contains a small number of additional comprador-flagged bankers but is not used for the percentages reported here. A banker is coded as Christianity-related if the biography indicates Christian schooling, Christian family background, Christian affiliation, or work for a Christian organization. The founder measure is based on a keyword scan of the biography text for founding-action verbs near banking-institution keywords. These statistics are descriptive and are not interpreted causally.

Several patterns are consistent with the idea that compradors could act as intermediaries between foreign and Chinese finance. First, Christianity-related bankers are more common among comprador-background bankers than among other bankers. Second, most comprador-background bankers worked at both foreign and Chinese banks. Third, a large share later founded or invested in Chinese financial institutions. This suggests that compradors may have transmitted institution-specific knowledge from foreign firms and banks into Chinese banking, complementing the education and publication channels emphasized in the main text.

Clan Institutions as an Alternative Explanation

One alternative explanation is that missionary activity weakened kin-based institutions, which then created space for modern banks. This is relevant because Confucian clan institutions and financial markets may operate as substitutes (Chen et al., 2022b). Missionary activity may also weaken kin-based institutions more generally (Henrich, 2020; Schulz et al., 2019). The exercises below suggest that the bank effect is not simply a clan-substitution result.

Figure E9: Missionary Presence and Modern Banks by Clan Intensity



Notes: The figure reports historical DiD estimates for the effect of missionary presence on the total number of modern banks, separately for prefectures above and below the median of clan intensity among prefectures with positive clan records. Never-treated prefectures are included as controls in both panels.

Table E5: Missionary Exposure and Clan Intensity: Horse Race

VARIABLES	(1) Modern banks in 1936	(2) Modern banks in 1936	(3) Modern banks in 1936	(4) Modern banks in 1936
Missionary Count	0.252*** (0.063)	0.245*** (0.065)	1.634* (0.917)	1.678* (0.952)
Clan Intensity (per 100 books)		0.315 (0.261)		-0.695 (1.080)
Estimator	OLS	OLS	2SLS	2SLS
Clan Control	No	Yes	No	Yes
Railway Controls	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Obs.	247	247	247	247
R sq.	0.452	0.453	-0.423	-0.472
Mean Dep. Var.	5.850	5.850	5.850	5.850
S.D. Dep. Var.	13.59	13.59	13.59	13.59

Notes: The table reports cross-sectional regressions for modern banks in 1936. Missionary exposure is the total number of missionary institutions. Clan intensity is measured using clan records per 100 genealogy books. The IV columns use the railway-based missionary exposure instrument introduced in Section 4.2. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table E6: Missionary Exposure and Clan Intensity per Capita: Horse Race

VARIABLES	(1) Modern banks in 1936	(2) Modern banks in 1936	(3) Modern banks in 1936	(4) Modern banks in 1936
Missionary Count	0.252*** (0.063)	0.255*** (0.065)	1.634* (0.917)	1.675* (0.932)
Clan per 10,000 People		-1.029 (1.742)		-5.634 (5.134)
Estimator	OLS	OLS	2SLS	2SLS
Clan Control	No	Yes	No	Yes
Railway Controls	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Obs.	247	247	247	247
R sq.	0.452	0.452	-0.423	-0.463
Mean Dep. Var.	5.850	5.850	5.850	5.850
S.D. Dep. Var.	13.59	13.59	13.59	13.59

Notes: The table reports cross-sectional regressions for modern banks in 1936. Missionary exposure is the total number of missionary institutions. Clan intensity is measured per 10,000 people. The IV columns use the railway-based missionary exposure instrument introduced in Section 4.2. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Examples from Historical Narratives

- Zifeng Lin, the founder of Kiawah Savings Bank (It's now known as China CITIC Bank International Limited). He studied at the Christian-founded Baptist Jieyang Truth High School, Shantou Queshi High School and the Peking Union Medical College. He became a Christian at the age of 14. He was recruited because of his knowledge of English and the West and was sent to Hong Kong to help develop the local market for the Rice Company. He then recognized the opportunity to expand his banking business.
- The secondary school curriculum of the Tengchow College in Shandong recorded that in the sixth year of students' enrolment, the school would offer a course on the "policy of enriching the country", which taught Western economics. Calvin Wilson Mateer, founder of Tengchow College, argues that the Chinese are eager to learn what makes the West so powerful in their quest to make China rich and strong, and that studying economics is necessary.
- Yuan Cudeng (1874 - 1954), formerly known as Xianan, a native of Yin County, Zhejiang Province, affiliated with the Anglican Church and the YMCA, was a famous entrepreneur in Shanghai's commerce and industry. When he was young, his mother worked as a maid in the house of the British missionary Hai He De in the Ningbo Christian Association (the predecessor of the Methodist Church). Mrs. Hai sent Xian An to the Kai Ming Street Fidelity School, which was run by the Methodist Church, to study free of charge. Two years later, she also sent him to Fidei Secondary School on the north bank of the river, where he was baptized as a believer in Christ.
- Kung Hsiang-hsi, a famous banker and politician in the Republic of China, suffered from mumps in 1889, and was cured by Ding Jiali's Renjiu Hospital in Taigu. 1890, Kung was enrolled in the Chinese-American Public School run by the church, and after graduation, he went to the Luhe Academy run by the American Congregationalist Church of North China in Tongzhou, Zhili, and was baptized as a Christian in 1894.
- Feng Yuxiang, a Christian general and one of the leaders of the Kuomintang (Nationalist Party) in the Republic of China era, began to believe in Christianity in the second half of March 1918 when he was stationed at Changde with his troops, and under the influence of the doctor who treated him, he often went to church to listen to sermons by the pastor. In April 1925, he founded the Northwest Bank in Zhangjiakou, Hebei Province.

Figure E10: Zifeng Lin



- Kesheng Xu, born in Shanghai to a Supervisory Board Christian family. In his early years of business in Shanghai, he recognized the market potential of the emerging insurance industry and participated in the founding of Zhaotai Water and Fire Insurance Company, one of the earliest private insurance companies in China.
- Charles Sung, the father-in-law of Dr. Sun Yat-sen, President of the Republic of China, is engaged in church-related businesses, having established the Meihua Bookstore, specializing in the printing of printed Bibles, and a flour mill, among others.
- -Translate: Xiong (2010) argues that missionaries were the main protagonists in the spread of Western knowledge to China. Before the Opium War, missionaries are the only one who translated and printed science-related readings.
- -Newspapers: From 1880, Missionaries in China pursued religious goals through secular endeavors (Wang, 1997 p.18). Publication of newspapers and magazines and establishment of schools for the transformation of Chinese culture and the spread of Christian civilization (Wang, 1997 p.18)

Mechanism 2: Missionaries as Pioneers

Table E7: Effect of Total Missionary Institutions on Export Transactions in 1936

	IV	OLS
	(1)	(2)
Variables	Export Transactions	Export Transactions
Tot Miss Inst	0.865* (0.496)	0.289*** (0.0893)
RR Duration	0.00212 (0.146)	0.185** (0.0719)
RR Density	-595.0 (637.4)	-617.4 (491.6)
Province FE	Yes	Yes
Observations	231	231
Adj. R-squared	-0.214	0.385
Mean Dep. Var.	16.07	16.07
S.D. Dep. Var.	33.66	33.66
K-P F-Stat	11.68	NA

Notes: OLS and IV regression from the 2SLS of Equation 8, introduced in Section 4.2. Outcome: total export transactions, explanatory variables: total missionary institutions, length of connection to the railway, and railway density.

- Example of Hunan province. During Qing dynasty, Hunan is one of China's relatively closed inland provinces. According to Hunan Gazetteer and records from missionaries, the Hunanese are more xenophobic and resistant to foreigners entering the province. Griffith John, a missionary of the London Society, and B.H. Alexander, a British minister of the American Missionary Church, were the first foreigners to enter Changsha. B.H. Alexander went into the city to sell books and preach during the day and stayed on the ship at night, persisting in his activities for half a year. After years of this activity, the doors of Changsha were gradually opened for the cause of Christ.
- Example of Yunan Province. Missionaries were the first foreigners to enter Yunnan. In 1888, Samuel Pollard, a Methodist missionary from the United Kingdom, made the long and arduous journey overland as well as on foot into Yunnan, a landlocked region of southwestern China. Pollard not only succeeded in converting the local Miao people to Christianity, but also built a series of hospitals, schools, and other public facilities in the area, and managed to improve local customs and introduce various

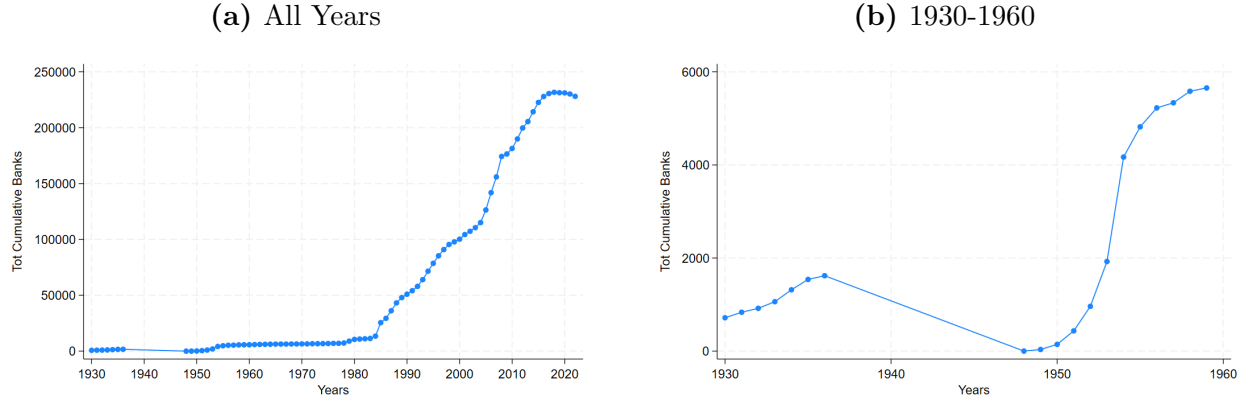
Figure E11: Yunan Province Example



sports and cultural activities, making the remote Miao village of Shimenkan one of the most advanced and civilized areas in southwestern China at one time.

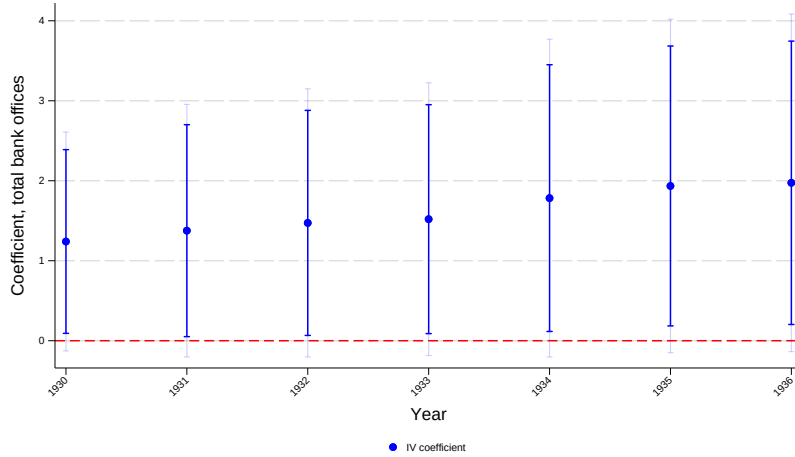
Appendix F: Long-Term Analysis

Figure F1: Raw Total-Bank Dynamics, 1930–1936 and 1949–2023



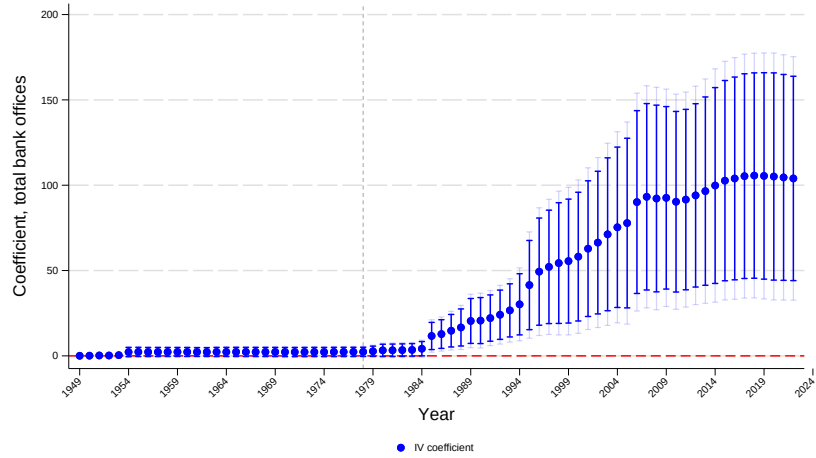
Notes: By 1936, the data record more than 2,000 modern banks. Following the closures, mergers, and nationalizations around 1949, the number of banks rises again; by 1960, it is approximately three times the 1936 count. The series contains no observations from 1937 to 1948.

Figure F2: 2SLS Estimates: Total Modern Banks, 1930–1936



Notes: The figure reports coefficients from separate year-specific regressions based on Equation 8, using the historical modern-bank data from the difference-in-differences design. The outcome and all IV-side variables use the 1st–99th percentile convention. The average post-treatment ATT in the historical design is 0.22. The dependent-variable means are 2.4 in 1930 and 5.8 in 1936. The IV estimates correspond to increases of approximately 51% and 34%, respectively. Dark and light intervals show 90% and 95% confidence intervals.

Figure F3: 2SLS Estimates: Total Banks, 1949–2022



Notes: The figure reports coefficients from separate year-specific regressions based on Equation 8. Modern county- and district-level bank counts are assigned to 1911 prefectures by each modern unit's centroid, and the annual outcome and all IV-side variables use the 1st–99th percentile convention. Mean total-bank counts are 145 in 1990 and 707 in 2022. The 1990 and 2022 estimates correspond to increases of approximately 14.2% and 14.7%, respectively. Dark and light intervals show 90% and 95% confidence intervals. Source: China Financial Supervision and Administration Bureau.

Table F1: Headline IV Estimates

	(1) Std. modern banks 1936	(2) Std. total banks 2020	(3) GDP 2019
Missionary institutions	0.152* (0.083)	0.145*** (0.051)	7,581*** (2,745)
RR Duration	-0.0125 (0.0095)	-0.0046 (0.0065)	-304.1 (330.5)
RR Density	105.5** (48.0)	47.50 (32.14)	1,972,945 (1,574,214)
Province FE	Yes	Yes	Yes
Observations	247	247	242
K–P F-stat	14.87	14.87	15.01
Mean dep. var.	0.000	0.000	46,051
SD dep. var.	1.000	1.000	52,206

Notes: The table reports 2SLS estimates from Equation 8. Treatment is total missionary institutions, instrumented with predicted missionary exposure from the railway-flow instrument. Column (1) uses standardized modern banks in 1936, the last pre-war year in the historical panel. Column (2) uses standardized 2020 bank counts constructed by assigning modern counties and districts to 1911 prefectures by their centroids. Column (3) uses 2019 GDP in millions of 2017 US dollars, constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons. The mission stock, instrument, railway controls, and outcomes use the 1st–99th percentile convention. RR Duration is years connected to the railway network as of 2000. RR Density is cumulative railway stock divided by prefecture area. Robust standard errors are in parentheses. K–P F-stat is the Kleibergen–Paap first-stage F-statistic. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Population Controls and Per-Capita Outcomes

Population adjustment produces a similar long-run bank result. Table F2 adds two alternative population measures to both stages of the IV specification or uses population to construct banks-per-capita outcomes. The estimated effect remains positive across all four specifications. Our preferred population-control and per-capita checks use the same county-centroid Census 2015 scale, and Figure F4 shows the corresponding year-by-year per-capita estimates. The post-reform coefficients remain positive and are statistically distinguishable from zero in recent years.

These checks are not the baseline specifications because population need not be predetermined with respect to missionary exposure. Missions and the institutions that followed them may have affected subsequent population growth, migration, or urban concentration. Controlling for later population can therefore condition on a potential outcome of treatment, while dividing by it changes the estimand. We report both approaches as robustness checks rather than treating population adjustment as conceptually neutral.

Table F2: Long-Run Bank Results with Population Controls and Per-Capita Outcomes

	Population controls		Per-capita outcomes	
	1880 population	Census 2015 county-centroid scale	1880 denominator	Census 2015 denominator
Missionary institutions	0.146** (0.069)	0.143* (0.085)	0.042 (0.027)	0.052** (0.024)
Outcome	Standardized 2020 bank presence		Standardized banks per million	
Province fixed effects	Yes	Yes	Yes	Yes
Railway controls	Yes	Yes	Yes	Yes
Observations	247	231	247	231
First-stage F-statistic	8.59	4.18	14.87	15.67

Notes: The table reports IV estimates from Equation 8. The 2020 bank outcome assigns modern county- and district-level bank counts to historical prefectures according to the modern unit's centroid. The mission stock, instrument, railway controls, and bank outcomes use the 1st–99th percentile convention. The first two specifications add the indicated log population measure or scale to both stages. The last two use bank counts per million denominator units and standardize the resulting outcome. The 1880 measure is the historical baseline population. The Census 2015 measure is constructed by summing normalized person weights within modern counties and districts and assigning each modern unit to its historical prefecture by its centroid; it is therefore a weighted relative population scale rather than a literal full-population count. All specifications include province fixed effects and the railway controls used in the baseline IV design. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table F3 applies the same four-specification structure to the direct-grid GDP outcome in 2019. The estimated effect is positive under every population adjustment. It is significant at the 5% level when 1880 population is added as a control and when GDP per capita uses either

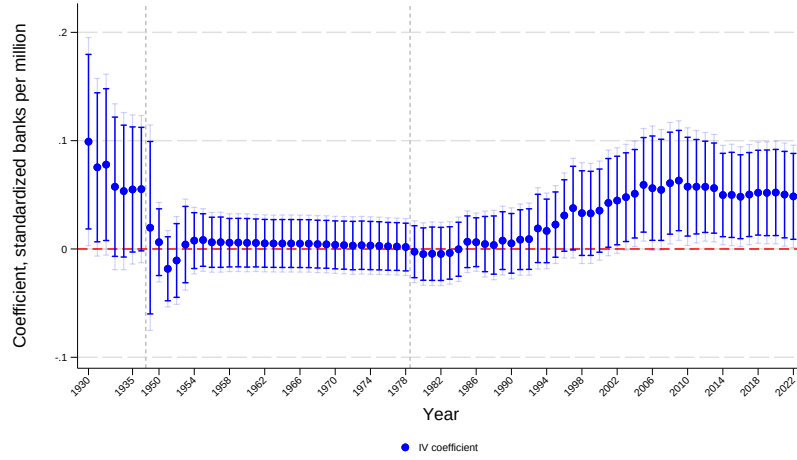
denominator, and at the 1% level when GDP per capita uses the Census 2015 denominator.

Table F3: Long-Run GDP Results with Population Controls and Per-Capita Outcomes

	Population controls		Per-capita outcomes	
	1880 population	Census 2015 county-centroid scale	1880 denominator	Census 2015 denominator
Missionary institutions	8,358** (3,922)	8,190 (5,049)	0.055** (0.025)	0.070*** (0.024)
Outcome	GDP, 2019 (millions of 2017 USD)		Standardized GDP per million	
Province fixed effects	Yes	Yes	Yes	Yes
Railway controls	Yes	Yes	Yes	Yes
Observations	242	231	242	231
First-stage F-statistic	8.64	4.18	15.01	15.67

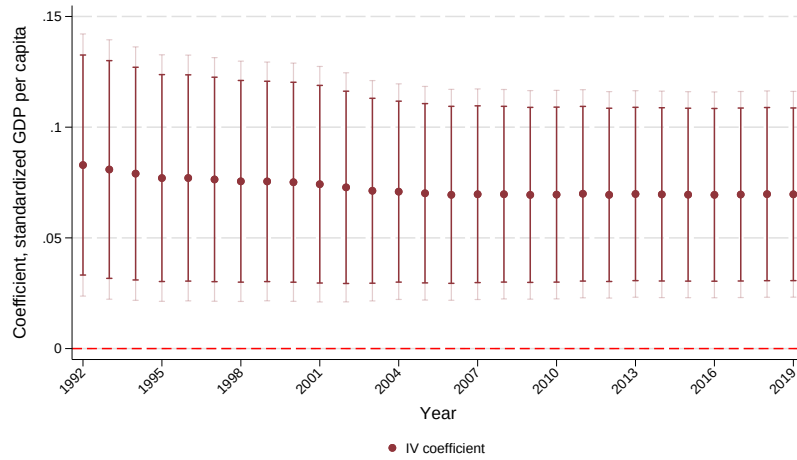
Notes: The table reports IV estimates from Equation 8. The 2019 GDP outcome sums 1-km grid-cell GDP directly within accepted 1911 prefecture polygons. The mission stock, instrument, railway controls, and GDP outcome use the 1st–99th percentile convention. The first two specifications add the indicated log population measure or scale to both stages. The last two express GDP per million denominator units and standardize the resulting outcome. The 1880 measure is the historical baseline population. The Census 2015 measure is constructed by summing normalized person weights within modern counties and districts and assigning each modern unit to its historical prefecture by its centroid; it is therefore a weighted relative population scale rather than a literal full-population count. All specifications include province fixed effects and the railway controls used in the baseline IV design. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Figure F4: 2SLS Estimates: Banks Per Capita, 1930–1936 and 1949–2022



Notes: The figure reports year-specific estimates from Equation 8. Modern county- and district-level bank counts are assigned to 1911 prefectures by each modern unit's centroid. The outcome is bank counts per million county-centroid Census 2015 scale units, winsorized at the 1st and 99th percentiles and standardized within year; the same Census 2015 denominator is used in every year. Blue points are IV estimates, and dark and light intervals show 90% and 95% confidence intervals. The red dashed horizontal line marks zero, and gray vertical dashed lines mark 1949 and 1978. The coefficient is 0.052 in 2020 ($p = 0.032$) and 0.049 in 2022 ($p = 0.045$); both estimates use 231 historical prefectures.

Figure F5: 2SLS Estimates: GDP Per Capita, 1992–2019 (Fixed Census 2015 Denominator)



Notes: The figure reports year-specific estimates from Equation 8. GDP is constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons and dividing by the same county-centroid Census 2015 population scale in every year. The ratio is winsorized at the 1st and 99th percentiles and standardized within year. Maroon points and dark/light intervals show the IV estimates and 90%/95% confidence intervals; the red dashed horizontal line marks zero. All regressions use the 231-prefecture Census 2015 sample, province fixed effects, and the baseline railway controls.

Industrial Enterprises and External Finance Dependence

We complement the bank and GDP results with industrial-enterprise outcomes from the Annual Survey of Industrial Production (ASIP). The outcome is the number of industrial enterprises observed in the survey, not sales, employment, output, borrowing, or the universe of registered firms. We report all ASIP enterprises and private ASIP enterprises, where private enterprises are identified using the survey’s paid-in-capital ownership classification. The yearly outcomes are expressed per million residents and standardized within year before estimating Equation 8.

To study whether the enterprise response is related to financial intermediation, we construct an industry-level measure of external finance dependence following the logic of [Rajan and Zingales \(1998\)](#). For firm i in three-digit industry j , we first calculate

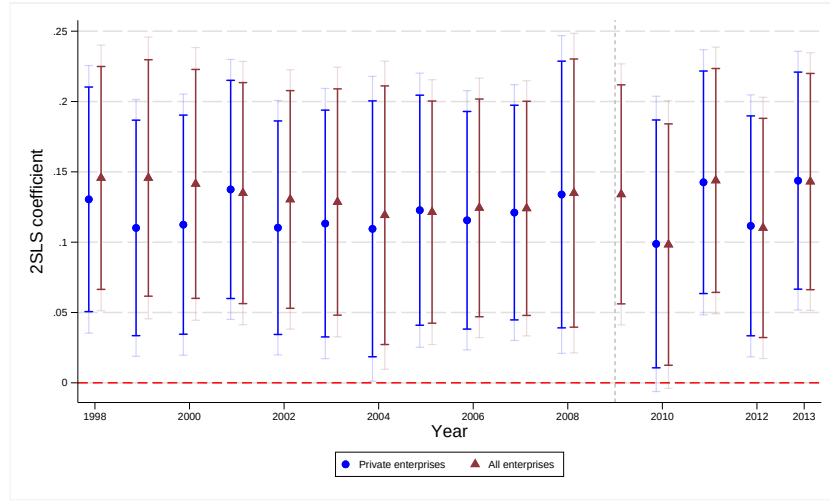
$$\text{EFD}_{it} = \frac{\text{Investment}_{it} - \text{Operating Cash Flow}_{it}}{\text{Investment}_{it}}$$

over 2004–2006, requiring positive investment. Investment is the ASIP real-investment field. Operating cash flow is the direct operating-cash-flow field when available and operating cash inflows minus outflows otherwise. We then take the median of this ratio across firm-year observations within each three-digit industry. “High finance dependence” denotes industries at or above the median of these industry values; “low finance dependence” denotes industries below the median. This classification is therefore a predetermined industry characteristic measuring the need for outside finance, not a firm-level measure of realized borrowing.

We assign geocoded ASIP firm locations directly to accepted 1911 prefecture polygons. For records without a usable direct match, we use the modern county or district centroid and then an area-weighted modern-city crosswalk as a final fallback. Firm counts are aggregated to historical prefecture-year cells before the per-capita transformation, 1st–99th percentile winsorization, and within-year standardization.

Together, the enterprise results indicate that the long-run effect is not limited to the number of banks. Missionary-exposed prefectures have more later industrial activity, including private industrial enterprises. The high-minus-low finance-dependence outcome is also positive, although the independently standardized high- and low-dependence coefficients are similar in magnitude. We interpret this as complementary evidence consistent with a finance channel, while recognizing that ASIP covers surveyed industrial enterprises above the applicable size threshold.

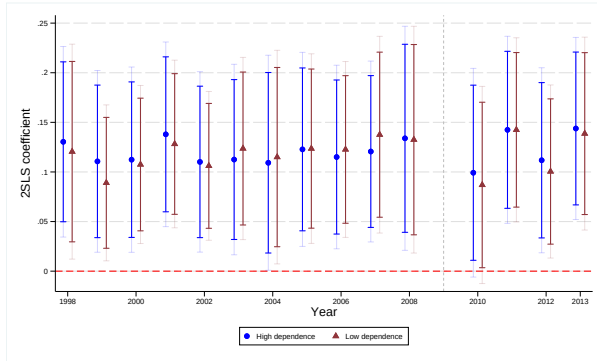
Figure F6: Historical Missionary Exposure and Industrial Enterprises



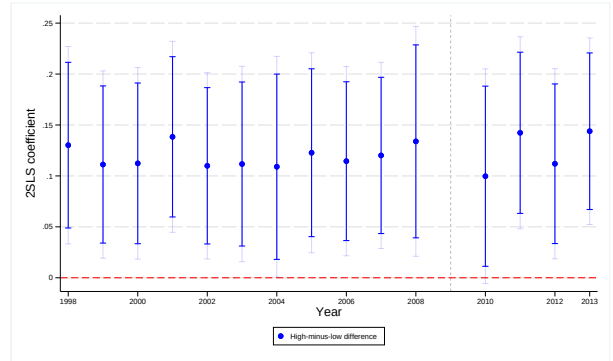
Notes: The figure reports year-specific 2SLS estimates from Equation 8. Geocoded ASIP firms are assigned directly to accepted 1911 prefectures, with county-centroid and area-weighted city fallbacks when a direct match is unavailable. Outcomes are private and all ASIP industrial enterprises per million residents, winsorized at the 1st and 99th percentiles and standardized within year. Dark and light intervals show 90% and 95% confidence intervals. In the 2007–2013 period-average specification, the coefficient is 0.128 standard deviations for private enterprises and 0.129 for all enterprises. The corresponding raw private-enterprise estimate is approximately 40 firms per million residents, or 15.8% of the sample mean. The private-enterprise outcome is omitted in 2009 because the ASIP ownership field mechanically assigns every observation to the same category in that year; the all-enterprise outcome remains defined.

Figure F7: Historical Missionary Exposure and Private Enterprises by External Finance Dependence

(a) High- and low-dependence industries



(b) High-minus-low difference



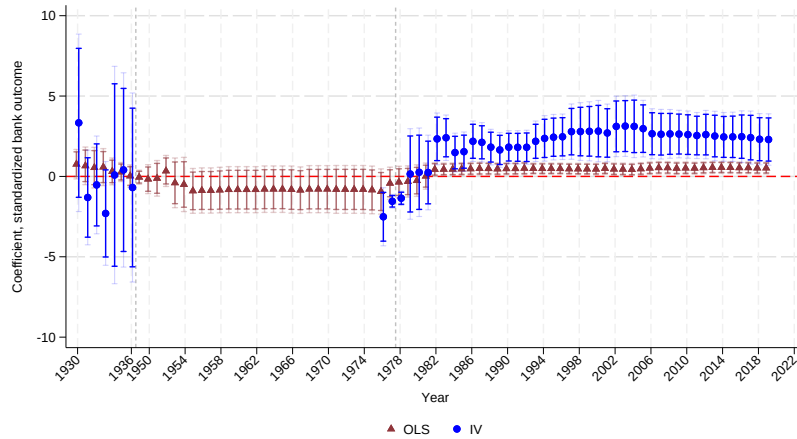
Notes: The figure reports year-specific 2SLS estimates from Equation 8. Geocoded ASIP firms are assigned directly to accepted 1911 prefectures, with county-centroid and area-weighted city fallbacks when a direct match is unavailable. The outcomes are private ASIP industrial enterprises per million residents in high- and low-external-finance-dependence industries, winsorized at the 1st and 99th percentiles and standardized within year. Panel B directly estimates the high-minus-low difference. Dark and light intervals show 90% and 95% confidence intervals. In the 2007–2013 period-average specification, the high-minus-low estimate is 0.128 standard deviations; the corresponding raw estimate is approximately 37 private enterprises per million residents, or 15.8% of the sample mean of the difference outcome. The private-enterprise outcome is omitted in 2009 because of the ASIP ownership-coding anomaly described in Figure F6.

Long-Run Mechanism Evidence

This section reports the detailed evidence summarized in Section 6. The exercises use several units of observation and therefore cannot be reduced to one identical regression. Wherever the unit is a prefecture, we retain the baseline long-run specification: railway controls and province fixed effects in OLS, with historical missionary exposure instrumented by predicted missionary exposure in IV. Individual-level exercises add the fixed effects and controls appropriate to the underlying biographies or surveys. Shaanxi exercises use counties nested in 1911 prefectures, absorb historical-prefecture fixed effects, and instrument pre-1950 mission presence with predicted missionary exposure. We report these results as a structured set of mechanism checks rather than as independent causal estimates.

Shaanxi County-Level Long-Run Replication

Figure F8: Historical Missionary Exposure and Long-Run Financial Institutions in Shaanxi



Notes: The figure reports year-specific OLS and IV estimates for standardized modern banks or financial institutions in Shaanxi counties. Blue circles show IV estimates and maroon triangles show OLS estimates; light and dark intervals are 95% and 90% confidence intervals. OLS includes 1911 historical-prefecture fixed effects. IV instruments pre-1950 mission presence with predicted missionary exposure and adds railway duration and cumulative railway-area exposure. Early post-1949 financial-institution coverage is sparse, so Mao-period IV estimates are generally not estimable.

Table F4: Shaanxi County-Level Bank Estimates by Period

Period	OLS	OLS SE	IV	IV SE	First-stage F	Obs.
1930–1936	0.452	0.407	-0.144	1.966	15.88	30
1949–1976	-0.763	0.665	—	—	—	—
1978–2022	0.430**	0.165	2.047***	0.671	14.93	87

Notes: The outcome is the standardized number of modern banks or financial institutions, averaged within county over the indicated period. Specifications follow Figure F8. The Mao-period IV is not estimable because early post-1949 bank coverage is too sparse after fixed effects and railway controls. The historical county-bank series records cumulative establishments rather than a complete active-stock panel. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Returning Talent and Elite Networks

Table F5: Mission-Exposed Origins and the Technical Diaspora in Taiwan

Origin-place exposure	Any Taiwan record		Observed in Taiwan, 1974 or later	
	(1)	(2)	(3)	(4)
Any mission institution by 1880	2.73* (1.65)		2.45** (1.18)	
Any mission school by 1900		6.23* (3.33)		5.36** (2.37)
Native-province FE	Yes	Yes	Yes	Yes
Historical controls	Yes	Yes	Yes	Yes
Observations	11,657	11,657	11,657	11,657
Origin-prefecture clusters	205	205	205	205
Mean dependent variable	5.1%	5.1%	3.0%	3.0%

Notes: Each observation is an engineer/scientist origin record. Coefficients are percentage-point changes from origin-prefecture OLS regressions. The 1974-or-later outcome is the stricter near-reform Taiwan measure. Historical controls include coastal location, main-river access, treaty-port and provincial-capital status, city size, log population, river length, 1820 tax per capita, *jinshi*, and official counts. Standard errors are clustered by origin prefecture. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table F7: Elite-Biography Locations and Long-Run Banks

	Standardized number of banks, 2022	
	OLS	IV
Mission count	0.027*** (0.006)	0.082** (0.033)
Mission count $\times$ elite-biography prefecture	0.018 (0.011)	0.068* (0.038)
Elite-biography prefecture	0.203* (0.115)	-0.255 (0.344)
Railway duration	0.005** (0.002)	-0.010 (0.008)
Railway density	17.334 (17.352)	50.808* (30.425)
Province FE	Yes	Yes
Observations	247	247
R-squared	0.464	
K-P F-statistic		6.65

Notes: The unit is a 1911 prefecture. Elite-biography prefecture equals one if the prefecture is represented by at least one Republican-era elite biography by 1936. The IV uses predicted historical mission count and its interaction with the elite-biography indicator as instruments. Because the interaction specification has a K-P F-statistic below 10, the IV heterogeneity estimate should be interpreted cautiously. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Commercial Persistence and Foreign Investment

Table F8: Historical Channels and Long-Run Banks: Sequential Horse Race

	Baseline	Publications	Education	Institutions	Trade/open	Market
Mission count	0.045*** (0.009)	0.032*** (0.005)	0.030*** (0.005)	0.027*** (0.010)	0.025*** (0.006)	0.017*** (0.007)
Share absorbed	—	27.6%	32.5%	38.7%	44.0%	61.1%
Railway controls	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	247	247	247	247	247	247

Notes: The unit is a 1911 prefecture and the outcome is the standardized number of banks in 2022. Each column adds one block to the baseline OLS. Publications include publication counts and mission-press exposure. Education includes mission education and higher-education institutions. Institutions include *Piaohao* stock, historical deposits, and historical bank size. Trade/open includes trade instances, treaty ports, concessions, and foreign-trade enterprises. Market includes historical banks and enterprises. The exercise is descriptive because these controls may themselves be outcomes of missionary exposure. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table F9: Historical Missionary Exposure and Foreign Direct Investment

	Log FDI, 2017		Log FDI per capita, 2017	
	OLS	IV	OLS	IV
Mission count	0.066*** (0.011)	0.129** (0.051)	0.018* (0.009)	0.060* (0.034)
Railway controls	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Observations	240	240	240	240
R-squared	0.569		0.483	
First-stage F-statistic		14.22		14.22
Mean dependent variable	9.799	9.799	4.306	4.306
SD dependent variable	2.557	2.557	1.727	1.727

Notes: The unit is a 1911 prefecture. Modern-city FDI is assigned to historical prefectures using area-overlap weights. Raw total and per-capita FDI are winsorized at the 1st and 99th percentiles before applying the reported $\log(1 + x)$ transformations. Mission count is the historical number of missionary institutions. IV instruments mission count with predicted missionary exposure; the mission stock, instrument, and railway controls use the same 1st–99th percentile convention. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Religious and Cultural Change

Table F10: Historical Missionary Exposure and Contemporary Christian Affiliation

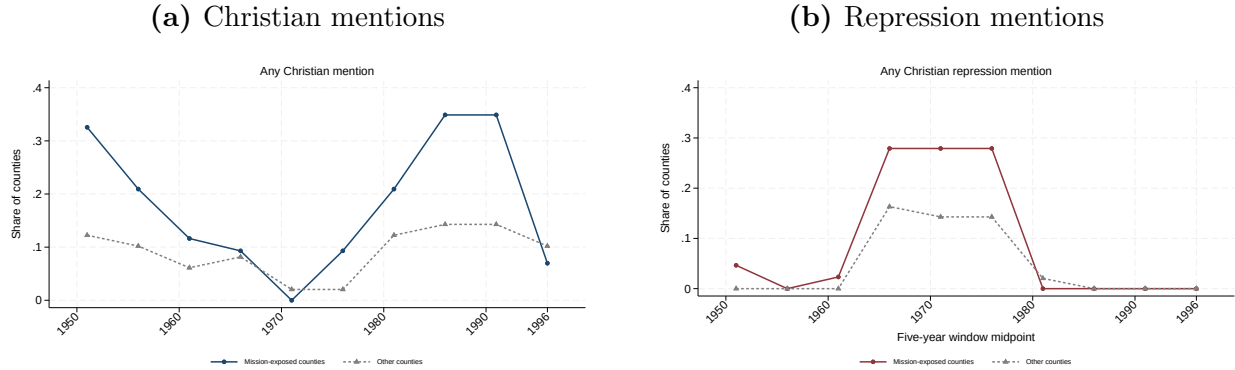
<i>Panel A. China Family Panel Studies</i>			
	Christian belief	Protestant belief	Catholic belief
Mission count	0.0002 (0.0002)	0.0005** (0.0002)	-0.0001 (0.0002)
1911 prefecture FE	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Observations	97,509	97,509	97,509
Clusters	122	122	122
Mean dependent variable	0.035	0.028	0.013
<i>Panel B. Chinese Spiritual Life Survey</i>			
	Christian belief	Protestant belief	Catholic belief
Mission count	0.0029* (0.0017)	0.0020 (0.0017)	0.0010 (0.0009)
1911 province FE	Yes	Yes	Yes
Railway controls	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Observations	5,660	5,660	5,660
Clusters	40	40	40
Mean dependent variable	0.028	0.026	0.003

Notes: The CFPS unit is a respondent-wave located to a 1911 county; specifications include 1911 prefecture fixed effects, wave fixed effects, and individual controls, with standard errors clustered by 1911 county. The Spiritual Life Survey uses the high-quality geocoded sample; specifications include 1911 province fixed effects, railway controls, and individual controls, with standard errors clustered by assigned 1911 prefecture. Mission count is the historical missionary-institution count divided by 10. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table F11: Historical Missionary Exposure and Contemporary Religiosity

	China Family Panel Studies			Chinese Spiritual Life Survey		
	Religious group member	Religion important	Religious activity frequency	Believes in God	Ever prays	Donated to religion
Mission count	0.0000 (0.0003)	0.0002 (0.0011)	-0.0011 (0.0038)	0.0026 (0.0048)	-0.0038 (0.0065)	0.0020 (0.0076)
Geographic FE	Prefecture	Prefecture	Prefecture	Province	Province	Province
Wave FE	Yes	Yes	Yes	No	No	No
Railway controls	No	No	No	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	67,883	74,561	74,885	5,362	5,660	5,706
Mean dependent variable	0.025	0.218	0.598	0.083	0.114	0.226

Notes: Sample construction, mission-count scaling, and clustering follow Table F10. The table shows little systematic association between historical missionary exposure and the intensity of contemporary religious practice. Robust standard errors are in parentheses.

Figure F10: Christianity and Repression in Shaanxi Local Histories

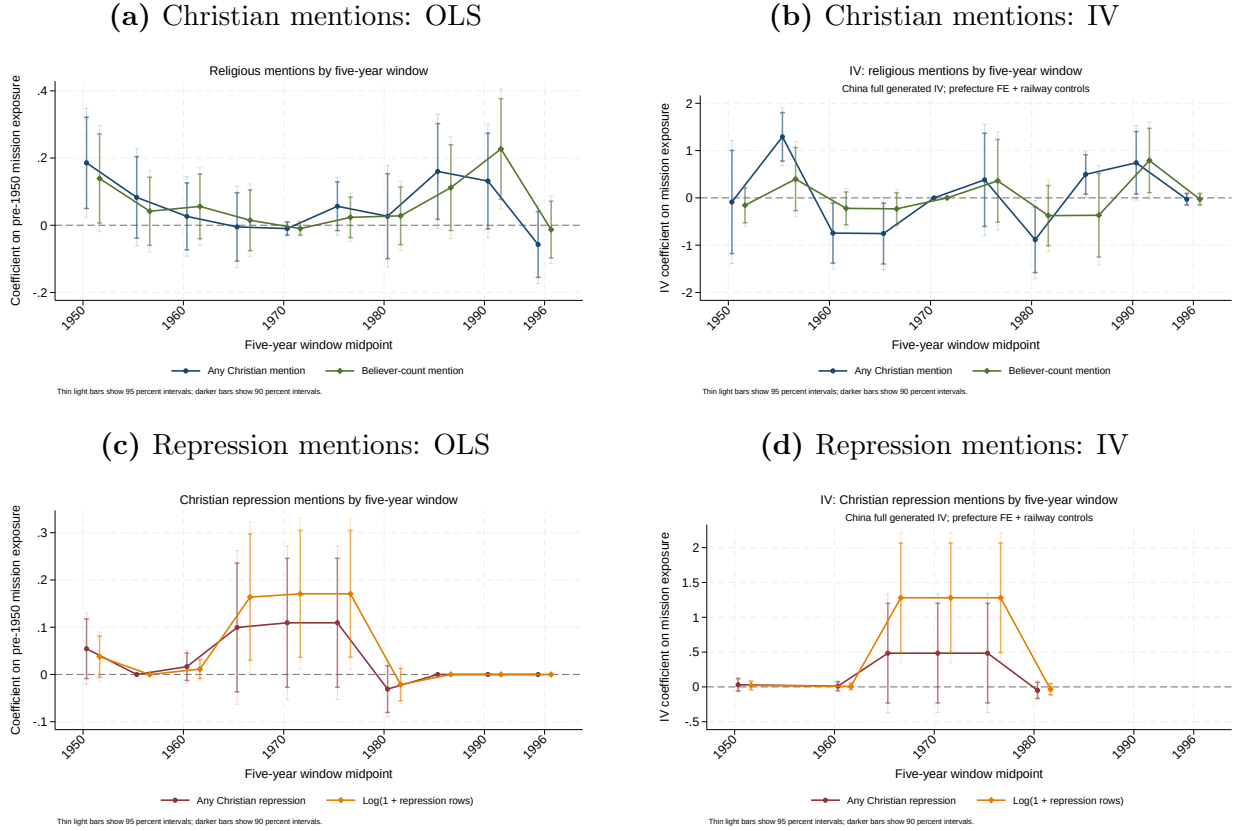
Notes: The figures plot county-window mention shares through 1998 for Shaanxi counties with and without a recorded pre-1950 missionary institution. Repression mentions are explicit local-history records of suppression, closure, cessation, persecution, or Cultural Revolution disruption involving Christian institutions or communities.

Table F12: Historical Missionary Exposure and Trust

	Trust people generally	Trust family	Trust religious groups
Mission count	0.0377** (0.0162)	0.0028 (0.0146)	0.0487 (0.0382)
1911 province FE	Yes	Yes	Yes
Railway controls	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Observations	4,389	5,753	2,472
Mean dependent variable	1.781	3.821	1.877

Notes: The data are from the high-quality geocoded Chinese Spiritual Life Survey sample. Mission count is the historical missionary-institution count divided by 10. Standard errors are clustered by assigned 1911 prefecture. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Figure F11: Dynamic Estimates for Christian and Repression Mentions in Shaanxi



Notes: Each point is estimated separately by time window. OLS specifications include 1911 historical-prefecture fixed effects. IV specifications instrument the pre-1950 mission-presence indicator with predicted missionary exposure and add railway controls. Bars show 90% and 95% confidence intervals.

Table F13: Historical Missionary Exposure and Financial Behavior

	Asset ownership			Risk attitudes and knowledge		
	Has savings	Has financial asset	Has non-deposit asset	Risk tolerance (1–4)	Willing to take investment risk	Knows investment risk
Mission count	0.0020** (0.0010)	0.0019** (0.0009)	0.0051*** (0.0014)	0.0047 (0.0031)	0.0018 (0.0015)	0.0028 (0.0024)
1911 prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	131,549	130,797	130,797	6,476	6,476	6,525
Mean dependent variable	0.732	0.770	0.215	1.993	0.576	0.856

Notes: The unit is a CFPS respondent-wave located to a 1911 county. Mission count is the historical missionary-institution count divided by 10. Standard errors are clustered by 1911 county. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Direct Persistence of Mission Education and Publications

Table F14: Mission Schools and Long-Run Banks

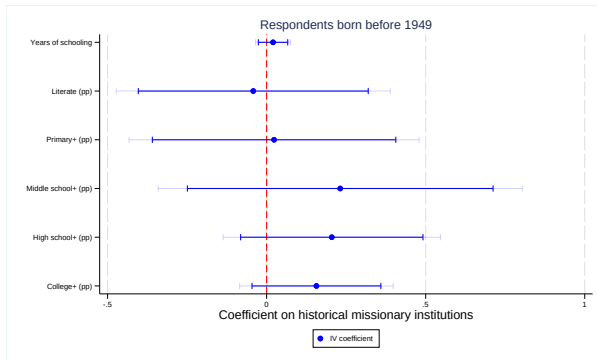
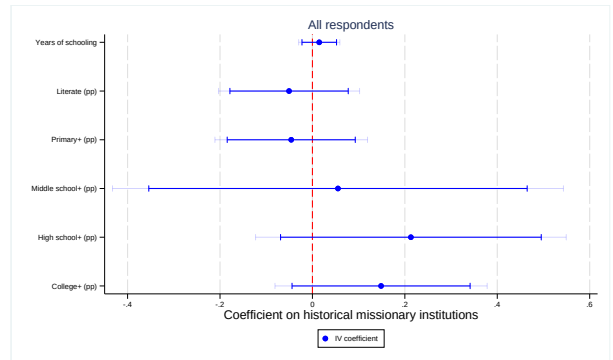
	Standardized number of banks, 2022	
	OLS	IV
Mission count	0.027*** (0.005)	0.093 (0.060)
Mission count $\times$ school prefecture	0.018 (0.014)	0.082 (0.064)
School prefecture	0.038 (0.171)	-1.312 (0.825)
Railway duration	0.004** (0.002)	-0.010 (0.008)
Railway density	18.524 (17.048)	55.597* (32.877)
Province FE	Yes	Yes
Observations	247	247
R-squared	0.467	
K-P F-statistic		4.23

Notes: The unit is a 1911 prefecture. School prefecture equals one for locations with any historical missionary school before 1937. The IV uses predicted mission count and its interaction with school-prefecture status as instruments. The interaction IV is weak, so the estimates are reported as a low-powered test rather than evidence of no effect. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table F15: Mission Publications and Long-Run Banks

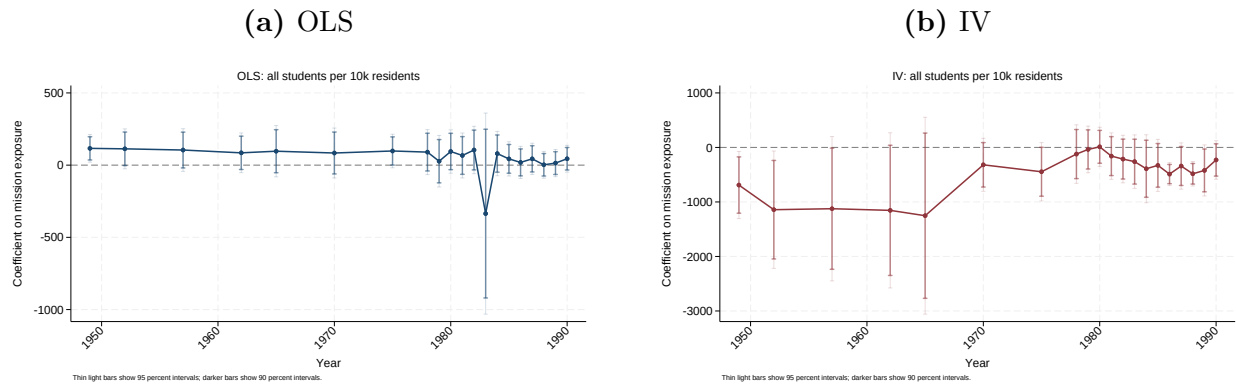
	Standardized number of banks, 2022	
	OLS	IV
Mission count	0.038*** (0.007)	0.124 (0.113)
Mission count $\times$ publication prefecture	0.012 (0.020)	0.083 (0.155)
Publication prefecture	-0.161 (0.336)	-2.510 (2.259)
Railway duration	0.004** (0.002)	-0.010 (0.009)
Railway density	18.602 (17.768)	57.500 (37.147)
Province FE	Yes	Yes
Observations	247	247
R-squared	0.461	
K-P F-statistic		1.52

Notes: The unit is a 1911 prefecture. Publication prefecture equals one for locations with any historical mission-publication record before 1937. The IV uses predicted mission count and its interaction with publication-prefecture status as instruments. The interaction IV is weak, so the estimates are reported as a low-powered test rather than evidence of no effect. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Figure F12: Historical Missionary Exposure and Education in the 2000 Census**(a) Respondents born before 1949****(b) All respondents**

Notes: The unit is a 1911 prefecture. Census counties are assigned to historical prefectures by their centroids. Each point is the IV coefficient on historical missionary institutions from a separate regression. The mission stock, instrument, and railway controls use the 1st–99th percentile convention. All specifications include province fixed effects, railway total, cumulative railway-area exposure, and robust standard errors. Dark and light bars show 90% and 95% confidence intervals.

Figure F13: Historical Missionary Exposure and Students in Shaanxi



Notes: The outcome is all students per 10,000 residents in the Shaanxi yearbook county panel. OLS absorbs 1911 historical-prefecture fixed effects. IV instruments the pre-1950 mission-presence indicator with predicted missionary exposure and adds railway controls. Bars show 90% and 95% confidence intervals.

State Capacity and Infrastructure

Table F16: Historical Missionary Exposure, State Capacity, and Market Activity in Shaanxi

<i>Panel A. Fiscal capacity</i>		
Outcome	1949–1990 IV	1978–1990 IV
Log fiscal expenditure per capita	-1.130*** (0.370)	-0.616*** (0.144)
Log fiscal revenue per capita	-0.845 (0.534)	0.082 (0.286)
Health workers per 10,000 residents	-4.813 (15.494)	-0.294 (17.961)
<i>Panel B. Investment and infrastructure</i>		
Outcome	1949–1975 IV	1978–1990 IV
Log fixed-asset investment per capita	-1.182 (1.231)	-1.509 (1.673)
Log basic-construction investment per capita	-1.167 (1.226)	-1.562 (1.623)
Log machinery power per 10,000 residents	-1.320* (0.735)	-2.327** (0.885)
Log rural electricity per capita	-1.513*** (0.365)	-0.405 (0.431)
Log health institutions per 10,000 residents	-0.385 (0.286)	-0.565* (0.334)
Log health-institution beds per 10,000 residents	-0.726 (0.941)	-0.864 (0.817)
<i>Panel C. Market-facing outcomes</i>		
Outcome	1949–1975 IV	1978–1990 IV
Log industrial enterprise units	1.516*** (0.442)	1.560*** (0.466)
Log state-owned enterprise units	0.275 (0.491)	-1.436 (1.705)
Log retail sales per capita	0.970** (0.421)	0.486*** (0.157)
Log township enterprise employment	–	1.415*** (0.485)
Log township enterprise units	–	1.022* (0.539)

Notes: The unit is a 1911 Shaanxi county. Each cell reports the IV coefficient on pre-1950 mission presence from a separate regression. Mission presence is instrumented with predicted missionary exposure. All specifications absorb 1911 historical-prefecture fixed effects and include railway duration and cumulative railway-area exposure. Panel A uses 89 observations and a first-stage F-statistic of 15.00. Panel B uses 85–88 observations in 1949–1975 and 89 observations in 1978–1990; first-stage F-statistics range from 13.00 to 15.00. Panel C uses 88 observations for the displayed Mao-period outcomes and 89 after reform. Township-enterprise outcomes are available only in the reform period. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Recentered IV with Simulated Missionary Flows

This section reports a robustness check for the exposure structure of the long-run IV. The concern is that prefectures with earlier or denser railway histories may be more exposed to any national flow process through transport networks. In this setting, average missionary flows interacted with railway access could therefore proxy for generic railway-mediated exposure

rather than the realized timing of missionary inflows. Following [Borusyak and Hull \(2023\)](#), we compute the expected instrument under simulated counterfactual missionary-flow histories and subtract that expected component from the original instrument.

Let F_t be the national missionary flow in year t , and let $R_{i,t-1}$ be prefecture i 's lagged cumulative railway stock. The original instrument is $Z_i = \sum_t \hat{\beta} F_t R_{i,t-1}$. For each simulation, we permute the national missionary-flow series across years, keep each prefecture's railway path fixed, and compute the instrument implied by this counterfactual timing. Averaging over 1,000 simulated flow histories gives μ_i , the component of the instrument mechanically predicted by the prefecture's railway exposure profile under average missionary-flow timing. The recentered instrument is $\tilde{Z}_i = Z_i - \mu_i$. We use two versions: a full-year permutation and a 10-year block permutation that preserves within-decade serial structure in missionary flows.

Table F17: Recentered IV with Simulated Missionary Flows

Outcome	Instrument	Coef.	SE	K-P F	Obs.
Std. total banks, 2020	Original IV	0.145***	0.051	14.87	247
Std. total banks, 2020	Recentered, full permutation	0.127***	0.043	16.72	247
Std. total banks, 2020	Recentered, 10-year blocks	0.128***	0.044	16.70	247
Direct-grid GDP, 2019	Original IV	7,581***	2,745	15.01	242
Direct-grid GDP, 2019	Recentered, full permutation	6,548***	2,273	16.87	242
Direct-grid GDP, 2019	Recentered, 10-year blocks	6,616***	2,318	16.84	242

Notes: The table reports headline 2SLS estimates for the original long-run IV and two recentered instruments under the 1st–99th percentile convention. The full permutation treats the realized annual missionary-flow values as exchangeable across years. The 10-year block permutation randomly reorders decade-sized blocks while preserving the within-block sequence of flows. All specifications use the same controls as the baseline long-run IV: railway duration, railway density, and province fixed effects. Modern bank counts are assigned from counties and districts to 1911 prefectures by their centroids. GDP is measured in millions of 2017 US dollars and constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons. [Table F18](#) reports the corresponding first and second stages. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table F18: First and Second Stages for Recentered Missionary-Flow IV

	Original IV	Recentered, full-year	Recentered, 10-year blocks
<i>Panel A. First stage: missionary institutions</i>			
Instrument	3.400*** (0.882)	5.923*** (1.448)	5.884*** (1.440)
RR Duration	0.0690** (0.0279)	0.0667** (0.0282)	0.0667** (0.0281)
RR Density	-685.3*** (128.9)	-665.5*** (116.0)	-667.3*** (117.3)
Province FE	Yes	Yes	Yes
Observations	247	247	247
K-P F	14.87	16.72	16.70
<i>Panel B. Second stage: standardized total banks, 2020</i>			
Missionary institutions	0.145*** (0.051)	0.127*** (0.043)	0.128*** (0.044)
RR Duration	-0.0046 (0.0065)	-0.0025 (0.0055)	-0.0026 (0.0056)
RR Density	47.5 (32.1)	41.2 (29.6)	41.7 (29.8)
Province FE	Yes	Yes	Yes
Observations	247	247	247
K-P F	14.87	16.72	16.70
Mean dep. var.	0.000	0.000	0.000
SD dep. var.	1.000	1.000	1.000

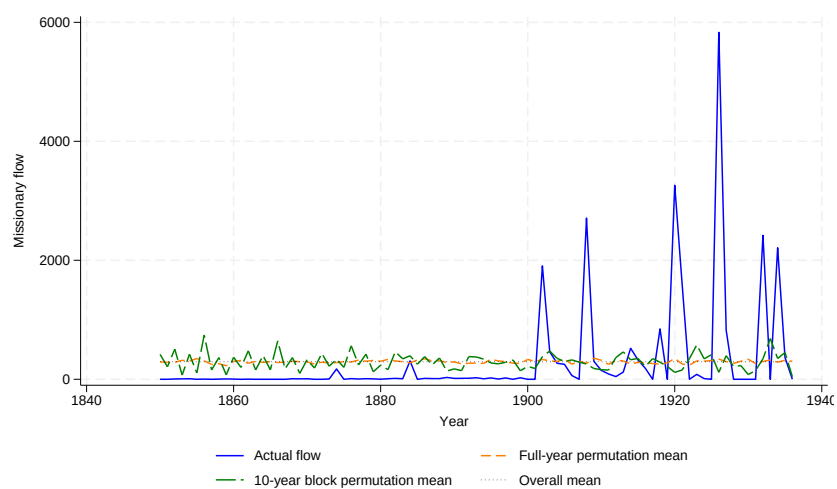
Notes: Robust standard errors are in parentheses. The first-stage dependent variable is missionary institutions in the outcome year. Modern county- and district-level bank counts are assigned to 1911 prefectures by their centroids. The mission stock, instrument, railway controls, and outcome use the 1st–99th percentile convention. All specifications include railway duration, railway density, and province fixed effects. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table F18: First and Second Stages for Recentered Missionary-Flow IV (continued)

	Original IV	Recentered, full-year	Recentered, 10-year blocks
<i>Panel C. First stage: missionary institutions, direct-grid GDP sample</i>			
Instrument	3.441*** (0.888)	5.985*** (1.457)	5.946*** (1.449)
RR Duration	0.0675** (0.0280)	0.0653** (0.0283)	0.0653** (0.0282)
RR Density	-694.7*** (131.6)	-674.0*** (119.0)	-675.7*** (120.4)
Province FE	Yes	Yes	Yes
Observations	242	242	242
K-P F	15.01	16.87	16.84
<i>Panel D. Second stage: direct-grid GDP, 2019</i>			
Missionary institutions	7,581*** (2,745)	6,548*** (2,273)	6,616*** (2,318)
RR Duration	-304.1 (330.5)	-187.2 (271.5)	-194.9 (276.9)
RR Density	1,972,945 (1,574,214)	1,618,686 (1,397,870)	1,641,973 (1,411,742)
Province FE	Yes	Yes	Yes
Observations	242	242	242
K-P F	15.01	16.87	16.84
Mean dep. var.	46,051	46,051	46,051
SD dep. var.	52,206	52,206	52,206

Notes: Robust standard errors are in parentheses. The first-stage dependent variable is missionary institutions in the outcome year. A separate GDP first stage is reported because the direct-grid GDP and bank outcomes contain different prefectures. The mission stock, instrument, railway controls, and GDP outcome use the 1st–99th percentile convention. All specifications include railway duration, railway density, and province fixed effects. GDP is measured in millions of 2017 US dollars and constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

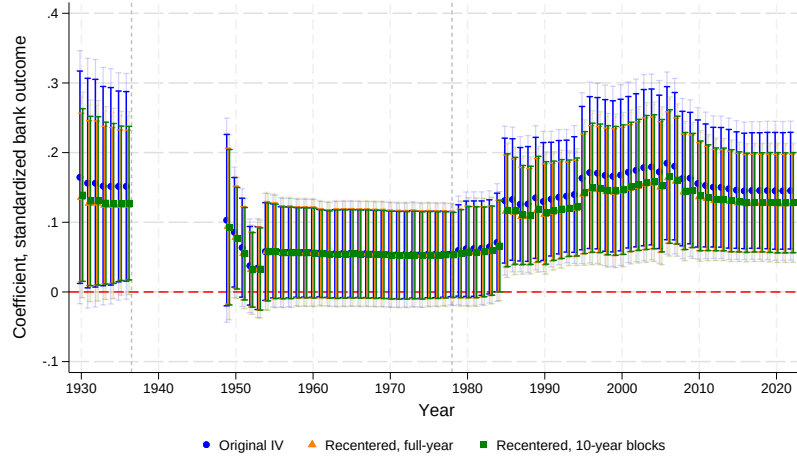
Figure F14: Actual and Simulated Missionary-Flow Histories



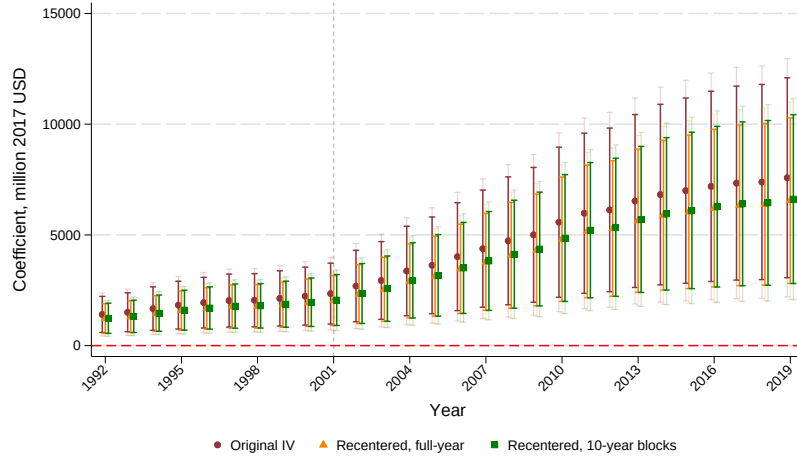
Notes: The figure compares the realized 1st–99th percentile national missionary-flow series with the average counterfactual flow series generated by full-year and 10-year block permutations. These simulated flow histories are used to compute the expected instrument that is subtracted from the original IV.

Figure F15: Long-Run IV Estimates with Recentered Instruments

(a) Standardized total banks



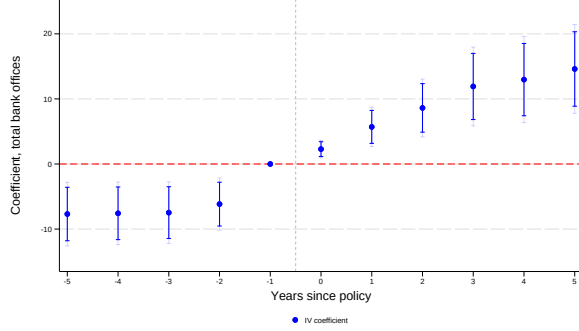
(b) Direct-grid GDP



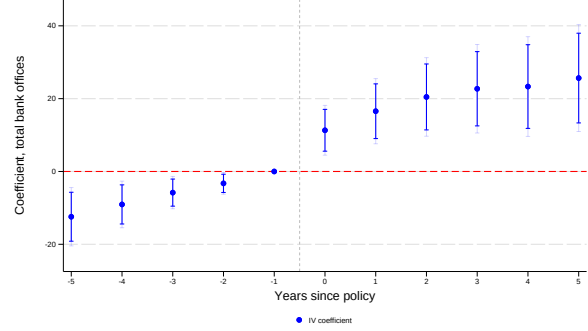
Notes: The figure compares the baseline long-run IV estimates with the recentered-IV estimates based on simulated missionary-flow histories. All inputs and outcomes use the 1st–99th percentile convention. The bank panel uses county- and district-centroid assignments to 1911 prefectures. The GDP panel uses the direct-grid outcome constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons. The point estimates are slightly smaller after recentering but follow the same time pattern as the baseline estimates.

Figure F16: Missionary Exposure and Banking Regulations: Total Banks

(a) Minimal Government Control (1886)

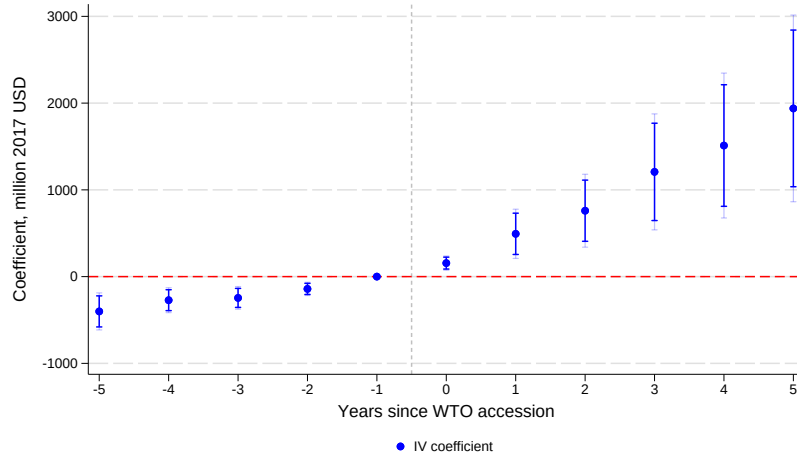


(b) WTO Transition Period (2006)



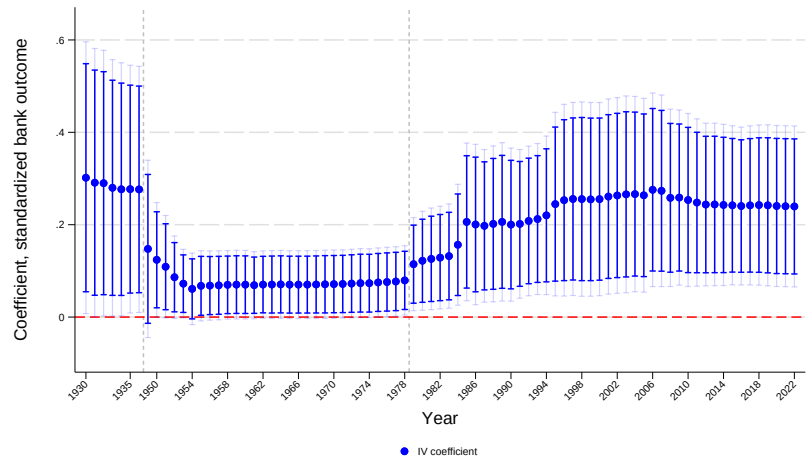
Notes: The figure reports event-study estimates from Equation 9 for the five years before and after the policy change, using total banks as the outcome. Dark and light intervals show 90% and 95% confidence intervals. Standard errors are clustered at the prefecture level. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid; the bank outcome and regression inputs use the 1st–99th percentile convention.

Figure F17: Missionary Exposure and WTO Accession: GDP



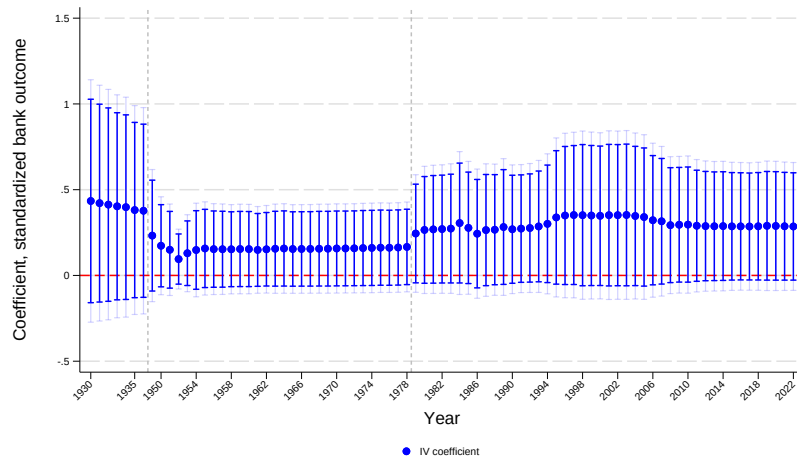
Notes: The figure reports event-study estimates from Equation 9 for the five years before and after the policy change, using GDP as the outcome. Dark and light intervals show 90% and 95% confidence intervals. Standard errors are clustered at the prefecture level. GDP is constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons; the outcome and regression inputs use the 1st–99th percentile convention.

Figure F18: Long-Run 2SLS Robustness: Railway Density and Two Railway Bins



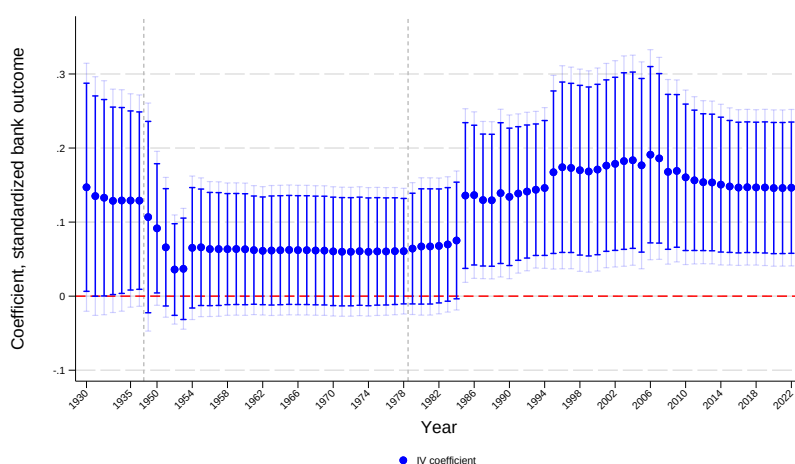
Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.

Figure F19: Long-Run 2SLS Robustness: Railway Density and Six Railway Bins



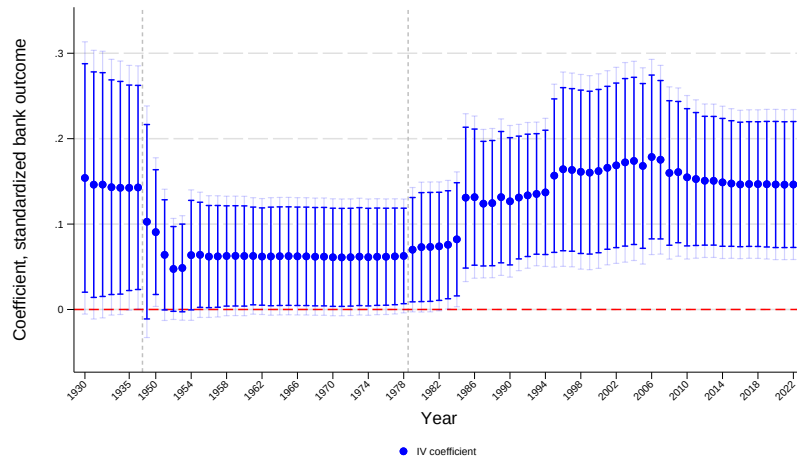
Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.

Figure F20: Long-Run 2SLS Robustness: Railway Mileage per Square Kilometer



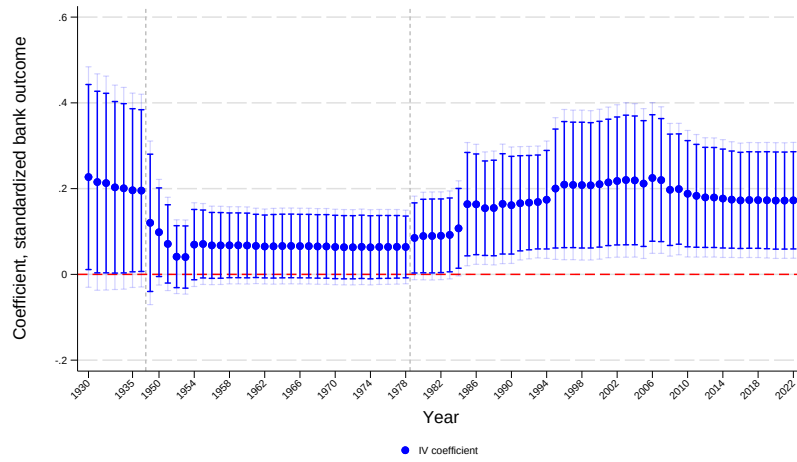
Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.

Figure F21: Long-Run 2SLS Robustness: Latitude and Longitude Controls



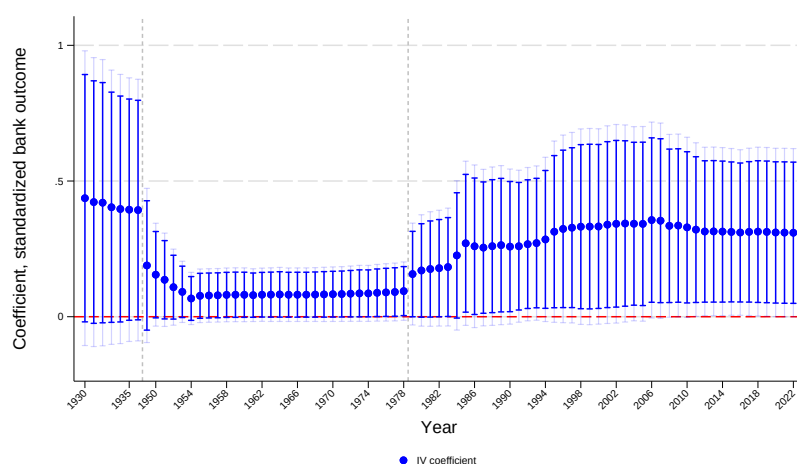
Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.

Figure F22: Long-Run 2SLS Robustness: Railway Count in 1936



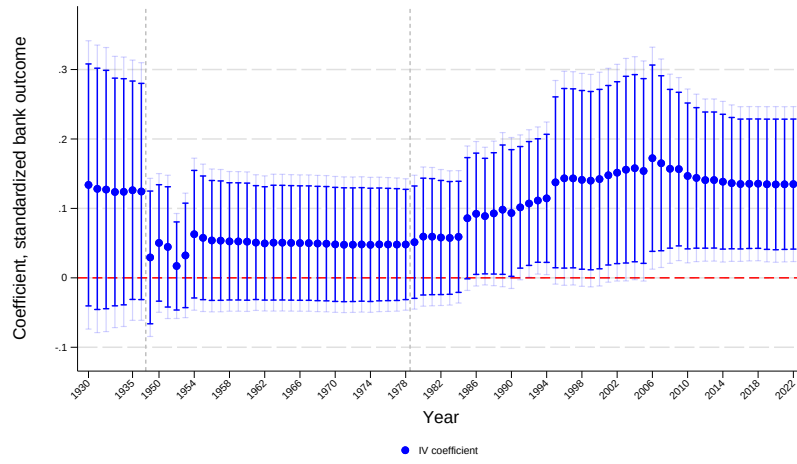
Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.

Figure F23: Long-Run 2SLS Robustness: Railway Duration and Two Railway Bins



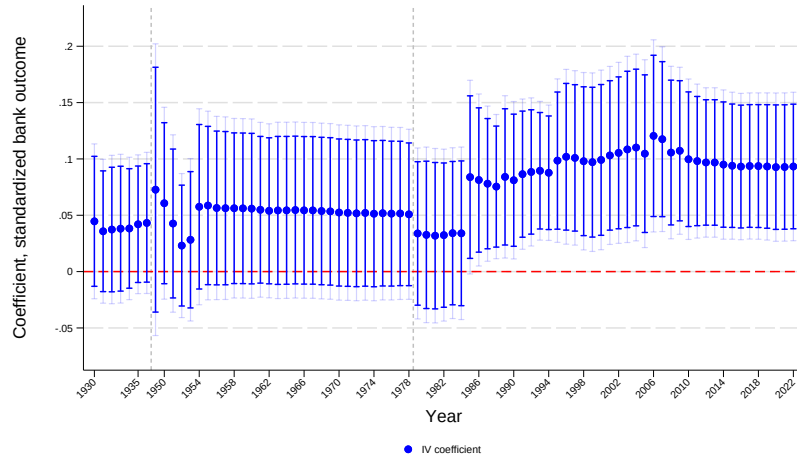
Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.

Figure F24: Long-Run 2SLS Robustness: Excluding Treaty Ports



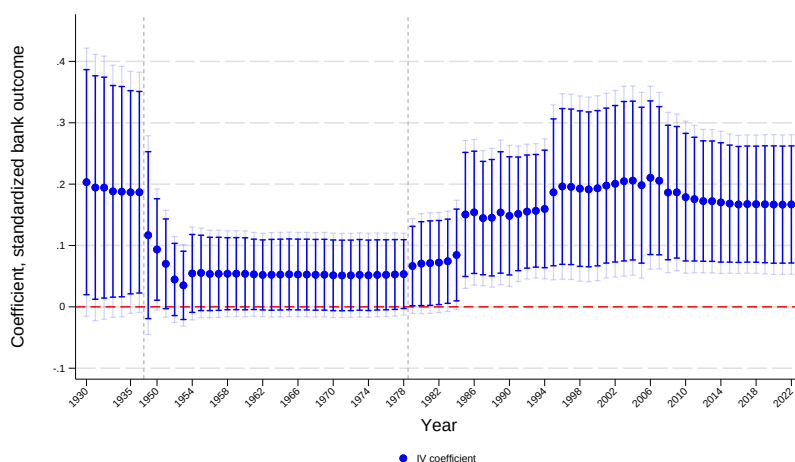
Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.

Figure F25: Long-Run 2SLS Robustness: Excluding the Three Largest Railway Outliers



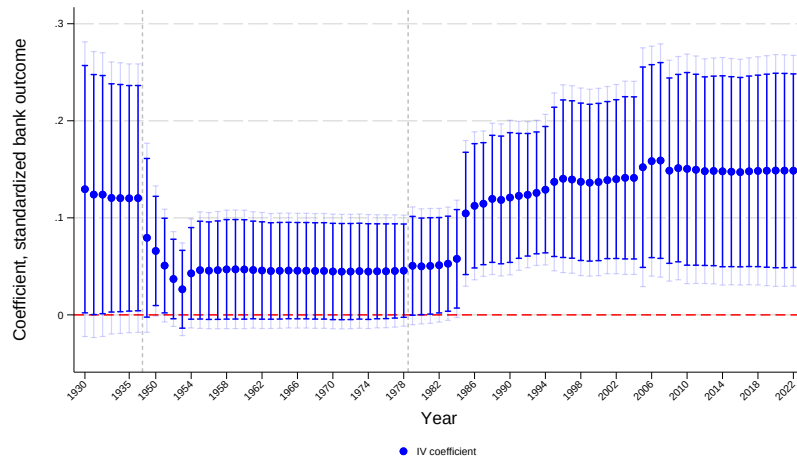
Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.

Figure F26: Long-Run 2SLS Robustness: Alternative Zero-Stage Window, 1850–1950



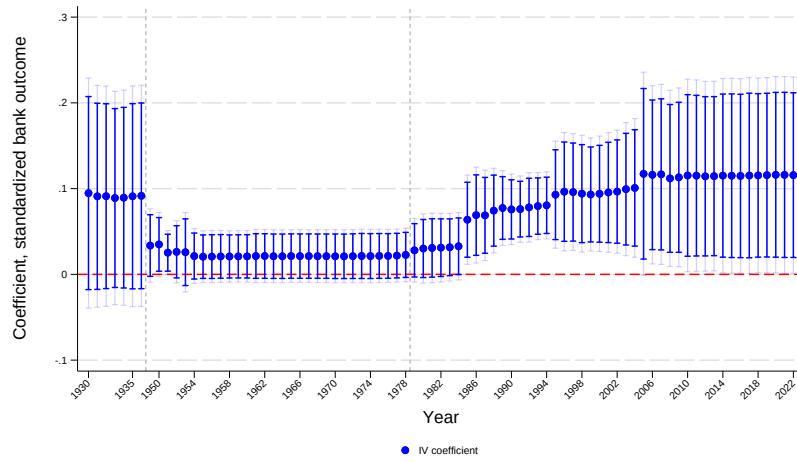
Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.

Figure F27: Long-Run 2SLS Robustness: Alternative Winsorization at the 0.5th and 99.5th Percentiles



Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.

Figure F28: Long-Run 2SLS Robustness: No Winsorization



Notes: The outcome is the standardized number of total banks in 1930–1936 and 1949–2022. Each point is the coefficient from a separate year-specific regression based on Equation 8, using both the historical modern-bank data from the difference-in-differences design and the recent bank data. Modern county- and district-level bank counts are assigned to 1911 prefectures by centroid. The baseline uses the 1st–99th percentile convention; Figure F27 instead uses the 0.5th–99.5th percentile convention; Figure F28 reports the fully unwinsorized specification. Dark and light intervals show 90% and 95% confidence intervals. The red dashed line marks zero; gray dashed lines mark 1949 and 1978. Source: China Financial Supervision and Administration Bureau.